

Table 7.8 Symbols Used in the Gene Map of the *E. coli* Chromosome

Genetic Symbols	Mutant Character	Enzyme or Reaction Affected
<i>araD</i>	Cannot use the sugar arabinose as a carbon source	L-Ribulose-5-phosphate-4-epimerase
<i>araA</i>		L-Arabinose isomerase
<i>araB</i>		L-Ribulokinase
<i>araC</i>	Requires the amino acid arginine for growth	N-Acetylglutamate synthetase N-Acetyl-γ-glutamokinase N-Acetylglutamic-γ-semialdehyde dehydrogenase Acetylornithine- <i>d</i> -transaminase Acetylornithinase Ornithine transcarbamylase Argininosuccinic acid synthetase Argininosuccinase
<i>argB</i>		
<i>argC</i>		
<i>argH</i>		
<i>argG</i>		
<i>argA</i>		
<i>argD</i>		
<i>argE</i>		
<i>argF</i>		
<i>argR</i>		
<i>aroA, B, C</i>	Arginine operon regulator	Shikimic acid to 3-Enolpyruvyl-shikimate-5-phosphate Biosynthesis of shikimic acid
<i>aroD</i>	Requires several aromatic amino acids and vitamins for growth	
<i>azi</i>	Resistant to sodium azide	
<i>bio</i>	Requires the vitamin biotin for growth	Carbamate kinase
<i>carA</i>	Requires uracil and arginine	
<i>carB</i>		
<i>chlA-E</i>	Cannot reduce chlorate	Nitrate-chlorate reductase and hydrogen lysase
<i>cysA</i>	Requires the amino acid cysteine for growth	3-Phosphoadenosine-5-phosphosulfate to sulfide
<i>cysB</i>		Sulfate to sulfide; four known enzymes
<i>cysC</i>		
<i>dapA</i>	Requires the cell-wall component diaminopimelic acid	Dihydrodipicolinic acid synthetase
<i>dapB</i>		N-Succinyl-diaminopimelic acid deacylase
<i>dap + bom</i>		Aspartic semialdehyde dehydrogenase
<i>dnaA-Z</i>	Requires the amino acid precursor homoserine and the cell-wall component diaminopimelic acid for growth	DNA biosynthesis D-Serine deaminase
<i>Dsd</i>	Mutation, DNA replication	
<i>fla</i>	Cannot use the amino acid D-serine as a nitrogen source	Galactokinase
<i>galA</i>	Flagella are absent	
<i>galB</i>	Cannot use the sugar galactose as a carbon source	
<i>galD</i>	Requires glycine	Galactose-1-phosphate uridyl transferase
<i>glyA</i>		Uridine-diphosphogalactose-4-epimerase
<i>gua</i>		Serine hydroxymethyl transferase
<i>H</i>	Requires the purine guanine for growth	Ten known enzymes*
<i>bis</i>	The H antigen is present	
<i>bsdR</i>	Requires the amino acid histidine for growth	
<i>ile</i>	Host restriction	Endonuclease R
<i>ilvA</i>	Requires the amino acid isoleucine for growth	Threonine deaminase
<i>ilvB</i>	Requires the amino acids isoleucine and valine for growth	α-Hydroxy-β-keto acid rectoisomerase
<i>ilvC</i>		α, β-Dihydroxyisovaleric dehydrase*
<i>ind</i> (indole)		Transaminase B
<i>λ</i> (<i>attλ</i>)	Cannot grow on tryptophan as a carbon source	Tryptophanase
	Chromosomal location where prophage λ is normally inserted	
<i>lacI</i>	<i>Lac</i> operon regulator	Galactoside permease β-Galactosidase
<i>lacY</i>	Unable to concentrate β-galactosides	
<i>lacZ</i>	Cannot use the sugar lactose as a carbon source	

continued

Table 7.8 continued

Genetic Symbols	Mutant Character	Enzyme or Reaction Affected
<i>lacO</i>	Constitutive synthesis of lactose operon proteins	Defective operator
<i>leu</i>	Requires the amino acid leucine for growth	Three known enzymes*
<i>lip</i>	Requires lipoate	
<i>lon</i> (long form)	Filament formation and radiation sensitivity are affected	
<i>lys</i>	Requires the amino acid lysine for growth	Diaminopimelic acid decarboxylase
<i>lys + met</i>	Requires the amino acids lysine and methionine for growth	
<i>λrec, malt</i>	Resistant to phage λ and cannot use the sugar maltose	Regulator for two operons
<i>maltK</i>	Cannot use the sugar maltose as a carbon source	Maltose permease
<i>man</i>	Cannot use mannose sugar	Phosphomannose isomerase
<i>mela</i>	Cannot use melibiose sugar	Alpha-galactosidase
<i>met A-M</i>	Requires the amino acid methionine for growth	Ten or more genes
<i>mtl</i>	Cannot use the sugar mannitol as a carbon source	Two enzymes
<i>muc</i>	Forms mucoid colonies	Regulation of capsular polysaccharide synthesis
<i>nalA</i>	Resistant to nalidixic acid	
<i>O</i>	The O antigen is present	
<i>pan</i>	Requires the vitamin pantothenic acid for growth	
<i>pabB</i>	Requires <i>p</i> -aminobenzoate	
<i>pbe A, B</i>	Requires the amino acid phenylalanine for growth	
<i>pbo</i>	Cannot use phosphate esters	Alkaline phosphatase
<i>pil</i>	Has filaments (pili) attached to the cell wall	
<i>plsB</i>	Deficient phospholipid synthesis	Glycerol 3-phosphate acyltransferase
<i>polA</i>	Repairs deficiencies	DNA polymerase I
<i>proA</i>	Requires the amino acid proline for growth	
<i>proB</i>		
<i>proC</i>		
<i>ptsI</i>	Defective phosphotransferase system	Pts-system enzyme I
<i>purA</i>	Requires certain purines for growth	Adenylosuccinate synthetase
<i>purB</i>		Adenylosuccinase
<i>purC, E</i>		5-Aminoimidazole ribotide (AIR) to 5-aminoimidazole-4-(N-succino carboximide) ribotide
<i>purD</i>		Biosynthesis of AIR
<i>pyrB</i>	Requires the pyrimidine uracil for growth	Aspartate transcarbamylase
<i>pyrC</i>		Dihydroorotase
<i>pyrD</i>		Dihydroorotic acid dehydrogenase
<i>pyrE</i>		Orotidylic acid pyrophosphorylase
<i>pyrF</i>		Orotidylic acid decarboxylase
<i>R gal</i>	Constitutive production of galactose	Repressor for enzymes involved in galactose production
<i>R1 pbo, R2 pbo</i>	Constitutive synthesis of phosphatase	Alkaline phosphatase repressor
<i>R try</i>	Constitutive synthesis of tryptophan	Repressor for enzymes involved in tryptophan synthesis
<i>RC</i> (RNA control)	Uncontrolled synthesis of RNA	
<i>recA</i>	Cannot repair DNA radiation damage or recombine	
<i>rhaA-D</i>	Cannot use the sugar rhamnose as a carbon source	Isomerase, kinase, aldolase, and regulator
<i>rpoA-D</i>	Problems of transcription	Subunits of RNA polymerase
<i>serA</i>	Requires the amino acid serine for growth	3-Phosphoglycerate dehydrogenase
<i>serB</i>		Phosphoserine phosphatase
<i>str</i>	Resistant to or dependent on streptomycin	
<i>suc</i>	Requires succinic acid	

continued

Table 7.8 *continued*

Genetic Symbols	Mutant Character	Enzyme or Reaction Affected
<i>supB</i>	Suppresses ochre mutations	t-RNA
<i>tonA</i>	Resistant to phages T1 and T5 (mutants called B/1, 5)	T1, T5 receptor sites absent
<i>tonB</i>	Resistant to phage T1 (mutants called B/1)	T1 receptor site absent
<i>T6, colK rec</i>	Resistant to phage T6 and colicine K	T6 and colicine receptor sites absent
<i>T4 rec</i>	Resistant to phage T4 (mutants called B/4)	T4 receptor site absent
<i>tsx</i>	T6 resistance	
<i>thi</i>	Requires the vitamin thiamine for growth	
<i>tolC</i>	Tolerance to colicine E1	
<i>tbr</i>	Requires the amino acid threonine for growth	
<i>thy</i>	Requires the pyrimidine thymine for growth	Thymidylate synthetase
<i>trpA</i>	Requires the amino acid tryptophan for growth	Tryptophan synthetase, A protein
<i>trpB</i>		Tryptophan synthetase, B protein
<i>trpC</i>		Indole-3-glycerolphosphate synthetase
<i>trpD</i>		Phosphoribosyl anthranilate transferase
<i>trpE</i>		Anthranilate synthetase
<i>tyrA</i>	Requires the amino acid tyrosine for growth	Chorismate mutase T-prephenate dehydrogenase
<i>tyrR</i>		Regulates three genes
<i>uvrA-E</i>	Resistant to ultraviolet radiation	Ultraviolet-induced lesions in DNA are reactivated
<i>valS</i>	Cannot charge Valyl-tRNA	Valyl-tRNA synthetase
<i>xyl</i>	Cannot use the sugar xylose as a carbon source	

Source: B. J. Bachmann and K. B. Low, "Linkage map of *Escherichia coli* K-12," *Microbiological Reviews*, 44:1–56. Copyright © 1990 American Society for Microbiology, Washington, D.C. Reprinted by permission.

* Denotes enzymes controlled by the homologous gene loci of *Salmonella typhimurium*.

S U M M A R Y

STUDY OBJECTIVE 1: To define bacteria and bacterial viruses and learn about methods of studying them 149–154

Prokaryotes (bacteria) usually have a single circular chromosome of double-stranded DNA. A bacteriophage consists of a chromosome wrapped in a protein coat. Its chromosome can be DNA or RNA. Phenotypes of bacteria include colony morphology, nutritional requirements, and drug resistance. Phage phenotypes include plaque morphology and host range. Replica-plating is a rapid screening technique for assessing the phenotype of a bacterial clone.

STUDY OBJECTIVE 2: To study life cycles and sexual processes in bacteria and bacteriophages 154–166

In transformation, a competent bacterium can take up relatively large pieces of DNA from the medium. This DNA can be incorporated into the bacterial chromosome.

During the process of conjugation, the fertility factor, F, is passed from an F⁺ to an F[−] cell. If the F factor integrates

into the host chromosome, an Hfr cell results that can pass its entire chromosome into an F[−] cell. The F factor is the last region to cross into the F[−] cell.

In transduction, a phage protein coat containing some of the host chromosome passes to a new host bacterium. Again, recombination with this new chromosomal segment can take place.

STUDY OBJECTIVE 3: To make use of the sexual processes of bacteria and their viruses to map their chromosomes 155–171

We can map the phage chromosome by measuring recombination after a bacterium has been simultaneously infected by two strains of the virus carrying different alleles. In *E. coli*, mapping is most efficiently accomplished via interrupted mating and transduction. The former provides information on general gene arrangement and the latter provides finer details.

S O L V E D P R O B L E M S

PROBLEM 1: A wild-type strain of *B. subtilis* is transformed by DNA from a strain that cannot grow on galactose (*gal*[−]) and also needs biotin for growth (*bio*[−]). Transformants are isolated by exposing the transformed cells to minimal medium with penicillin, killing the wild-type cells. After the penicillin is removed, replica-plating is used to establish the genotypes of 30 transformants:

Class 1 <i>gal</i> [−] <i>bio</i> [−]	17
Class 2 <i>gal</i> [−] <i>bio</i> ⁺	4
Class 3 <i>gal</i> ⁺ <i>bio</i> [−]	9

What is the relative co-occurrence of these two loci?

Answer: The three classes of colonies represent the three possible transformant groups. Classes 2 and 3 are single transformants and class 1 is the double transformant. We are interested in the relative co-occurrence of the two loci. Therefore we divide the number of double transformants by the total: $r = 17/(17 + 4 + 9) = 0.57$. This is a relative value inverse to a map distance; the larger it is, the closer the loci are to each other.

PROBLEM 2: A *gal*[−] *bio*[−] *attλ*[−] strain of *E. coli* is transduced by P22 phages from a wild-type strain. Transductants are selected for by growing the cells with galactose as the sole energy source. Replica-plating and testing for lysogenic ability gives the genotypes of 106 transformants:

Class 1 <i>gal</i> ⁺ <i>bio</i> [−] <i>attλ</i> [−]	71
Class 2 <i>gal</i> ⁺ <i>bio</i> ⁺ <i>attλ</i> [−]	0
Class 3 <i>gal</i> ⁺ <i>bio</i> [−] <i>attλ</i> ⁺	9
Class 4 <i>gal</i> ⁺ <i>bio</i> ⁺ <i>attλ</i> ⁺	26

What is the gene order, and what are the relative cotransduction frequencies?

Answer: We have selected all transductants that are *gal*⁺. Class 2 is in the lowest frequency (0) and therefore represents the quadruple crossover between the transducing DNA and the host chromosome. From this, we see that *attλ* must be in the middle because this low-probability event is the one that would have switched only the middle locus. In other words, the two end loci would be recombinant, and the middle locus would have the host allele. We can only calculate two cotransduction frequencies because these are selected data. Note that in class 1, there is no cotransduction between *gal* and either of the other two loci; class 2 would show the cotransduction of *gal* and *bio*; class 3 represents the cotransduction of *gal* and *attλ*; and class 4 represents the cotransduction of *gal* and both other loci. Therefore, cotransduction values are

$$\begin{aligned} gal-att\lambda &= (9 + 26)/106 = 35/106 = 0.33 \\ gal-bio &= (0 + 26)/106 = 26/106 = 0.25. \end{aligned}$$

E X E R C I S E S A N D P R O B L E M S *

BACTERIA AND BACTERIAL VIRUSES
IN GENETIC RESEARCH

1. What is the nature and substance of prokaryotic chromosomes and viral chromosomes? Are viruses alive?

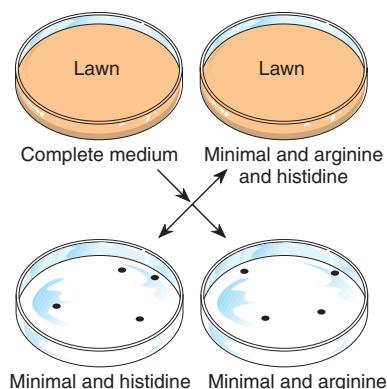
TECHNIQUES OF CULTIVATION

2. What are the differences between a heterotroph and an auxotroph? a minimal and a complete medium? an enriched and a selective medium?
3. What are the differences between a plaque and a colony?

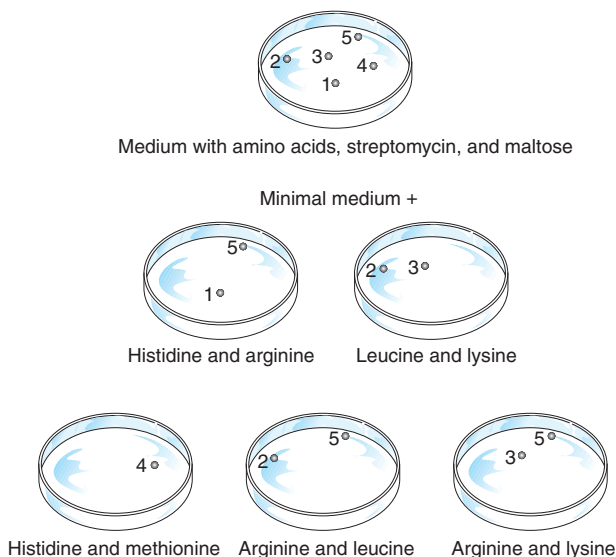
BACTERIAL PHENOTYPES

4. What genotypic notation indicates alleles that make a bacterium

- a. resistant to penicillin?
 - b. sensitive to azide?
 - c. require histidine for growth?
 - d. unable to grow on galactose?
 - e. able to grow on glucose?
 - f. susceptible to phage T1 infection?
5. An *E. coli* cell is placed on a petri plate containing λ phages. It produces a colony overnight. By what mechanisms might it have survived?
 6. An *E. coli* lawn is formed on a petri plate containing complete medium. Replica-plating is used to transfer material to plates containing minimal medium and combinations of the amino acids arginine and histidine (see the figure). Give the genotype of the original strain as well as the genotypes of the odd colonies found growing on the plates.



7. Prototrophic Hfr *E. coli* strain G11, sensitive to streptomycin and *malt*⁺ (can use maltose) is used in a conjugation experiment. The *str* locus is one of the last to be transferred, whereas the *malt* locus is one of the first. This strain is mated to an F⁻ strain resistant to streptomycin, *malt*⁻ (cannot utilize maltose), and requiring five amino acids (histidine, arginine, leucine, lysine, and methionine). Recombinants are selected for by plating on a medium with streptomycin, with maltose as the sole carbon source, and all five amino acids present. Thus, all recombinant F⁻ cells will grow irrespective of their amino acid requirements. Five colonies are grown on the original plate with streptomycin, maltose, and all five amino acids in question (see the figure). These colonies are replica-plated onto minimal medium containing various amino acids. What are the genotypes of each of the five colonies?



8. A petri plate with complete medium has six colonies growing on it after one of the conjugation experiments described earlier. The colonies are numbered, and the plate is used as a master to replicate onto plates of glucose-containing selective (minimal) medium with various combinations of additives. From the following data, which show the presence (+) or absence (-) of growth, give your best assessment of the genotypes of the six colonies.

	Colony					
On Minimal Medium +	1	2	3	4	5	6
Nothing	-	-	+	-	-	-
Xylose + arginine	+	-	+	+	-	-
Xylose + histidine	-	-	+	-	-	-
Arginine + histidine	-	+	+	+	-	-
Galactose + histidine	-	-	+	-	-	+
Threonine + isoleucine	-	-	+	-	+	-
Threonine + valine	-	-	+	-	-	-
Threonine + valine + lactose	-	-	+	-	-	-

VIRAL PHENOTYPES

9. Give possible genotypes of an *E. coli*-phage T1 system in which the phage cannot grow on the bacterium. Give genotypes for a T1 phage that can grow on the bacterium.

SEXUAL PROCESSES IN BACTERIA AND BACTERIOPHAGES

10. What is a plasmid? How does one integrate into a host's chromosome? How does it leave?
11. In conjugation experiments, one Hfr strain should carry a gene for some sort of sensitivity (e.g., *azi*^s or *str*^s) so that the Hfr donors can be eliminated on selective media after conjugation has taken place. Should this locus be near to or far from the origin of transfer point of the Hfr chromosome? What are the consequences of either alternative?
12. How does a geneticist doing interrupted mating experiments know that the locus for the drug-sensitivity allele, used to eliminate the Hfr bacteria after conjugation, has crossed into the F⁻ strain?
13. Diagram the step-by-step events required to integrate foreign DNA into a bacterial chromosome in each of the three processes outlined in the chapter (transformation, conjugation, transduction). Do the same for viral recombination. (See also TRANSDUCTION)

14. The DNA from a prototrophic strain of *E. coli* is isolated and used to transform an auxotrophic strain deficient in the synthesis of purines (*purB*⁻), pyrimidines (*pyrC*⁻), and the amino acid tryptophan (*trp*⁻). Tryptophan was used as the marker to determine whether transformation had occurred (the selected marker). What are the gene order and the relative co-occurrence frequencies between loci, given these data:

<i>trp</i> ⁺ <i>pyrC</i> ⁺ <i>purB</i> ⁺	86
<i>trp</i> ⁺ <i>pyrC</i> ⁺ <i>purB</i> ⁻	4
<i>trp</i> ⁺ <i>pyrC</i> ⁻ <i>purB</i> ⁺	67
<i>trp</i> ⁺ <i>pyrC</i> ⁻ <i>purB</i> ⁻	14

15. Using the data in figure 7.16, draw a tentative map of the *E. coli* chromosome.
16. Three Hfr strains of *E. coli* (P4X, KL98, and Ra-2) are mated individually with an auxotrophic F⁻ strain using interrupted mating techniques. Using the following data, construct a map of the *E. coli* chromosome, including distances in minutes.

Donor Loci	Approximate Time of Entry		
	Hfr P4X	Hfr KL98	Hfr Ra-2
<i>gal</i> ⁺	11	67	70
<i>thr</i> ⁺	94	50	87
<i>xyl</i> ⁺	73	29	8
<i>lac</i> ⁺	2	58	79
<i>bis</i> ⁺	38	94	43
<i>ilv</i> ⁺	77	33	4
<i>argG</i> ⁺	62	18	19

How many different petri plates and selective media are needed?

17. Design an experiment using interrupted mating and create a resulting possible data set that would correctly map five of the loci on the *E. coli* chromosome (fig. 7.27).
18. Lederberg and his colleagues (Nester, Schafer, and Lederberg, 1963, *Genetics* 48:529) determined gene order and relative distance between genes using three markers in the bacterium *Bacillus subtilis*. DNA from a prototrophic strain (*trp*⁺ *bis*⁺ *tyr*⁺) was used to transform the auxotroph. The seven classes of transformants, with their numbers, are tabulated as follows:

<i>trp</i> ⁺	<i>trp</i> ⁻	<i>trp</i> ⁻	<i>trp</i> ⁺	<i>trp</i> ⁺	<i>trp</i> ⁻	<i>trp</i> ⁺
<i>bis</i> ⁻	<i>bis</i> ⁺	<i>bis</i> ⁻	<i>bis</i> ⁺	<i>bis</i> ⁻	<i>bis</i> ⁺	<i>bis</i> ⁺
<i>tyr</i> ⁻	<i>tyr</i> ⁻	<i>tyr</i> ⁺	<i>tyr</i> ⁻	<i>tyr</i> ⁺	<i>tyr</i> ⁺	<i>tyr</i> ⁺
2,600	418	685	1,180	107	3,660	11,940

Outline the techniques used to obtain these data. Taking the loci in pairs, calculate co-occurrences. Construct the most consistent linkage map of these loci.

19. In a transformation experiment, an *a*⁺ *b*⁺ *c*⁺ strain is used as the donor and an *a*⁻ *b*⁻ *c*⁻ strain as the recipient. One hundred *a*⁺ transformants are selected and then replica-plated to determine whether *b*⁺ and *c*⁺ are present. What can you conclude about the relative positions of the genes, based on the listed genotypes?

<i>a</i> ⁺ <i>b</i> ⁻ <i>c</i> ⁻	21
<i>a</i> ⁺ <i>b</i> ⁻ <i>c</i> ⁺	69
<i>a</i> ⁺ <i>b</i> ⁺ <i>c</i> ⁻	3
<i>a</i> ⁺ <i>b</i> ⁺ <i>c</i> ⁺	7

20. In a transformation experiment, an *a*⁺ *b*⁺ *c*⁻ strain is used as donor and an *a*⁻ *b*⁻ *c*⁺ strain as recipient. If you select for *a*⁺ transformants, the least frequent class is *a*⁺ *b*⁺ *c*⁺. What is the order of the genes?
21. A mating between *bis*⁺, *leu*⁺, *thr*⁺, *pro*⁺, *str*^s cells (Hfr) and *bis*⁻, *leu*⁻, *thr*⁻, *pro*⁻, *str*^r cells (F⁻) is allowed to continue for twenty-five minutes. The mating is stopped, and the genotypes of the recombinants are determined. What is the first gene to enter, and what is the probable gene order, based on the following data?

Genotype	Number of Colonies
<i>bis</i> ⁺	0
<i>leu</i> ⁺	12
<i>thr</i> ⁺	27
<i>pro</i> ⁺	6

22. a. In a transformation experiment, the donor is *trp*⁺ *leu*⁺ *arg*⁺, and the recipient is *trp*⁻ *leu*⁻ *arg*⁻. The selection process is for *trp*⁺ transformants, which are then further tested. Forty percent are *trp*⁺ *arg*⁺; 5% are *trp*⁺ *leu*⁺. In what two possible orders could the genes be arranged?
- b. You can do only one more transformation to determine gene order. You must use the same donor and recipient, but you can change the selection procedure for the initial transformants. What should you do, and what results should you expect for each order you proposed in a?
23. DNA from a bacterial strain that is *a*⁺ *b*⁺ *c*⁺ is used to transform a strain that is *a*⁻ *b*⁻ *c*⁻. The numbers of each transformed genotype appear. What can we say about the relative position of the genes?

Genotype	Number
<i>a</i> ⁺ <i>b</i> ⁻ <i>c</i> ⁻	214
<i>a</i> ⁻ <i>b</i> ⁺ <i>c</i> ⁻	231
<i>a</i> ⁻ <i>b</i> ⁻ <i>c</i> ⁺	206
<i>a</i> ⁺ <i>b</i> ⁺ <i>c</i> ⁻	11
<i>a</i> ⁺ <i>b</i> ⁺ <i>c</i> ⁺	6
<i>a</i> ⁺ <i>b</i> ⁻ <i>c</i> ⁺	93
<i>a</i> ⁻ <i>b</i> ⁺ <i>c</i> ⁺	14

24. An Hfr strain that is $a^+ b^+ c^+ d^+ e^+$ is mated with an F^- strain that is $a^- b^- c^- d^- e^-$. The mating is interrupted every five minutes, and the genotypes of the F^- recombinants are determined. The results appear following. (A *plus* indicates appearance; a *minus* the lack of the locus.) Draw a map of the chromosome and indicate the position of the F factor, the direction of transfer, and the minutes between genes.

Time	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
5	—	—	—	—	—
10	+	—	—	—	—
15	+	—	—	—	—
20	+	—	—	—	—
25	+	—	—	—	—
30	+	—	—	+	—
35	+	—	—	+	—
40	+	+	—	+	—
45	+	+	—	+	—
50	+	+	—	+	—
55	+	+	—	+	—
60	+	+	—	+	—
65	+	+	+	+	—
70	+	+	+	+	—
75	+	+	+	+	+

25. A bacterial strain that is $lys^+ his^+ val^+$ is used as a donor, and $lys^- his^- val^-$ as the recipient. Initial transformants are isolated on minimal medium + histidine + valine.
- What genotypes will grow on this medium?
 - These colonies are replicated to minimal medium + histidine, and 75% of the original colonies grow. What genotypes will grow on this medium?
 - The original colonies are also replicated to minimal medium + valine, and 6% of the colonies grow. What genotypes will grow on this medium?
 - Finally, the original colonies are replicated to minimal medium. No colonies grow. From this information, what genotypes will grow on minimal medium + histidine and on minimal medium + valine?
 - Based on this information, which gene is closer to *lys*?
 - The original transformation is repeated, but the original plating is on minimal medium + lysine + histidine. Fifty colonies appear. These colonies are replicated to determine their genotypes, with these results:

$val^+ his^+ lys^+$	0
$val^+ his^- lys^+$	37
$val^+ his^+ lys^-$	3

Based on all the results, what is the most likely gene order?

LIFE CYCLES OF BACTERIOPHAGES

26. Define *prophage*, *lysate*, *lysogeny*, and *temperate phage*.
27. Outline an experiment to demonstrate that two phages do not undergo recombination until a bacterium is infected simultaneously with both.
28. Doermann (1953, *Cold Spr. Harb. Symp. Quant. Biol.* 18:3) mapped three loci of phage T4: minute, rapid lysis, and turbid. He infected *E. coli* cells with both the triple mutant ($m^- r^- tu^-$) and the wild-type ($m^+ r^+ tu^+$) and obtained the following data:

<i>m</i>	m^+	<i>m</i>	<i>m</i>	m^+	m^+	<i>m</i>	m^+
<i>r</i>	<i>r</i>	r^+	<i>r</i>	r^+	<i>r</i>	r^+	r^+
<i>tu</i>	<i>tu</i>	<i>tu</i>	tu^+	<i>tu</i>	tu^+	tu^+	tu^+
3,467	474	162	853	965	172	520	3,729

What is the linkage relationship among these loci? In your answer include gene order, relative distance, and coefficient of coincidence.

29. Wild-type phage T4 (r^+) produce small, turbid plaques, whereas *rII* mutants produce large, clear plaques. Four *rII* mutants (*a–d*) are crossed. (Assume, for the purposes of this problem, that *a–d* are four closely linked loci. The actual structure of the *rII* region is presented in chapter 12. Here, assume that $a \times b$ means $a^- b^+ c^+ d^+ \times a^+ b^- c^+ d^+$.) These percentages of wild-type plaques are obtained in crosses:

$a \times b$	0.3
$a \times c$	1.0
$a \times d$	0.4
$b \times c$	0.7
$b \times d$	0.1
$c \times d$	0.6

Deduce a genetic map of these four mutants.

30. A phage cross is performed between $a^+ b^+ c^+$ and $a^- b^- c^-$ phage. Based on these results, derive a complete map:

$a^+ b^+ c^+$	1,801
$a^+ b^+ c^-$	954
$a^+ b^- c^+$	371
$a^+ b^- c^-$	160
$a^- b^+ c^+$	178
$a^- b^+ c^-$	309
$a^- b^- c^+$	879
$a^- b^- c^-$	1,850
	6,502

31. The *rII* mutants of T4 phage will grow and produce large plaques on strain B; *rII* mutants will not grow on strain K12. Certain crosses are performed in strain B. (As with question 29, assume that the three mutants are of three separate loci in the *rII* region.) By diluting and plating on strain B, it is determined that each experiment generates about 250×10^7 phage. By dilution, approximately 1/10,000 of the progeny are plated on K12 to generate these wild-type recombinants (plaques on K12):

1×2	50
1×3	25
2×3	75

Draw a map of these three mutants (1, 2, and 3) and indicate the distances between them.

TRANSDUCTION

32. Define and illustrate *specialized* and *generalized transduction*.
33. In *E. coli*, the three loci *ara*, *leu*, and *ilvH* are within 1/2-minute map distance apart. To determine the exact order and relative distance, the prototroph (*ara*⁺ *leu*⁺ *ilvH*⁺) was infected with transducing phage P1. The lysate was used to infect the auxotroph (*ara*⁻ *leu*⁻ *ilvH*⁻). The *ara*⁺ classes of transductants were selected to produce the following data:

<i>ara</i> ⁺	<i>ara</i> ⁺	<i>ara</i> ⁺	<i>ara</i> ⁺
<i>leu</i> ⁻	<i>leu</i> ⁺	<i>leu</i> ⁻	<i>leu</i> ⁺
<i>ilvH</i> ⁻	<i>ilvH</i> ⁻	<i>ilvH</i> ⁺	<i>ilvH</i> ⁺
32	9	0	340

Outline the specific techniques used to isolate the various transduced classes. What is the gene order and what are the relative cotransduction frequencies between genes? Why do some classes occur so infrequently?

34. Consider this portion of an *E. coli* chromosome:

thr *ara leu*

Three *ara* loci, *ara-1*, *ara-2*, and *ara-3*, are located in the *ara* region. A mutant of each locus (*ara-1*⁻, *ara-2*⁻, and *ara-3*⁻) was isolated, and their order with respect to *thr* and *leu* was analyzed by transduction. The donor was always *thr*⁺ *leu*⁺ and the recipient was always *thr*⁻ *leu*⁻. Each *ara* mutant was used as a donor in one cross and as a recipient in another; *ara*⁺ transductants were selected in each case. The *ara*⁺ transductants were then scored for *leu*⁺ and *thr*⁺. Based on the following results, determine the order of the *ara*⁻ mutants with respect to *thr* and *leu*.

Cross	Recipient	Donor	Ratio: $\frac{thr^- ara^+ leu^+}{thr^+ ara^+ leu^-}$
1	<i>ara-1</i> ⁻	<i>ara-2</i> ⁻	48.5
2	<i>ara-2</i> ⁻	<i>ara-1</i> ⁻	2.4
3	<i>ara-1</i> ⁻	<i>ara-3</i> ⁻	4.0
4	<i>ara-3</i> ⁻	<i>ara-1</i> ⁻	19.1
5	<i>ara-2</i> ⁻	<i>ara-3</i> ⁻	1.5
6	<i>ara-3</i> ⁻	<i>ara-2</i> ⁻	25.5

35. An *E. coli* strain that is *leu*⁺ *thr*⁺ *azi*^r is used as a donor in a transduction of a strain that is *leu*⁻ *thr*⁻ *azi*^s. Either *leu*⁺ or *thr*⁺ transductants are selected and then scored for unselected markers. The results are obtained:

Selected Marker	Unselected Markers
<i>leu</i> ⁺	48% <i>azi</i> ^r
<i>leu</i> ⁺	2% <i>thr</i> ⁺
<i>thr</i> ⁺	3% <i>leu</i> ⁺
<i>thr</i> ⁺	0% <i>azi</i> ^r

What is the order of the three loci?

CRITICAL THINKING QUESTIONS

- Consider the data from table 7.4. Is there another way to interpret the data other than coming from a circular bacterial chromosome?
- Why might transformation have evolved, given that the bacterium is importing DNA from a dead organism?

8

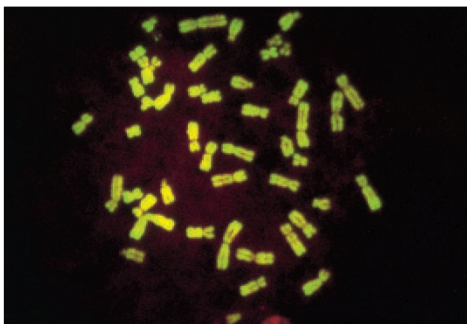
CYTOGENETICS

STUDY OBJECTIVES

1. To observe the nature and consequences of chromosomal breakage and reunion 178
2. To observe the nature and consequences of variation in chromosome numbers in human and nonhuman organisms 190

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Chromosomes of an individual with trisomy 21,
Down syndrome. (© Dr. Ram Verma /Phototake, NYC.)

Our understanding of the chromosomal theory of genetics grew primarily through mapping loci, using techniques that require alternative allelic forms, or mutations, of these loci. Changes in the genetic material also occur at a much coarser level—the level of **cytogenetics**, which is a level visible under the light microscope. The word *cytogenetics* combines the words *cytology* and *genetics*; cytology is the study of cells. Cytogenetics is thus defined as the study of cells from the perspective of genetics. In practice, it is the study of changes in the gross structure and number of chromosomes in cells. In this chapter, we investigate how these alterations happen and what their consequences are to the organism.

VARIATION IN CHROMOSOMAL STRUCTURE



In general, chromosomes can break due to ionizing radiation, physical stress, or chemical compounds. When a break occurs in the chromosome before DNA replication, during the S phase of the cell cycle (see fig. 3.6), the break itself is replicated. After the S phase, any breaks that occur affect single chromatids.

Every break in a chromatid produces two ends. These ends have been described as “sticky,” meaning simply that enzymatic processes of the cell tend to reunite them. Broken ends do not attach to the undamaged terminal ends of other chromosomes. (Normal chromosomal ends are capped with structures called *telomeres*—see chapter 15.) If broken ends are not brought together, they can remain broken. But, if broken chromatid ends are brought into apposition, they may rejoin in any of several ways. First, the two broken ends of a single chromatid can reunite. Second, the broken end of one chromatid can fuse with the broken end of another chromatid, resulting in an exchange of chromosomal material and a new combination of alleles. Multiple breaks can lead to a variety of alternative recombinations. These chromosomal aberrations have major genetic, evolutionary, and medical consequences. The types of breaks and reunions discussed in this chapter can be summarized as follows:

- I. Noncentromeric breaks
 - A. Single breaks
 - 1. Restitution
 - 2. Deletion
 - 3. Dicentric bridge
 - B. Two breaks (same chromosome)
 - 1. Deletion
 - 2. Inversion
 - C. Two breaks (nonhomologous chromosomes)

II. Centromeric breaks

- A. Fission
- B. Fusion

Single Breaks

If a chromosome breaks, the broken ends may rejoin. When the broken ends of a single chromatid rejoin (in a process called restitution), there is no consequence to the break. If they do not rejoin, the result is an **acentric fragment**, without a centromere, and a **centric fragment**, with a centromere. The centric fragment migrates normally during the division process because it has a centromere. The acentric fragment, however, is soon lost. It is subsequently excluded from the nuclei formed and eventually degrades. In other words, the viable, centric part of the chromosome has suffered a deletion. After mitosis, the daughter cell that receives the **deletion chromosome** may show several effects.

Pseudodominance is one possible effect. (This term was used in chapter 5 when we described alleles located on the X chromosome. With only one copy of the locus present, a recessive allele in males shows itself in the phenotype as if it were dominant—hence the term *pseudodominance*.) The normal chromosome homologous to the deletion chromosome has loci in the region, and recessive alleles show pseudodominance. A second possible effect is that, depending on the length of the deleted segment and the specific loci lost, the imbalance the deletion chromosome creates in the daughter cell may be lethal. If the deletion occurs before or during meiosis, it may be observed under the microscope. We discuss this event later in the chapter.

A single break can have yet another effect. Occasionally, the two centric fragments of a single chromosome may join, forming a two-centromere, or **dicentric, chromosome** and leaving the two acentric fragments to join or, alternatively, remain as two fragments (fig. 8.1). The acentric fragments are lost, as mentioned before. Because the centromeres are on sister chromatids, the dicentric fragment is pulled to opposite ends of a mitotic cell forming a bridge there; or, if meiosis is occurring, the dicentric fragment is pulled apart during the second meiotic division. The ultimate fate of this bridge is breakage as the spindle fibers pull the centromeres to opposite poles (or possibly exclusion from a new nucleus if the bridge is not broken).

The dicentric chromosome does not necessarily break in the middle, and subsequent processes exacerbate the imbalance created by an off-center break: duplications occur on one strand, whereas more deletions occur on the other (fig. 8.2). In addition, the “sticky” ends produced on both fragments increase the likelihood of repeating this **breakage-fusion-bridge cycle** in each generation. The great imbalances resulting from the duplications and deletions usually cause the cell line to die within several generations.

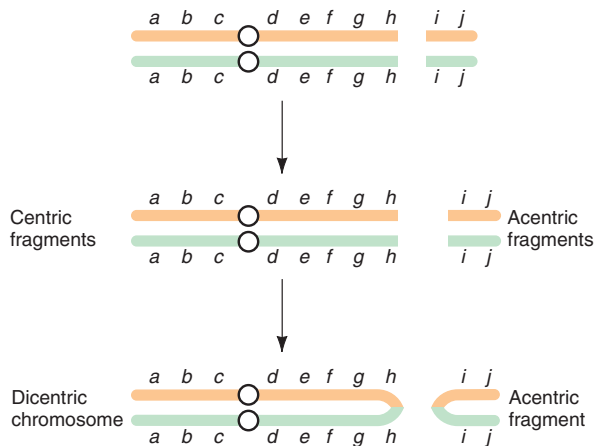


Figure 8.1 Chromosomal break with subsequent reunion to form a dicentric chromosome and an acentric fragment.

Two Breaks in the Same Chromosome



Deletion

Figure 8.3 shows two of the possible results when two breaks occur in the same chromosome. One alternative is a reunion that omits an acentric fragment, which is then lost. The centric piece, missing the acentric fragment (e-f-g in fig. 8.3), is a deletion chromosome. An organism

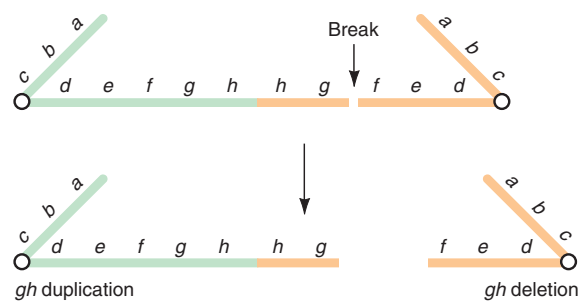


Figure 8.2 Breakage of a dicentric bridge causes duplications and additional deficiencies.

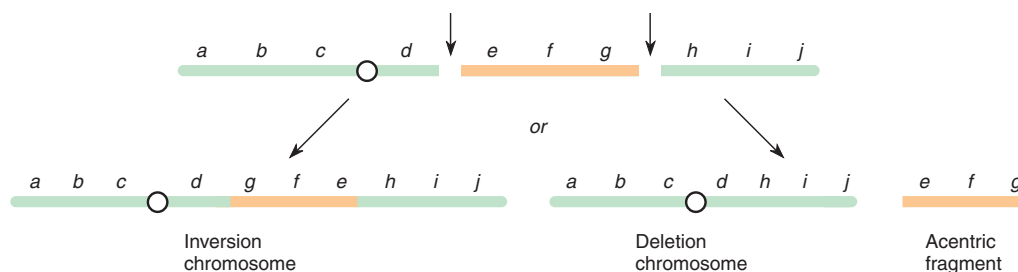


Figure 8.3 Two possible consequences of a double break (top arrows) in the same chromosome.

having this chromosome and a normal homologue will have, during meiosis, a bulge in the tetrad if the deleted section is large enough (fig. 8.4). The bulge also appears in the paired, polytene giant salivary gland chromosomes of *Drosophila*. (Note that when a bulge like that illustrated in figure 8.4 is seen in paired chromosomes, it indicates that one chromosome has a piece that is missing in the other. In our example, the bulge resulted from a deletion in one chromosome; it could also result from an insertion of a piece in the other chromosome.)

Inversion

Two breaks in the same chromosome can also lead to **inversion**, in which the middle section is reattached but in the inverted configuration (see fig. 8.3). An inversion has several interesting properties. To begin with, fruit flies homozygous for an inversion show new linkage relations when their chromosomes are mapped. One outcome of this new linkage arrangement is the possibility of a **position effect**, a change in the expression of a gene due to a changed linkage arrangement. Position effects are either stable, as in *Bar* eye of *Drosophila* (to be discussed), or variegated, as with *Drosophila* eye color. A normal female fly that is heterozygous (X^wX^+) has red eyes. If, however, the white locus is moved through an inversion so that it comes to lie next to heterochromatin (fig. 8.5), the fly shows a **variegation**—patches of the eye are white. This is presumably caused by a spread of the tight coiling of the heterochromatin, “turning off” the expression of the locus. In a heterozygote, if the turned-off allele is the wild-type, the cell will express the normally recessive white-eye allele. Depending on what happens in each cell, patches of red and white eye color result.

When synapsis occurs in an inversion heterozygote, either at meiosis or in the *Drosophila* salivary gland during endomitosis, a loop often forms to accommodate the point-for-point pairing process (figs. 8.6 and 8.7). An outcome of this looping tendency is **crossover suppression**. That is, an inversion heterozygote shows very little recombination of alleles within the inverted region. The reason is usually not that crossing over is actually

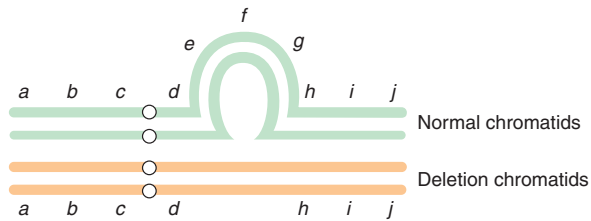


Figure 8.4 A bulge can occur in a meiotic tetrad if a large deletion has occurred.

suppressed, but rather that the products of recombination within a loop are usually lost. (Suppression can also occur in small inversions where loops don't form.) Figure 8.8 shows a crossover within a loop. The two nonsister chromatids not involved in a crossover in the loop will end up in normal gametes (carrying either the normal chromosome or the intact inverted chromosome). The products of the crossover, rather than being a simple recombination of alleles, are a dicentric and an acentric chromatid. The acentric chromatid is not incorporated into a gamete nucleus, whereas the dicentric chromatid begins a breakage-fusion-bridge cycle that creates a genetic imbalance in the gametes. The gametes thus carry chromosomes with duplications and deficiencies.

The inversion pictured in figure 8.8 is a **paracentric inversion**, one in which the centromere is outside the inversion loop. A **pericentric inversion** is one in which the inverted section contains the centromere. It, too, suppresses crossovers, but for slightly different reasons

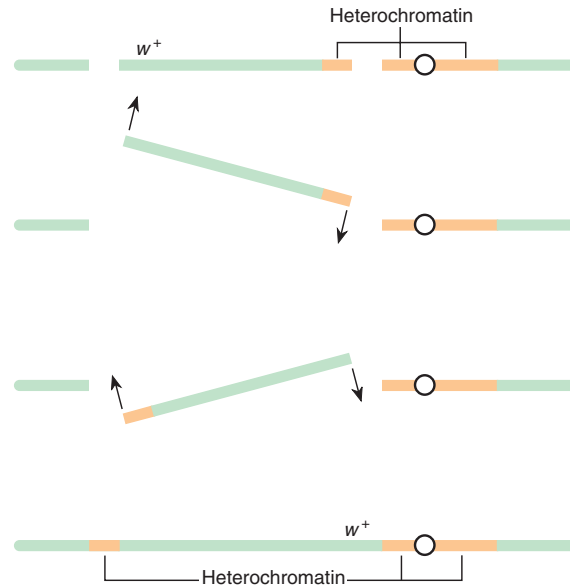


Figure 8.5 An inversion in the X chromosome of *Drosophila* produces a variegation in eye color in a female if her other chromosome is normal and carries the white-eye allele (X^w).

(fig. 8.9). All four chromatid products of a single crossover within the loop have centromeres and are thus incorporated into the nuclei of gametes. However, the two recombinant chromatids are unbalanced—they both have duplications and deficiencies. One has a duplication for

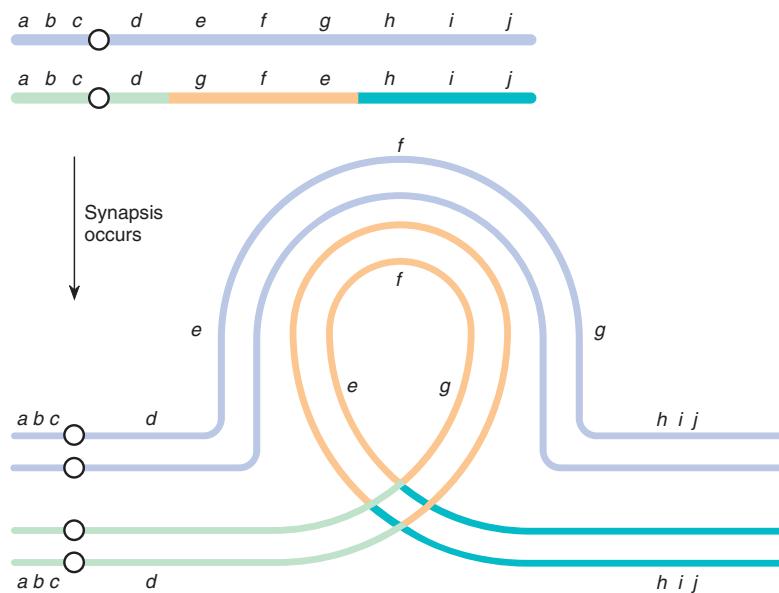


Figure 8.6 Tetrad at meiosis showing the loop characteristic of an inversion heterozygote.



Figure 8.7 A *Drosophila* heterozygous for an inversion will show a loop in the salivary gland chromosomes. (Compare with figure 8.6.)

a-b-c-d and is deficient for *b-i-j*, whereas the other is the reciprocal—deficient for *a-b-c-d* and duplicated for *b-i-j* (in fig. 8.9). These duplication-deletion gametes tend to form inviable zygotes. The result, as with the paracentric inversion, is the apparent suppression of crossing over.

Results of Inversion

Crossing over within inversion loops results in **semisterility**. Almost all gametes that contain dicentric or imbalanced chromosomes form inviable zygotes. Thus, a certain proportion of the progeny of inversion heterozygotes are not viable.

Inversions have several evolutionary ramifications. Those alleles originally together in the noninversion chromosome and those found together within the inver-

sion loop tend to stay together because of the low rate of successful recombination within the inverted region. If several loci affect the same trait, the alleles are referred to as a **supergene**. Until careful genetic analysis is done, the loci in a supergene could be mistaken for a single locus; they affect the same trait and are inherited apparently as a single unit. Examples include shell color and pattern in land snails and mimicry in butterflies (see chapter 21). Supergenes can be beneficial when they involve favorable gene combinations. However, at the same time, their inversion structure prevents the formation of new complexes. Supergenes, therefore, have evolutionary advantages and disadvantages. Chapter 21 discusses these evolutionary topics in more detail.

Sometimes the inversion process produces a record of the evolutionary history of a group of species. As species evolve, inversions can occur on preexisting inversions. This leads to very complex arrangements of loci. We can readily study these patterns in Diptera by noting the changed patterns of bands in salivary gland chromosomes. Since certain arrangements can only come about by a specific sequence of inversions, it is possible to know which species evolved from which. The same series of events can occur within the same species (box 8.1).

In summary then, inversions result in suppressed crossing over, semisterility, variegation position effects, and new linkage arrangements. All of these events have evolutionary consequences.

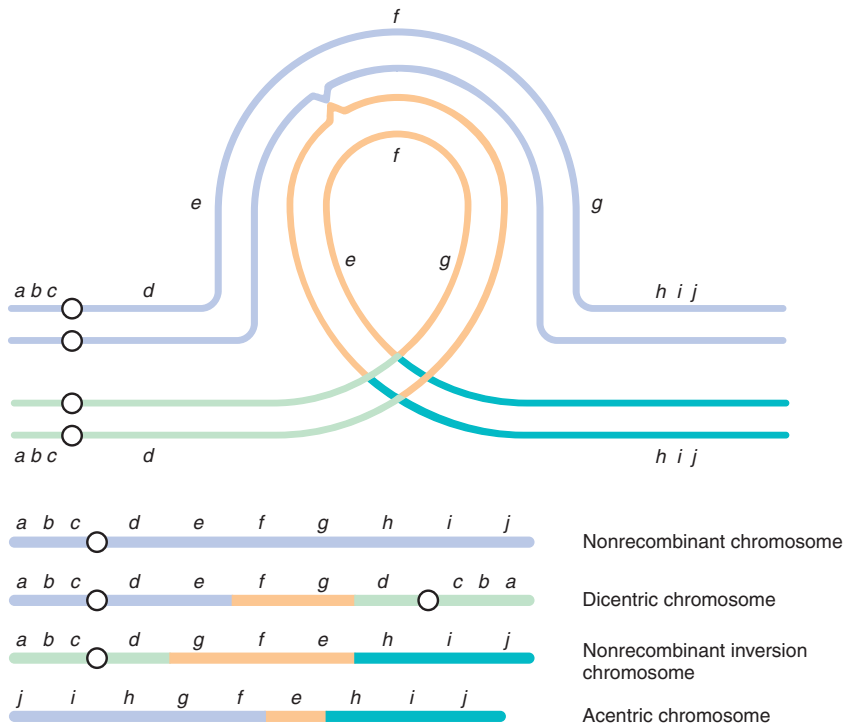


Figure 8.8 Consequences of a crossover in the loop region of a paracentric inversion heterozygote.

BOX 8.1

In 1966, David Futch published a study of the chromosomes of a fruit fly, *Drosophila ananassae*, an organism widely distributed throughout the tropical Pacific. The study was designed to determine something about the species status of various melanic forms of the fly. In the course of his work, Futch looked at the salivary gland chromosomes of flies from twelve different localities. He discovered twelve paracentric inversions, three pericentric inversions, and one translocation. Because of the precise banding patterns of these chromosomes, it was possible

Experimental Methods

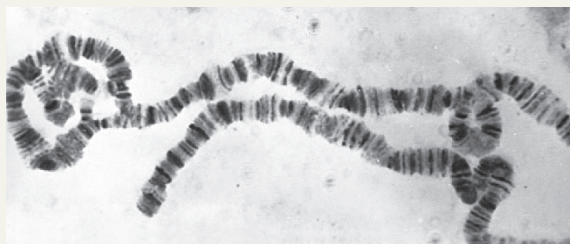
A Case History of the Use of Inversions to Determine Evolutionary Sequence

to determine the breakage points for each inversion.

Observation of several populations that have had sequential changes in their chromosomes makes

it possible to determine the sequence of successive changes. Once one knows the sequence of changes in different populations of *Drosophila ananassae*, along with the geographic locations of the populations, it is possible to determine the history of the way the flies colonized these tropical islands. *D. ananassae* is particularly suited to this type of work because it is believed to be a recent invader to most of the Pacific Islands that it occupies. It is of interest to know about the spread of this species as an adjunct to studies of human migration in the Pacific Islands

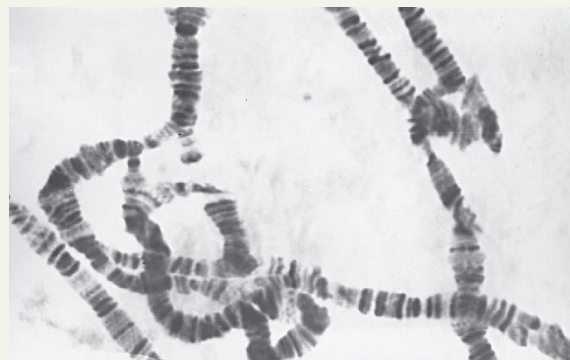
Figure 1 Photomicrographs of the left arm of chromosome 2 (2L) from larval *Drosophila ananassae* heterozygous for various complex gene arrangements. (a) Pairing when heterozygous for standard gene sequence and overlapping inversions (2LC; 2LD) and inversion 2LB (Standard × Tutuila light). (b) Pairing when heterozygous for standard gene sequence and single inversion 2LC and overlapping inversions (2LE; 2LB: Standard × New Guinea). (c) Pairing when heterozygous for overlapping inversions (2LD; 2LE; 2LF: Tutuila light × New Guinea). (From David G. Futch, "A study of speciation in South Pacific populations of *Drosophila ananassae*," in Marshall R. Wheeler, ed., *Studies in Genetics*, no. 6615 [Austin: University of Texas Press, 1966]. Reproduced by permission.)



(a)



(b)



(c)

Two Breaks in Nonhomologous Chromosomes

Breaks can occur simultaneously in two nonhomologous chromosomes. Reunion can then take place in various ways. The most interesting case occurs when the ends of two nonhomologous chromosomes are translocated to each other in a **reciprocal translocation** (fig. 8.10). The organism in which this has happened, a reciprocal translo-

cation heterozygote, has all the genetic material of the normal homozygote. Two outcomes of a reciprocal translocation, like those of an inversion, are new linkage arrangements in a homozygote—an organism with translocated chromosomes only—and variegation position effects.

During synapsis, either at meiosis or endomitosis, a point-for-point pairing in the translocation heterozygote

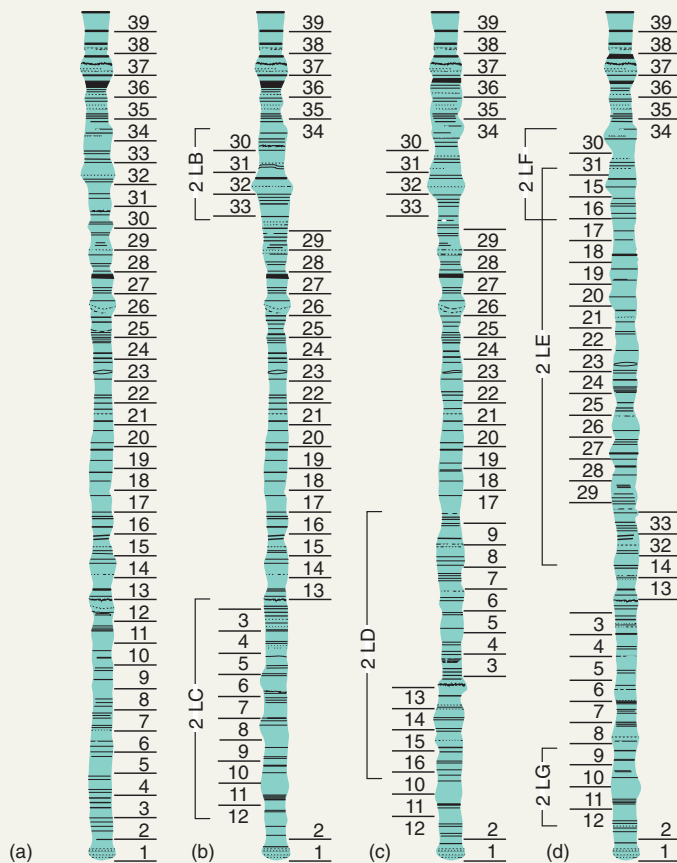


Figure 2 Chromosomal maps of 2L. (a) Standard gene sequence. (b) Ponape: breakpoints of 2LC and 2LB are indicated and the segments are shown inverted. (c) Tutuila light: breakpoints of 2LD are indicated. 2LC and 2LB are inverted. 2LD, which overlaps 2LC, is also shown inverted. (d) New Guinea: breakpoints of 2LE and 2LG are indicated. 2LC, 2LB, and 2LE are shown inverted. Note: only the breakpoints of 2LF and 2LG are shown; neither of these is inverted in the map. (From David G. Futch, "A study of speciation in South Pacific populations of *Drosophila ananassae*," in Marshall R. Wheeler, ed., *Studies in Genetics*, no. 6615 [Austin: University of Texas Press, 1966]. Reproduced by permission.)

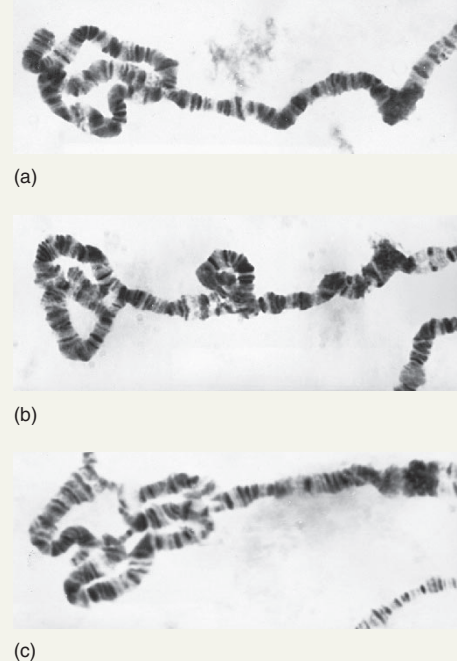


Figure 3 Photomicrographs of the right arm of chromosome 2 (2R) from larvae heterozygous for various complex gene arrangements. (a) Pairing when heterozygous for standard gene sequence and overlapping inversions (2RA; 2RB: Standard $\times$ Tutuila light). (b) Pairing when heterozygous for standard gene sequence and overlapping inversions (2RA; 2RC) and inversion 2RD (Standard $\times$ New Guinea). (c) Pairing when heterozygous for overlapping inversions 2RB, 2RC, and 2RD. Inversion 2RA is homozygous (Tutuila light $\times$ New Guinea). (From David G. Futch, "A study of speciation in South Pacific populations of *Drosophila ananassae*," in Marshall R. Wheeler, ed., *Studies in Genetics*, no. 6615 [Austin: University of Texas Press, 1966]. Reproduced by permission.)

continued

can be accomplished by the formation of a cross-shaped figure (fig. 8.10). Such a figure is diagnostic of a reciprocal translocation. A single crossover in a reciprocal translocation heterozygote will not produce chromatids that are further imbalanced, as it does in an inversion heterozygote. However, reciprocal translocation heterozygotes do produce nonviable progeny. Problems

can arise when centromeres separate at the first meiotic division.

Segregation After Translocation

Since two homologous pairs of chromosomes are involved, we have to keep track of the independent

BOX 8.1 CONTINUED

because *D. ananassae* is commensal with people.

Some of Futch's results are shown in figures 1–4, which diagram the left and right arms of the fly's second chromosome, as well as the synaptic patterns. We can see vividly the sequence of change in which one inversion occurs after a previous inver-

sion has already taken place. In figures 2 and 4, the standard (*a*) gave rise to (*b*), which then gave rise independently to (*c*) and (*d*). The standard is from Majuro in the Marshall Islands and is believed to be in the ancestral group of the species. Ponape is the home of (*b*), (*c*) is from Tutuila (eastern Samoa), and (*d*) is from New

Guinea. Thus, the sequence is Majuro to Ponape, and from there the same stock was transferred to Tutuila and New Guinea. This type of analysis has been useful in the *Drosophila* group throughout its range but especially in the Pacific Island populations and in the southwestern United States.

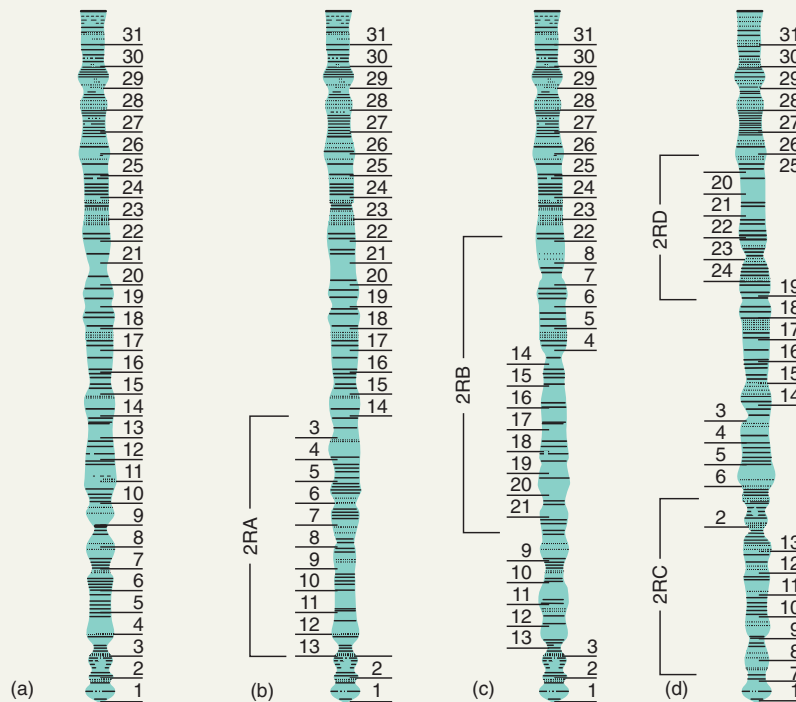


Figure 4 Chromosomal maps of 2R. (a) Standard gene sequence. (b) Ponape: breakpoints of 2RA are indicated and the segment is shown inverted. (c) Tutuila light: breakpoints of 2RB are indicated. 2RA is inverted and 2RB, which overlaps it, is also shown inverted. (d) New Guinea: breakpoints of 2RC and 2RD are indicated. 2RA is inverted; 2RC, which overlaps 2RA, and 2RD are shown inverted. (From David G. Futch, "A study of speciation in South Pacific populations of *Drosophila ananassae*," in Marshall R. Wheeler, ed., *Studies in Genetics*, no. 6615 [Austin: University of Texas Press, 1966]. Reproduced by permission.)

segregation of the centromeres of the two tetrads. There are two common possibilities and one that occurs less often (fig. 8.11). The first, called **alternate segregation**, occurs when the first centromere assort with the fourth centromere, leaving the second and third centromeres to go to the opposite pole. The result will be balanced

gametes, one with normal chromosomes and the other with a reciprocal translocation. Also likely is the **adjacent-1** type of segregation, in which the first and third centromeres segregate together in the opposite direction from the second and fourth centromeres. Here, both types of gametes are unbalanced, carrying duplica-

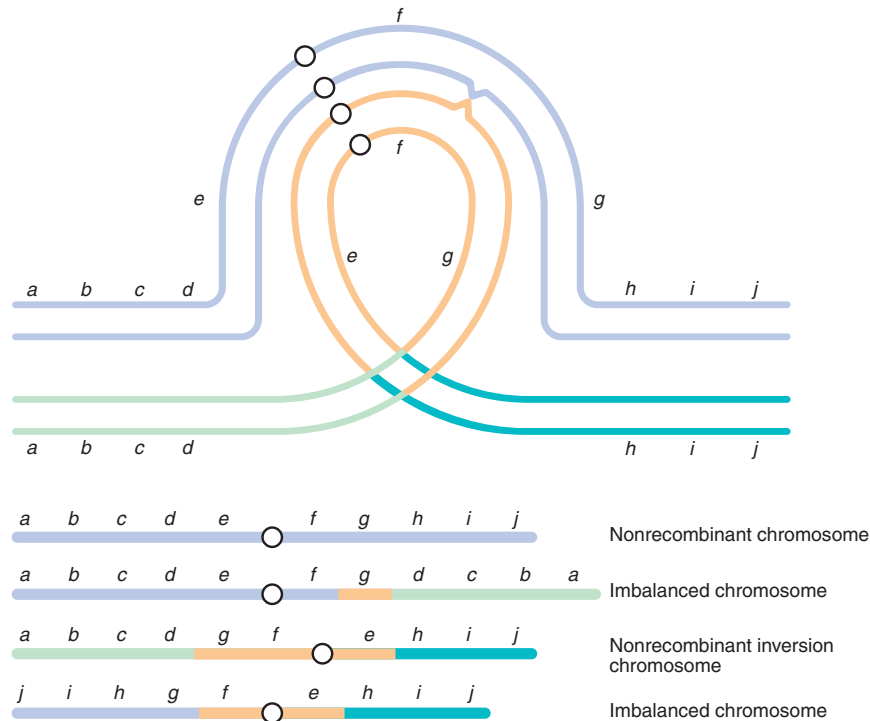


Figure 8.9 Consequences of a crossover in the loop region of a pericentric inversion heterozygote.

tions and deficiencies that are usually lethal. Since adjacent-1 segregation occurs at a relatively high frequency, a significant amount of sterility results from the translocation (as much as 50%).

An **adjacent-2** type of segregation (fig. 8.11), in which homologous centromeres go to the same pole (first with second, third with fourth), is a third possibility. This can result when the cross-shaped double tetrad opens into a circle in late prophase I. In the German cockroach, adjacent-2 patterns have been observed in 10 to 25% of meioses, depending upon which chromosomes are involved.

In summary, then, reciprocal translocations result in new linkage arrangements, variegated position effects, a cross-shaped figure during synapsis, and semisterility.

Centromeric Breaks

Another interesting variant of the simple reciprocal translocation occurs when two acrocentric chromosomes join at or very near their centromeres. The process, called a **Robertsonian fusion** after cytologist W. Robertson, produces a decrease in the number of chromosomes, although virtually the same amount of genetic material is maintained. Often, closely related species undergo Robertsonian fusions and end up with

markedly different chromosome numbers without any significant difference in the quantity of their genetic material. Therefore, cytologists frequently count the number of chromosomal arms rather than the number of chromosomes to get a more accurate picture of species affinities. The number of arms is referred to as the **fundamental number**, or **NF** (French: *nombre fondamentale*). In a similar fashion, **centromeric fission** increases the chromosome number without changing the fundamental number.

Duplications

Duplications of chromosomal segments can occur, as we have just seen, by the breakage-fusion-bridge cycle or by crossovers within the loop of an inversion. There is another way that duplications arise in small adjacent regions of a chromosome. We illustrate this with a particularly interesting example, the *Bar* eye phenotype in *Drosophila* (fig. 8.12). The wild-type fruit fly has about 800 facets in each eye. The *Bar* (*B*) homozygote has about 70 (a range of 20–120 facets). Another allele, *Doublebar* (*BB*; sometimes referred to as *Ultrabar*; *B^U*), brings the facet number of the eye down to about 45 when heterozygous and to about 25 when homozygous. Around 1920, researchers

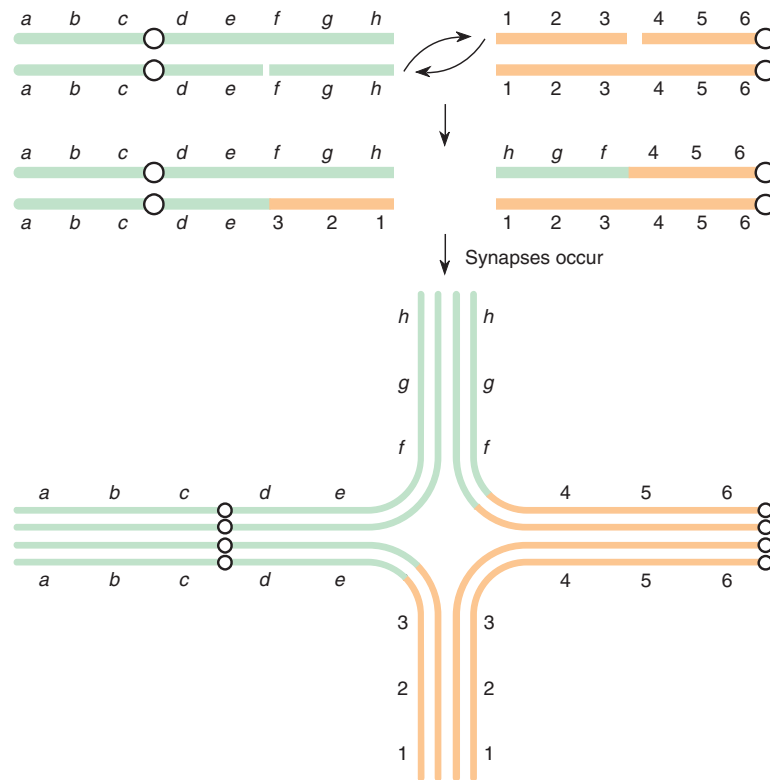


Figure 8.10 A reciprocal translocation heterozygote forms after breaks occur in nonhomologous chromosomes. Synapsis at meiosis forms a cross-shaped figure.

showed that about one progeny in 1,600 from homozygous *Bar* females is *Doublebar*. This is much more frequent than we expect from mutation.

Alfred Sturtevant found that in every *Doublebar* fly, a crossover had occurred between loci on either side of the *Bar* locus. He suggested that the change to *Doublebar* was due to **unequal crossing over** rather than to a simple mutation of one allele to another (fig. 8.13). If the homologous chromosomes do not line up exactly during synapsis, a crossover produces an unequal distribution of chromosomal material. Later, an analysis of the banding pattern of the salivary glands confirmed Sturtevant's hypothesis. It was found that *Bar* is a duplication of several bands in the 16A region of the X chromosome (fig. 8.14). *Doublebar* is a triplication of the segment.

A position effect also occurs in the *Bar* system. A *Bar* homozygote (B/B) and a *Doublebar*/wild-type heterozygote (BB/B^+) both have four copies of the 16A region. It would therefore be reasonable to expect that both genotypes would produce the same phenotype. However, the *Bar* homozygote has about seventy facets in each eye, whereas the heterozygote only has about forty-five. Thus, not only the amount of genetic material, but also its con-

figuration, determines the extent of the phenotype. *Bar* eye was the first position effect discovered.

Chromosomal Rearrangements in Human Beings

Several human syndromes and abnormalities are the result of chromosomal rearrangements, including deletions and translocations. The most common are described here. Keep three points in mind as you read. First, all of these disorders are rare. Second, the deletion syndromes are often caused by a balanced translocation in one of the parents. And third, about one in five hundred live births contains a balanced rearrangement of some kind, either a reciprocal translocation or inversion.

Fragile-X Syndrome

The most common cause of inherited mental retardation is the **fragile-X syndrome**. It occurs in about one in every 1,250 males and about one in every 2,000 females. Symptoms include mental retardation, altered speech patterns, and other physical attributes. The condition is

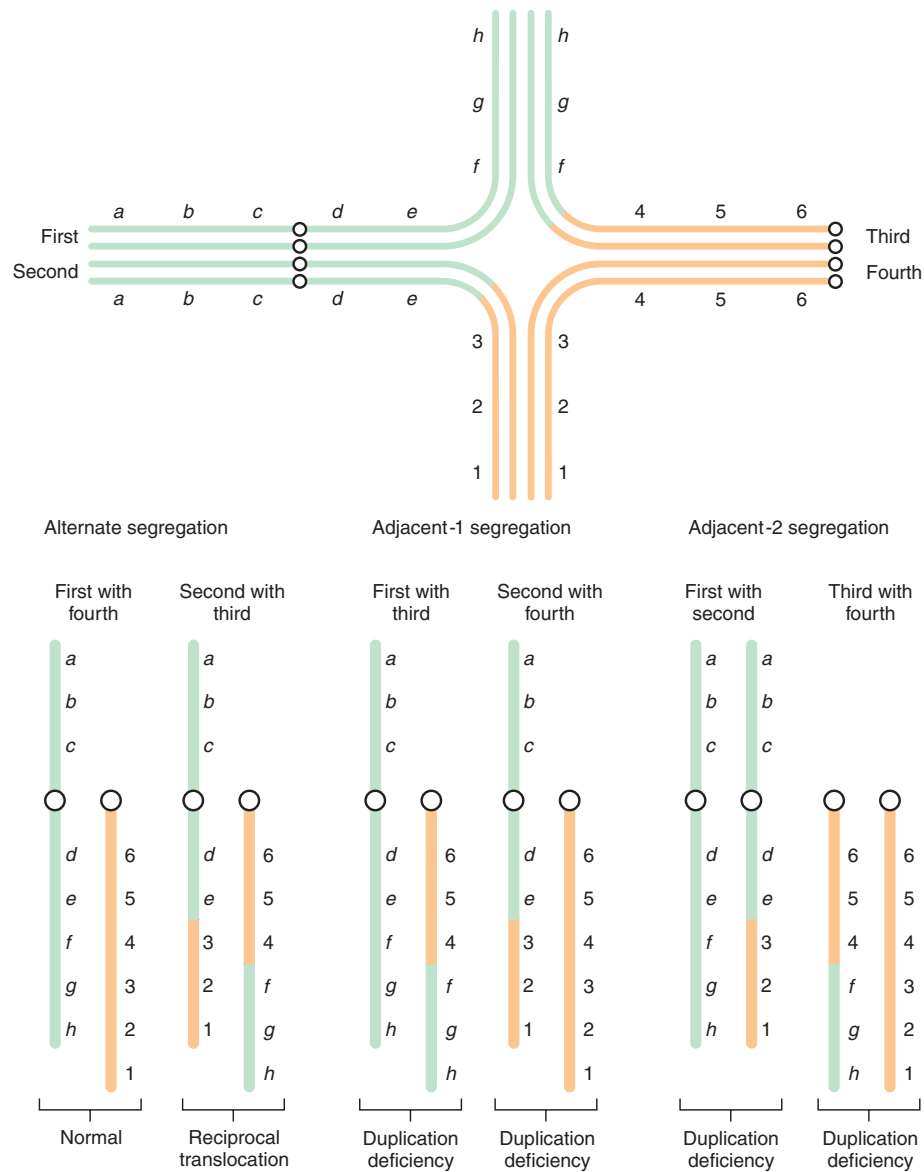


Figure 8.11 Three possible results of chromatid separation during meiosis in a reciprocal translocation heterozygote.

called the fragile-X syndrome because it is related to a region at the X chromosome tip that breaks more frequently than other chromosomal regions. However, the break is not required for the syndrome to occur, and the fragile-X chromosome is usually identified by the lack of chromatin condensation at the site; in fact, under the microscope, it appears that the tip of the chromosome is being held in place by a thread (fig. 8.15). The gene responsible for the syndrome is called *FMR-1*, for fragile-X mental retardation-1.

Fragile-X syndrome has a highly unusual pattern of inheritance: the chance of inheriting the disease increases through generations. This is so unusual a pattern that it was termed the Sherman Paradox. Approximately 20% of males with the fragile-X chromosome do not have symptoms but have grandchildren who do have the symptoms. The daughters of the symptomatic males also don't have symptoms, but obviously, they have another X chromosome to mask the symptoms. As generations proceed, the percentage of affected sons of

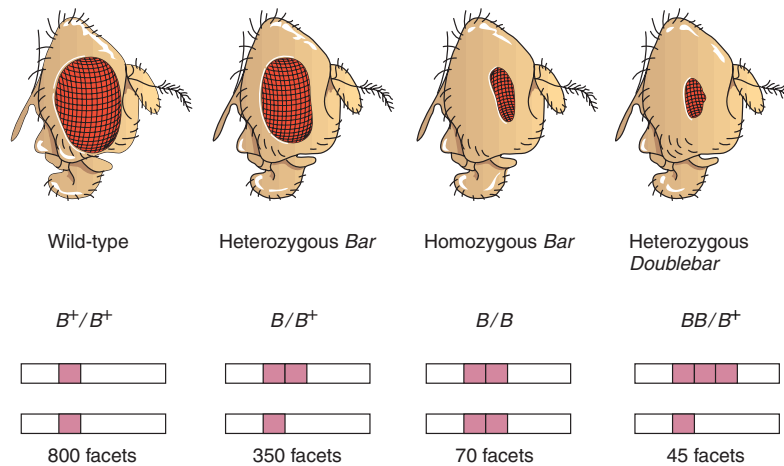


Figure 8.12 Bar eye in *Drosophila* females.

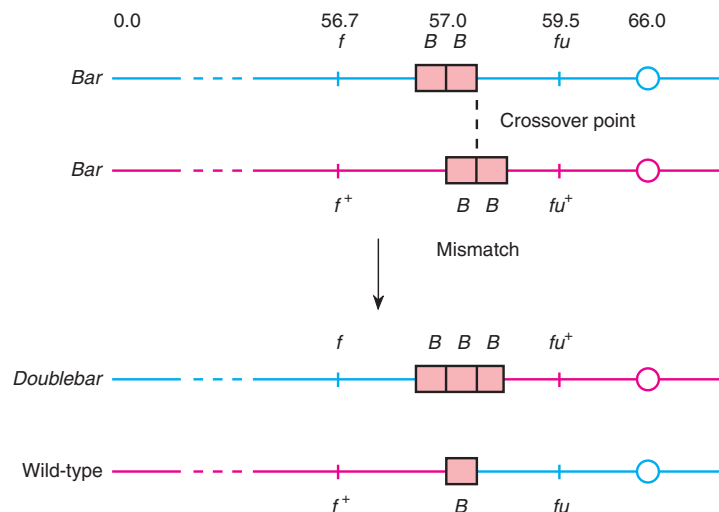


Figure 8.13 Unequal crossing over in a female Bar-eyed *Drosophila* homozygote as a result of improper pairing. A Doublebar chromosome (and concomitant wild-type chromosome) is produced by a crossover between forked (*f*) and fused (*fu*), two flanking loci.

carrier mothers increases. Molecular techniques, discussed in chapter 13, revealed the odd nature of this syndrome.

Basically, the *FMR-1* gene normally has between 6 and 50 copies of a three-nucleotide repeat, CCG. Chromosomes that have the fragile-site appearance have between 230 and 2,000 copies of the repeat. The number of repeats is very unstable; when carrier women transmit the chromosome, the number of repeats usually goes up. Repeat numbers above 230 inactivate the gene and thus cause the syndrome in men, who have only one copy of the X chromosome. The function of the gene is not currently known. This unusual form of inheritance, with un-

stable repeats in a gene, seems to be the mechanism in several other diseases as well, including muscular dystrophy and Huntington disease. We will discuss other unusual modes of inheritance in chapter 17.

Cri du Chat Syndrome, 46,XX or XY,5p-

The syndrome known as *cri du chat* (French: cry of the cat) is so called because of the catlike cry that about half the affected infants make. Microcephaly (an abnormally small head), congenital heart disease, and severe mental retardation are also common symptoms. This disorder arises from a deletion in chromosome 5 (fig. 8.16); most

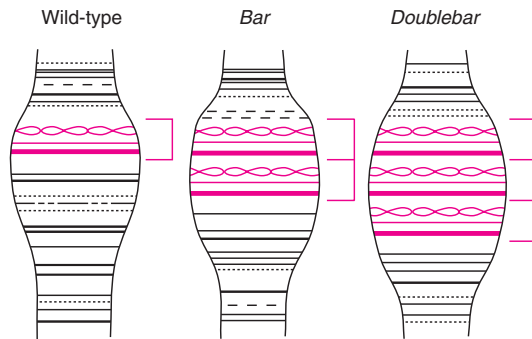


Figure 8.14 Bar region of the X chromosome of *Drosophila*.



Figure 8.15 Human metaphase chromosomes with the fragile-X site indicated by an arrow. (From Ian Craig, "Methylation and the Fragile X," *Nature* [1991] 349:742. Copyright © 1991 Macmillan Magazines, Ltd.)

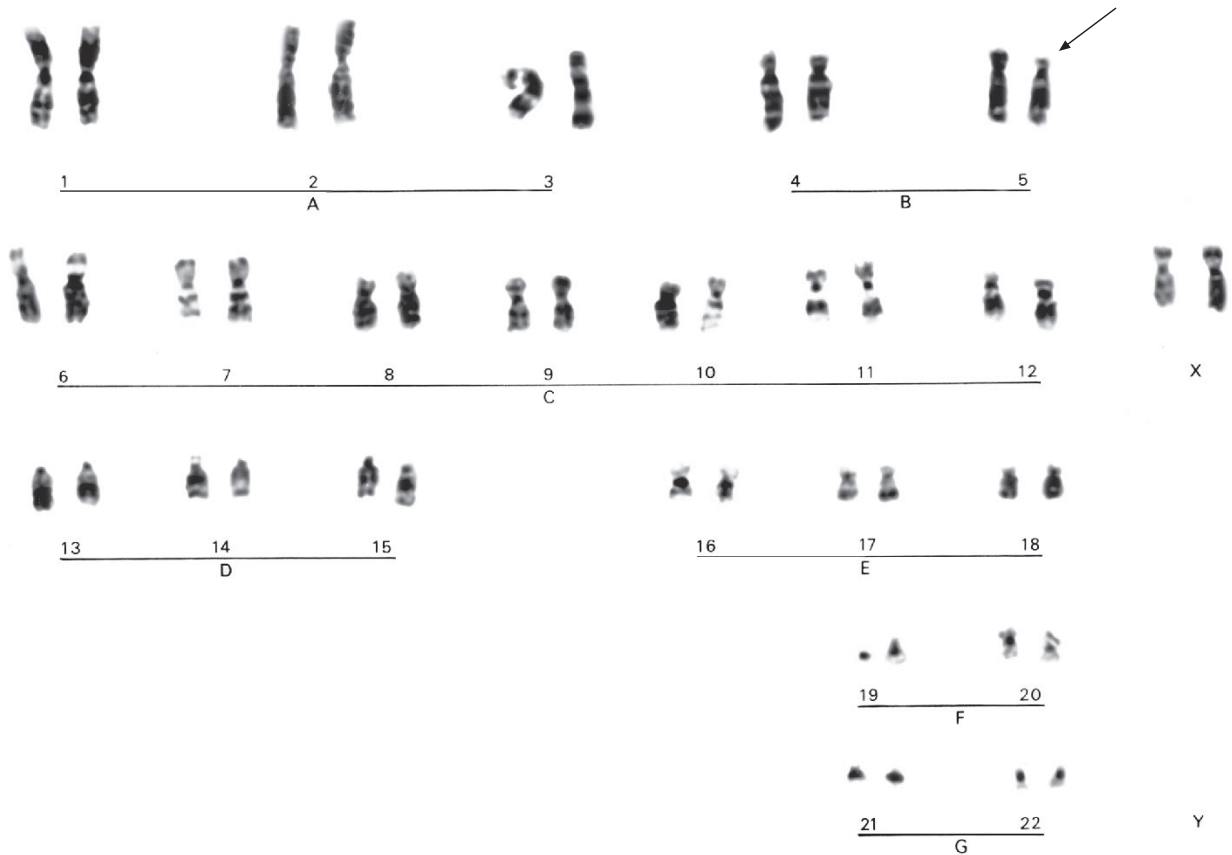


Figure 8.16 Karyotype of individual with cri du chat syndrome, due to a partial deletion of the short arm of chromosome 5 (5p-; arrow). (Reproduced courtesy of Dr. Thomas G. Brewster, Foundation for Blood Research, Scarborough, Maine.)

other deletions studied (4p–, 13q–, 18p–, 18q–) also result in microcephaly and severe mental retardation. The rarity of viable deletion heterozygotes is consistent with the fact that viable monosomics (having a single chromosome of a pair) are rare. An individual heterozygous for a deletion is, in effect, monosomic for the deleted region of the chromosome. Evidently, monosomy or heterozygosity for larger deleted regions of a chromosome is generally lethal in human beings.

VARIATION IN CHROMOSOME NUMBER

Anomalies of chromosome number occur as either **euploidy** or **aneuploidy**. Euploidy involves changes in whole *sets* of chromosomes; aneuploidy involves changes in chromosome *number* by additions or deletions of less than a whole set.

Aneuploidy

An explanation for the terminology of aneuploid change appears in table 8.1. A diploid cell missing a single chromosome is **monosomic**. A cell missing both copies of that chromosome is **nullisomic**. A cell missing two non-homologous chromosomes is a double monosomic. A similar terminology exists for extra chromosomes. For example, a diploid cell with an extra chromosome is **trisomic**. Aneuploidy results from nondisjunction in meiosis or by chromosomal lagging whereby one chromosome moves more slowly than the others during anaphase, is excluded from the telophase nucleus, and is thus lost. Here, nondisjunction is illustrated using the sex chromosomes in XY organisms such as human beings or fruit flies. Four examples are shown (fig. 8.17): nondisjunction in either the male or female at either the first or second meiotic divisions. Figure 8.18 shows the types of

zygotes that can result when these nondisjunctional gametes fuse with normal gametes. All of the offspring produced are chromosomally abnormal. The names and kinds of these imbalances in human beings are detailed later in this chapter.

Bridges first showed the occurrence of nondisjunction in *Drosophila* in 1916 with crosses involving the white-eye locus. When a white-eyed female was crossed with a wild-type male, typically the daughters were wild-type and the sons were white-eyed. However, occasionally (one or two per thousand), a white-eyed daughter or a wild-type son appeared. This could be explained most easily by a nondisjunctional event in the white-eyed females, where X^wX^w and 0 eggs (without sex chromosomes) were formed. Under this hypothesis, if a Y-bearing sperm fertilized an X^wX^w egg, the offspring would be an X^wX^wY white-eyed daughter. If a normal X^+ -bearing sperm fertilized the egg without sex chromosomes, the result would be an X^+0 wild-type son. Subsequently, these exceptional individuals were found by cytological examination to have precisely the predicted chromosomes (XXY daughters and X0 sons). The other types produced by this nondisjunctional event are the XX egg fertilized by an X-bearing sperm and the 0 egg fertilized by the Y-bearing sperm. The XXX zygotes are genotypically $X^wX^wX^+$, or wild-type daughters (which usually die), and Y0 flies (which always die).

Mosaicism

Rarely, an individual is made up of several cell lines, each with different chromosome numbers. These individuals are referred to as **mosaics** or **chimeras**, depending on the sources of the cell lines. Such conditions can be the result of nondisjunction or chromosomal lagging during mitosis in the zygote or in nuclei in the early embryo (mosaic). This is demonstrated, again for sex chromosomes, in figure 8.19. A lagging chromosome is shown in figure 8.20; in the figure, the X chromosome is lost in one

Table 8.1 Partial List of Terms to Describe Aneuploidy, Using *Drosophila* as an Example
(Eight Chromosomes: X, X, 2, 2, 3, 3, 4, 4)

Type	Formula	Number of Chromosomes	Example
Normal	$2n$	8	X, X, 2, 2, 3, 3, 4, 4
Monosomic	$2n - 1$	7	X, X, 2, 2, 3, 4, 4
Nullisomic	$2n - 2$	6	X, X, 2, 2, 4, 4
Double monosomic	$2n - 1 - 1$	6	X, X, 2, 3, 4, 4
Trisomic	$2n + 1$	9	X, X, 2, 2, 3, 3, 4, 4, 4
Tetrasomic	$2n + 2$	10	X, X, 2, 2, 3, 3, 3, 3, 4, 4
Double trisomic	$2n + 1 + 1$	10	X, X, 2, 2, 2, 3, 3, 3, 4, 4

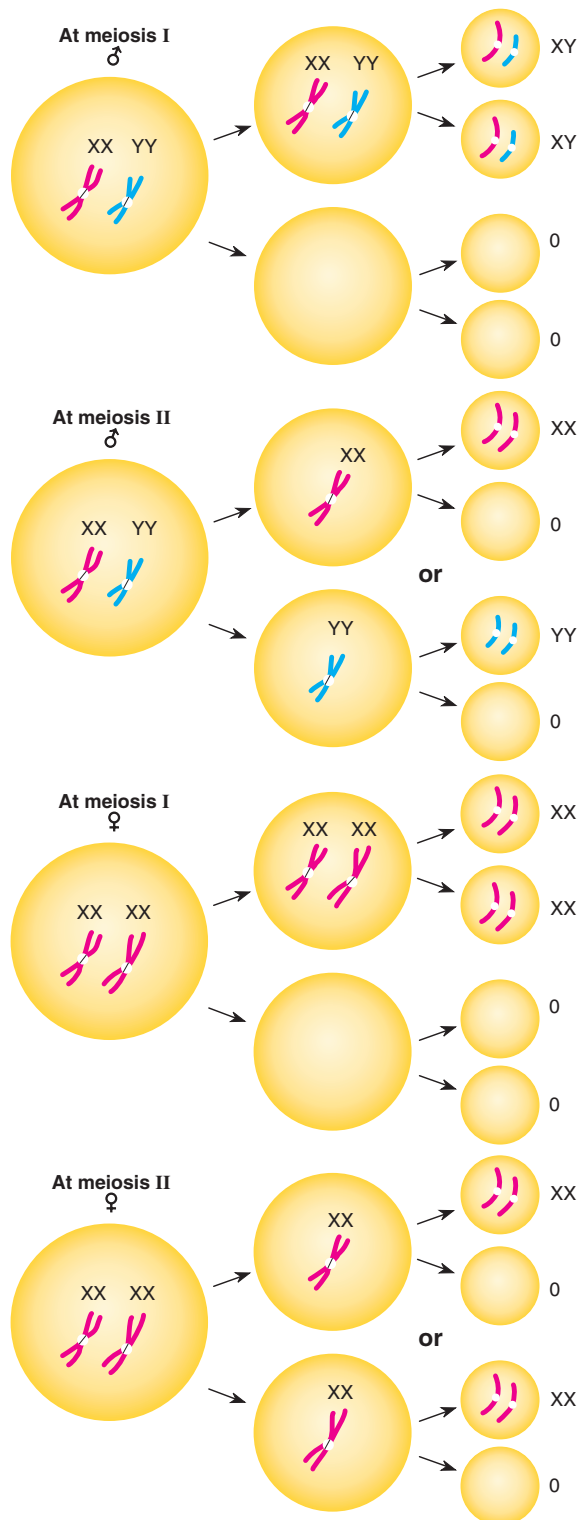


Figure 8.17 Nondisjunction of the sex chromosomes in *Drosophila* or human beings. "O" refers to the lack of sex chromosomes.

of the dividing somatic cells, resulting in an XX cell line and an XO cell line. In *Drosophila*, if this chromosomal lagging occurs early in development, an organism that is part male (XO) and part female (XX) develops. Figure 8.21 shows a fruit fly in which chromosomal lagging has occurred at the one-cell stage, causing the fly to be half male and half female. A mosaic of this type, involving male and female phenotypes, has a special name—**gynandromorph**. (A **hermaphrodite** is an individual, not necessarily mosaic, with both male and female reproductive organs.) Many sex-chromosomal mosaics are known in humans, including XX/X, XY/X, XX/XY, and XXX/X. At least one case is known of a human XX/XY chimera that resulted from the fusion of two zygotes, one

		♂				♀	
		XY	XX	YY	O	XX	O
Normal	♀ X	XXY	XXX	XYY	XO		
	♂ X					XXX	XO
	♂ Y					XXY	YO

Figure 8.18 Results of fusion of a nondisjunction gamete (top) with a normal gamete (side).

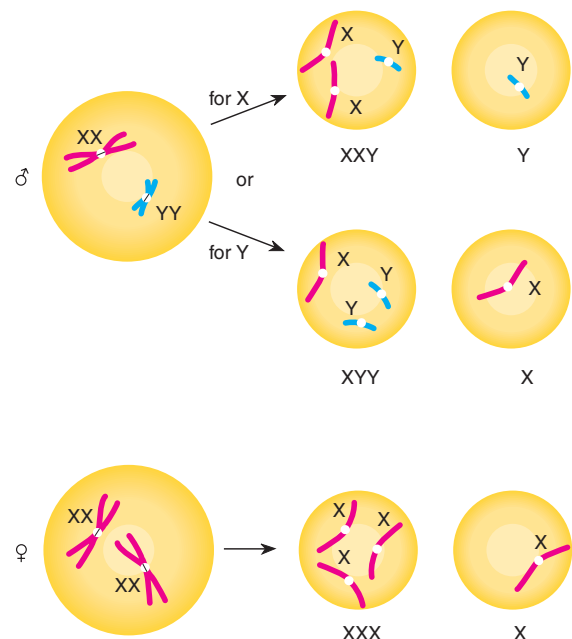


Figure 8.19 Mitotic nondisjunction of the sex chromosomes.

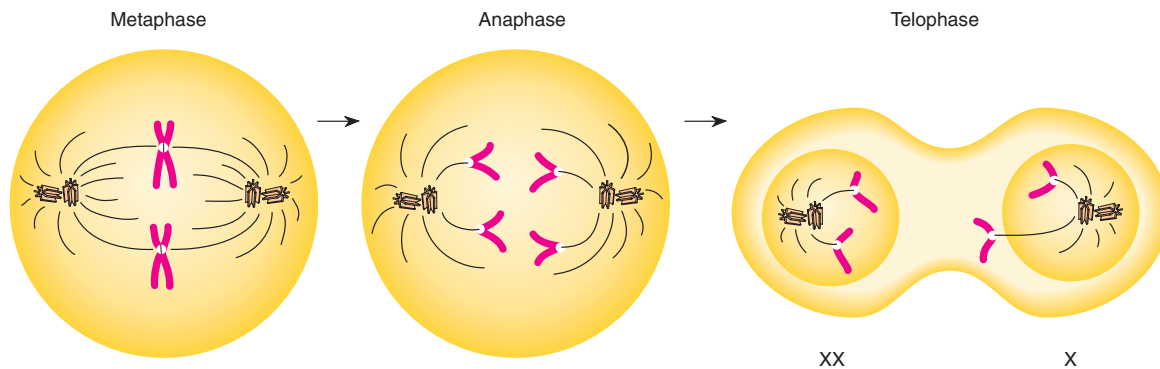


Figure 8.20 Chromosomal lagging at mitosis in the X chromosomes of a female *Drosophila*.

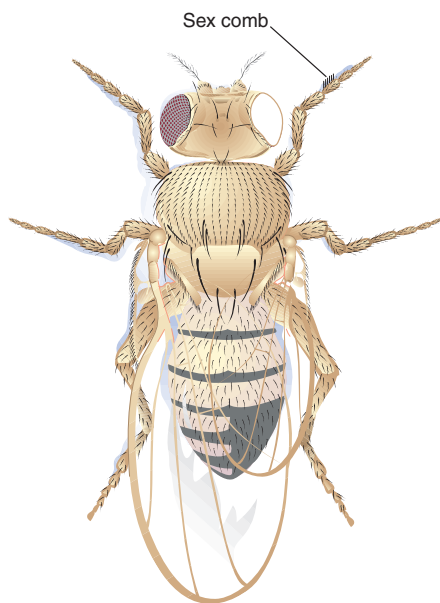


Figure 8.21 *Drosophila* gynandromorph. The left side is wild-type XX female; the right side is XO male, hemizygous for white eye and miniature wing.

formed by a sperm fertilizing an ovum and the other formed by a second sperm fertilizing a polar body of that ovum.

Aneuploidy in Human Beings

Approximately 50% of spontaneous abortions (miscarriages) among women in the United States involve fetuses with some chromosomal abnormality; about half of these are autosomal trisomics. About one in 160 live human births has some sort of chromosomal anomaly; most are balanced translocations, autosomal trisomics, or sex-chromosomal aneuploids.

In the standard system of nomenclature, a normal human chromosome complement is 46,XX for a female and 46,XY for a male. The total chromosome number appears first, then the description of the sex chromosomes, and, finally, a description of autosomes if some autosomal anomaly is evident. For example, a male with an extra X chromosome would be 47,XXY. A female with a single X chromosome would be 45,X. Since all the autosomes are numbered, we describe their changes by referring to their addition (+) or deletion (−). For example, a female with trisomy 21 would be 47,XX,+21. The short arm of a chromosome is designated p, the longer arm, q. When a change in part of the chromosome occurs, a + after the arm indicates an increase in the length of that arm, whereas a minus sign (−) indicates a decrease in its length. For example, a translocation (t) that transfers part of the short arm of chromosome 9 to the short arm of chromosome 18 would be 46,XX,t(9p−;18p+). The semicolon indicates that both chromosomes kept their centromeres.

Following are descriptions of viable human aneuploids who survive long enough after birth to have a named syndrome.

Trisomy 21 (Down Syndrome), 47,XX or XY,+21

Down syndrome (figs. 8.22 and 8.23) affects about one in seven hundred live births. Most affected individuals are mildly to moderately mentally retarded and have congenital heart defects and a very high (1/100) risk of acute leukemia. They are usually short and have a broad, short skull; hyperflexibility of joints; and excess skin on the back of the neck. The physician John Langdon Down first described this syndrome in 1866. (Modern convention is to avoid the possessive form of a name in referring to a syndrome.) Down syndrome was the first human syndrome attributed to a chromosomal disorder; Jérôme Lejeune, a physician in Paris, published this finding in

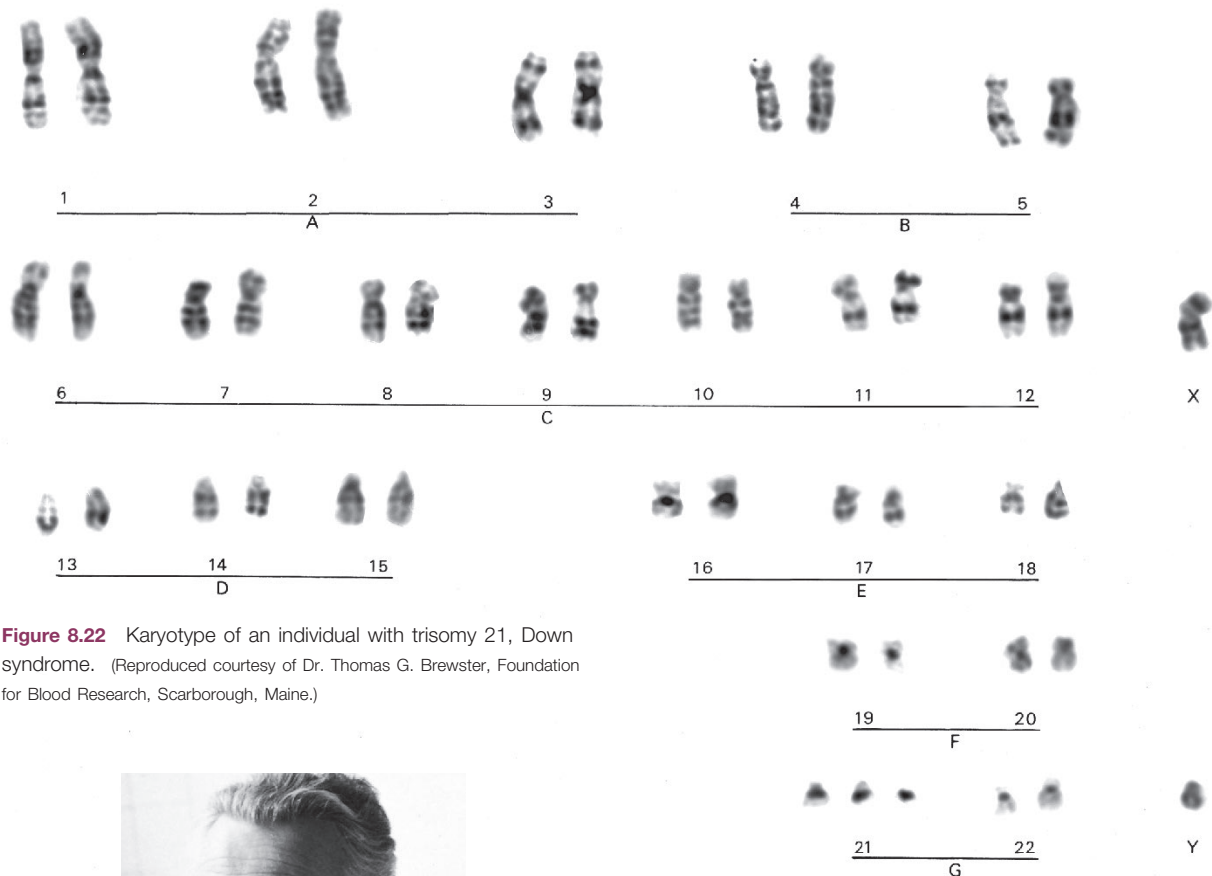


Figure 8.22 Karyotype of an individual with trisomy 21, Down syndrome. (Reproduced courtesy of Dr. Thomas G. Brewster, Foundation for Blood Research, Scarborough, Maine.)



Jérôme Lejeune (1926–1994).
(Courtesy of Dr. Jérôme Lejeune, Institut de
Progenese, Paris.)

1959. An interesting aspect of this syndrome is its increased incidence among children of older mothers (fig. 8.24), a fact known more than twenty-five years before the discovery of the cause of the syndrome. Since the future ova are in prophase I of meiosis (dictyotene) since before the mother's birth, all ova are the same age as the female. Presumably, older ova are more susceptible to nondisjunction of chromosome 21.

Recently, techniques of molecular genetics (chapter 13) have been used to identify the origins of the three copies of chromosome 21 in a large sample of individuals with Down syndrome. As expected, the overwhelming majority of the extra copies of chromosome 21 (95%) were of maternal origin. About 5% of the cases of Down syndrome were of mitotic origin, occurring either in the gonad of one of the parents (evenly split between mothers and fathers) or possibly postzygotically in the fetus.

Familial Down Syndrome

Down syndrome (trisomy 21), as described, is usually the result of either a nondisjunctional event during gametogenesis or, rarely, a mitotic event. It is a function of maternal age and is not inherited. (Although about half the children of a person with trisomy 21 will have trisomy 21 because of aneuploid gamete production, the possibility that an unaffected relative of the person will have abnormal children is no greater than for a person of the same age chosen at random from the general population.) However, about 4% of those with Down syndrome have been found to have a translocation of chromosome 21,



Figure 8.23 Individual with trisomy 21. (© Hattie Young/SPL/Photo Researchers.)

usually associated with chromosome 14, 15, or 22. The translocational and nontranslocational types of Down syndrome have identical symptoms; however, a balanced translocation can be passed on to offspring (see fig. 8.11). Alternate segregation of centromeres in the translocation heterozygote produces either a normal gamete or one carrying the balanced translocation. Adjacent segregation causes partial trisomy for certain chromosomal parts. When this occurs for most of chromosome 21, Down syndrome results.

It is worth mentioning that aside from trisomy and translocation, Down syndrome can come about through mosaicism, as mentioned earlier, or a centromeric event. About 2% of individuals with Down syndrome are mosaic for cells with both two and three copies of chromosome 21. Some evidence suggests that the original zygotes were trisomic, but then a daughter cell lost one of the copies of chromosome 21. The severity of the symptoms in these individuals relates to the percentage of trisomic cells they possess. Mosaicism increases with maternal age, just as trisomy in general does. In extremely rare cases, Down syndrome is caused by an abnormal chromosome 21 that has, rather than a short and long arm, two identical long arms attached to the centromere. This

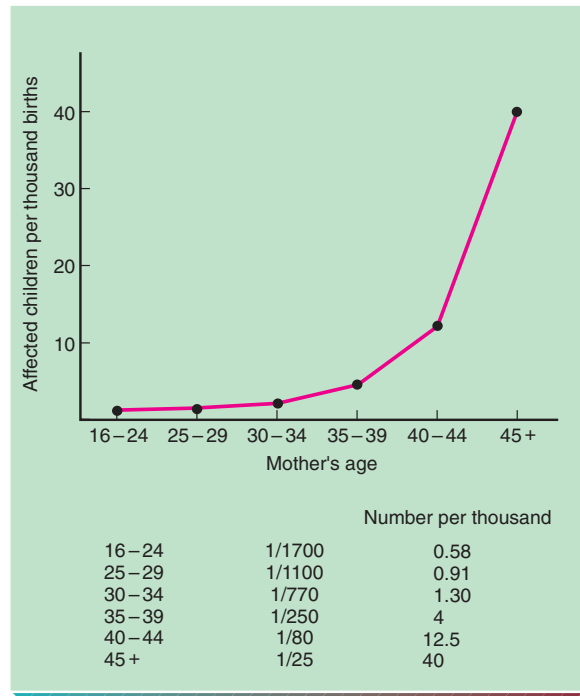


Figure 8.24 Increased risk of trisomy 21 attributed to the age of the mother. (From E. Hook, "Estimates of Maternal Age-Specific Risks of a Down-Syndrome Birth in Women age 34–41," *Lancet*, 2:33–34, Copyright © 1976 by The Lancet Ltd.)

type of chromosome, called an **isochromosome**, presumably occurs by an odd centromeric fission (fig. 8.25). Hence, a person with a normal chromosome 21 and an isochromosome 21 has three copies of the long arm of the chromosome and has Down syndrome.

Trisomy 18 (Edward Syndrome), 47,XX or XY,+ 18

Edward syndrome affects one in ten thousand live births (fig. 8.26). Most affected individuals are female, with 80 to 90% mortality by two years of age. The infant usually has an elfin appearance with small nose and mouth, a receding lower jaw, abnormal ears, and a lack of distal flexion creases on the fingers. The distal joints have limited motion, and the fingers display a characteristic posturing in which the little and index fingers overlap the middle two. The syndrome is usually accompanied by severe mental retardation.

Trisomy 13 (Patau Syndrome), 47,XX or XY,+ 13, and Other Trisomic Disorders

Patau syndrome affects one in twenty thousand live births. Diagnostic features are cleft palate, cleft lip, con-

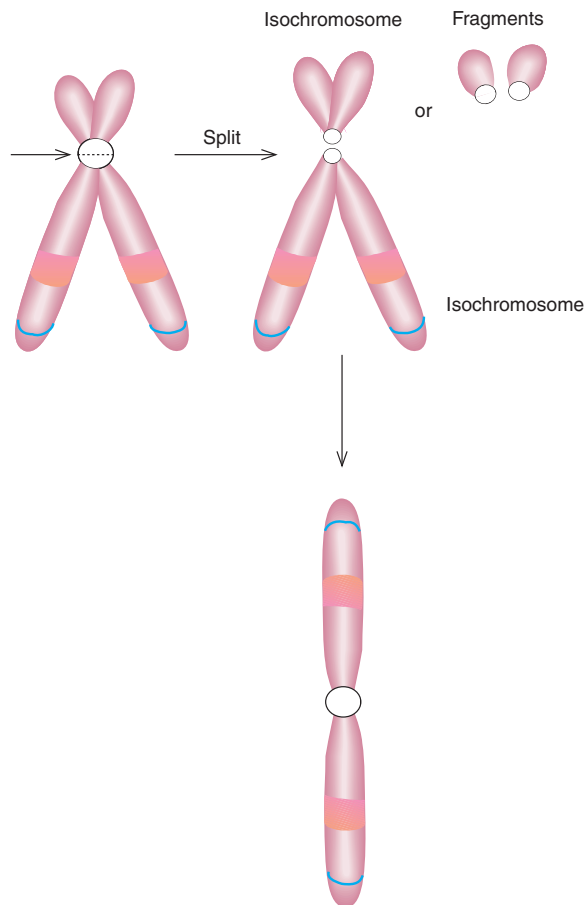


Figure 8.25 If the centromere of chromosome 21 breaks perpendicular to the normal division axis, it can form an isochromosome of the long arms and either an isochromosome of the short arms or two separate fragments. This can happen during anaphase of mitosis or meiosis II.

genital heart defects, polydactyly, and severe mental retardation. Mortality is very high in the first year of life.

Other autosomal trisomies are known but are extremely rare. These include trisomy 8 (47,XX or XY,+8) and cat's eye syndrome, a trisomy of an unidentified, small acrocentric chromosome (47,XX or XY,[+acrocentric]). Several aneuploids involving sex chromosomes are also known.

Turner Syndrome, 45,X

About one in ten thousand live female births is of an infant with Turner syndrome. This and 45,XX or XY,-21 and



Figure 8.26 Child with trisomy 18, Edward syndrome.
(Reproduced courtesy of Dr. Jérôme Lejeune, Institut de Progenese, Paris.)

45,XX or XY,-22 are the only nonmosaic, viable monosomies recorded in human beings (fig. 8.27), indicating the severe consequences monosomy has on all but the two smallest autosomes and a sex chromosome. Individuals with Turner syndrome usually have normal intelligence but underdeveloped ovaries, abnormal jaws, webbed necks, and shieldlike chests.

The symptoms of Turner syndrome have been logically deduced to be caused by a single dosage of genes that are normally present and active in two dosages. Thus, these genes would be located on both the X and Y chromosomes (pseudoautosomal) to provide two dosages in normal XY males and also be active in both X chromosomes in normal XX females. Therefore, they should be located on regions of the X chromosome that escape inactivation (see chapter 5). Studies of persons with small X-chromosomal deletions and molecular analyses of the X and Y chromosomes (outlined in chapter 13) have caused two genes to emerge as candidates: *ZFY* (on the Y chromosome, termed *ZFX* on the X chromosome) and *RPS4Y* (on the Y chromosome, termed *RPS4X* on the X chromosome). *ZFY* (zinc finger on the Y chromosome) was once believed to be the male-determining gene in mammals. *RPS4Y* encodes a ribosomal protein, one of the many proteins making up the ribosome.

It is interesting to note a dosage-compensation difference in people and mice, which have analogous genes termed *Zfx* and *Rps4x*. In mice, unlike in people, these genes are inactivated in the "Lyonized" X chromosome in females and have restricted activity in the Y

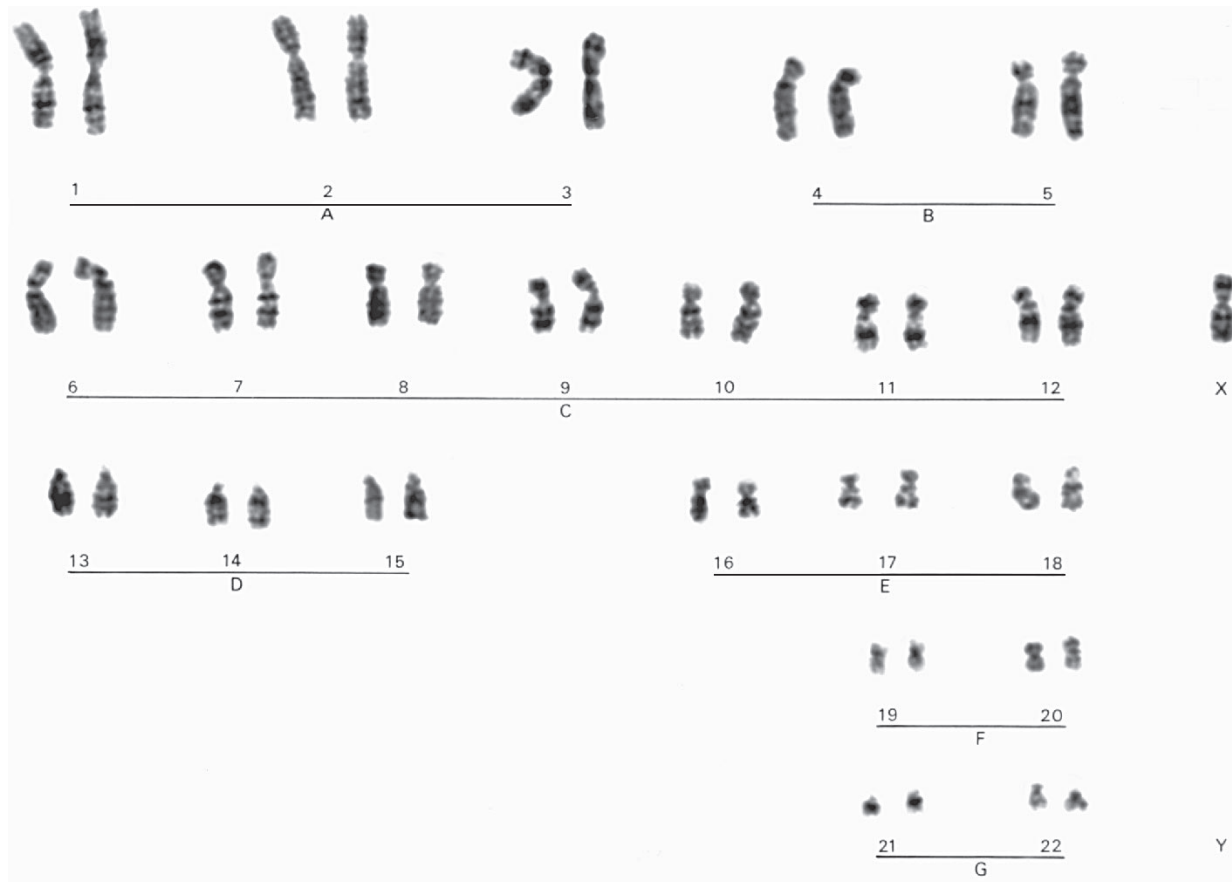


Figure 8.27 Karyotype of a person with Turner syndrome (X0). (Reproduced courtesy of Dr. Thomas G. Brewster, Foundation for Blood Research, Scarborough, Maine.)

chromosome. Hence, mouse cells seem normally to have only one copy of these genes functioning in normal XY males and XX females. Therefore, we would predict that the X0 genotype in mice would produce few if any negative effects as compared with a human X0 genotype, since mouse cells of both sexes normally only have one functional copy of each gene. In fact, human Turner syndrome fetuses have a 99% prenatal mortality rate, but virtually no prenatal mortality affects mouse fetuses with the X0 genotype (born of XX mothers). This confirms our predictions and points to differences between people and mice in dosage-compensation mechanisms for specific genes.

XXX Karyotype, 47,XXX

About one in one thousand live male births is of an individual with an XXX karyotype. (We avoid the term *syndrome* here because XXX men have no clearly defined series of attributes, other than often being taller

than normal.) Some controversy has surrounded this karyotype because it was once reported that it occurred in abundance in a group of mentally subnormal males in a prison hospital. Seven XXX males were found among 197 inmates, whereas only one in about two thousand control men were XXX. This study has subsequently been expanded and corroborated. Although it is now fairly well established that the incidence of XXX males in prison is about twentyfold higher than in society at large, the statistic is somewhat misleading: the overwhelming number of XXX men seem to lead normal lives. At most, about 4% of XXX men end up in penal or mental institutions, where they make up about 2% of the population.

There is some indication that the XXX men in prison had lower intelligence test levels. Thus, criminal tendency may be attributed to lower intelligence rather than a predisposition toward criminality caused by an extra Y chromosome. For the most part, expanded studies have indicated that XXX criminals do not commit violent crimes.

Research on this karyotype has produced its own problems. A research project at Harvard University on XYY males came under intense public pressure and was eventually terminated. The project, under the direction of Stanley Walzer (a psychiatrist) and Park Gerald (a geneticist), involved screening all newborn boys at the Boston Hospital for Women and following the development of those with chromosomal anomalies. The criticism of this work centered mainly on the necessity of informing parents that their sons had an XYY karyotype that might be associated with behavioral problems. Opponents of this work claimed that telling the parents could trigger a self-fulfilling prophecy; that is, parents who heard that their children were not normal and *might* cause trouble might then behave toward their children in a manner that would increase the probability that their children *would* cause trouble. The opponents claimed that the risks of this research outweighed the benefits. The project was terminated in 1975 primarily because of the harassment Walzer faced.

Klinefelter Syndrome, 47,XXY

The incidence of Klinefelter syndrome is about one in one thousand live births. Tall stature and infertility are common symptoms. Diagnosis is usually by buccal (cheek tissue) smear to ascertain the presence of a Barr body in a male, indicating an XXY karyotype. Some problems with behavior and speech development are associated with this syndrome.

Triple-X Female, 47,XXX, and Other Aneuploid Disorders of Sex Chromosomes

A triple-X female appears in about one in one thousand female live births. Fertility can be normal, but these individuals are usually mildly mentally retarded. Delayed growth, as well as congenital malformations, are also sometimes present. Other sex-chromosomal aneuploids, including XXXX, XXXXX, and XXXXY, are extremely rare. All seem to be characterized by mental retardation and growth deficiencies.

Euploidy

Euploid organisms have varying numbers of complete haploid chromosomal sets. We are already familiar with haploids (n) and diploids ($2n$). Organisms with higher numbers of sets, such as **triploids** ($3n$) and **tetraploids** ($4n$), are called polyploids. Three kinds of problems plague polyploids. First, the potential exists for a general imbalance in the organism due to the extra genetic material in each cell. For example, a triploid human fetus has about a one in a million chance to survive to birth, at which time death usually occurs due to problems in all

organ systems. Second, if there is a chromosomal sex-determining mechanism, it may be disrupted by polyploidy. And third, meiosis produces unbalanced gametes in many polyploids.

If the polyploid has an odd number of sets of chromosomes, such as triploid ($3n$), two of the three homologues will tend to pair at prophase I of meiosis, producing a bivalent and a univalent. The bivalent separates normally, but the third chromosome goes independently to one of the poles. This separation results in a 50% chance of aneuploidy in each of the n -different chromosomes, rapidly decreasing the probability of a balanced gamete as n increases. Therefore, as n increases, so does the likelihood of sterility. An alternative to the bivalent-univalent type of synapsis is the formation of trivalents, which have similar problems (fig. 8.28). Even-numbered polyploids, such as tetraploids ($4n$), can do better during meiosis. If the centromeres segregate two by two in each of the n meiotic figures, balanced gametes can result. Often, however, the multiple copies of the chromosomes form complex figures during synapsis, including monovalents, bivalents, trivalents, and quadrivalents, tending to result in aneuploid gametes and sterility.

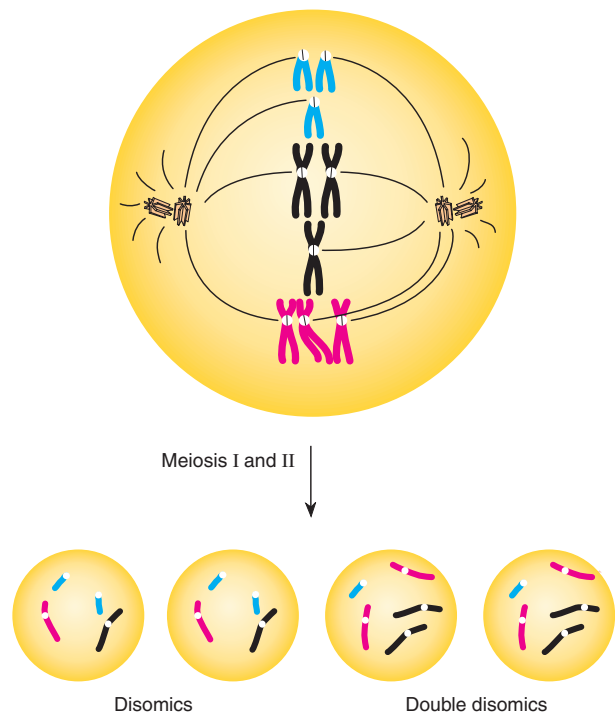


Figure 8.28 Meiosis in a triploid ($3n = 9$) and one possible resulting arrangement of gametes. The probability of a “normal” gamete is $(1/2)^n$ where n equals the haploid chromosome number. Here, $n = 3$ and $(1/2)^3 = 1/8$.

Some groups of organisms, primarily plants, have many polyploid members. An estimated 30 to 80% of all flowering plant species (angiosperms) are polyploids, as are 95% of ferns. (Polyploidy is apparently rare in gymnosperms and fungi.) For example, the genus of wheat, *Triticum*, has members with fourteen, twenty-eight, and forty-two chromosomes. Because the basic *Triticum* chromosome number is $n = 7$, these forms are $2n$, $4n$, and $6n$ species, respectively. Chrysanthemums have species of eighteen, thirty-six, fifty-four, seventy-two, and ninety chromosomes. With a basic number of $n = 9$, these species represent a $2n$, $4n$, $6n$, $8n$, and $10n$ series. In both these examples, the even-numbered polyploids are viable and fertile, but the odd-numbered polyploids are not.

Autopolyploidy

Polyploidy can come about in two different ways. In **autopolyploidy**, all of the chromosomes come from within the same species. In **allopolyploidy**, the chromosomes come from the hybridization of two different species (fig. 8.29). Autopolyploidy occurs in several different ways. The fusion of nonreduced gametes creates polyploidy. For example, if a diploid gamete fertilizes a normal haploid gamete, the result is a triploid. Similarly, if a diploid gamete fertilizes another diploid gamete, the result is a tetraploid. The equivalent of a nonreduced gamete comes about in meiosis if the parent cell is polyploid to begin with. For example, if one branch of a diploid plant is tetraploid, its flowers produce diploid gametes. These gametes are not the result of a failure to reduce chromosome numbers meiotically, but rather the result of successful meiotic reduction in a polyploid flower. The tetraploid tissue of the

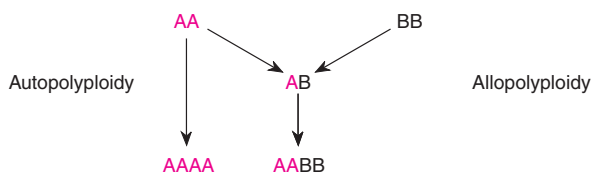


Figure 8.29 Autopolyploidy and allopolyploidy. If A and B are the haploid genomes of species A and B, respectively, then autopolyploidy produces a species with an AAAA karyotype, and allopolyploidy (with chromosome doubling) produces a species with AABB karyotype. If A represents seven chromosomes, then an AA diploid has fourteen chromosomes and an AAAA tetraploid has twenty-eight chromosomes. If B represents five chromosomes, then a BB diploid has ten chromosomes and an AABB allotetraploid has twenty-four chromosomes.

plant in this example can originate by the **somatic doubling** of diploid tissues.

Somatic doubling can come about spontaneously or be caused by anything that disrupts the normal sequence of a nuclear division. For example, colchicine induces somatic doubling by inhibiting microtubule formation. This prevents the formation of a spindle and thus prevents the chromosomes from moving apart during either mitosis or meiosis. The result is a cell with double the chromosome number. Other chemicals, temperature shock, and physical shock can produce the same effect.

Allopolyploidy

Allopolyploidy comes about by cross-fertilization between two species. The resulting offspring have the sum of the reduced chromosome number of each parent species. If each chromosome set is distinctly different, the new organisms have difficulty in meiosis because no two chromosomes are sufficiently homologous to pair. Then every chromosome forms a univalent (unpaired) figure, and they separate independently during meiosis, producing aneuploid gametes. However, if an organism can survive by vegetative growth until somatic doubling takes place in gamete precursor cells ($2n \rightarrow 4n$), or alternatively, if the zygote was formed by two unreduced gametes ($2n + 2n$), the resulting offspring will be fully fertile because each chromosome has a pairing partner at meiosis. We can draw an example from the work of Russian geneticist G. D. Karpechenko.

In 1928, Karpechenko worked with the radish (*Raphanus sativus*, $2n = 18$, $n = 9$) and cabbage (*Brassica oleracea*, $2n = 18$, $n = 9$). When these two plants are crossed, an F_1 results with $n + n = 18$ ($9 + 9$). This plant, which is an allodiploid, has characteristics intermediate between the two parental species (fig. 8.30). If somatic doubling takes place, the chromosome number is doubled to thirty-six, and the plant becomes an allopolyploid (an allotetraploid of $4n$). Since each chromosome has a homologue, this allotetraploid is also referred to as an **amphidiploid**. If we did not know its past history, this plant would simply be classified as a diploid with $2n = 36$. In this case, the new amphidiploid cannot successfully breed with either parent because the offspring are sterile triploids. It is, therefore, a new species and has been named *Raphanobrassica*. As an agricultural experiment, however, it was not a success because it did not combine the best features of the cabbage and radish.

Polyploidy in Plants and Animals

Although polyploids in the animal kingdom are known (in some species of lizards, fish, invertebrates, and a

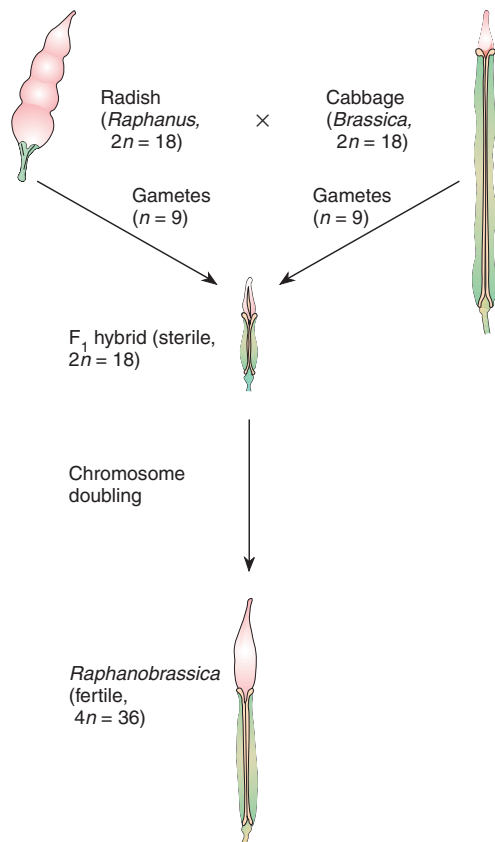


Figure 8.30 Hybridization of cabbage and radish, showing the resulting hybrid fruiting structures.

tetraploid mammal, the red viscacha rat), polyploidy as a successful evolutionary strategy is primarily a plant phenomenon. There are several reasons for this. To begin with, many more animals than plants have chromosomal sex-determining mechanisms. Polyploidy severely disrupts these mechanisms. For example, Bridges discovered a tetraploid female fruit fly, but it has not been possible to produce a tetraploid male. The tetraploid female's progeny were triploids and intersexes. A second reason why polyploidy is more common in plants is because plants can generally avoid the meiotic problems of polyploidy longer than most animals. Some plants can exist vegetatively, allowing more time for the rare somatic doubling event to occur that will produce an amphidiploid; animal life spans are more precisely defined, allowing less time for a somatic doubling. And third, many plants depend on the wind or insect pollinators to fertilize them and thus have more of an opportunity for hybridization. Many animals have relatively elaborate courting rituals that tend to restrict hybridization.

Polyploidy has been used in agriculture to produce “seedless” as well as “jumbo” varieties of crops. Seedless watermelon, for example, is a triploid. Its seeds are mostly sterile and do not develop. It is produced by growing seeds from the cross between a tetraploid variety and a diploid variety. Jumbo Macintosh apples are tetraploid.

S U M M A R Y

STUDY OBJECTIVE 1: To observe the nature and consequences of chromosomal breakage and reunion 178–190

Variation can occur in the structure and number of chromosomes in the cells of an organism. When chromosomes break, the ends become “sticky”; they tend to reunite with other broken ends. A single break can lead to deletions or the formation of acentric or dicentric chromosomes. Dicentrics tend to go through breakage-fusion-bridge cycles, which result in duplications and deficiencies.

Two breaks in the same chromosome can yield deletions and inversions. Variegation position effects, as well as new linkage arrangements, can result. Inversion heterozygotes produce loop figures during synapsis, which can form either at meiosis or in polytene chromosomes. Heterozygosity for an inversion suppresses crossovers; organisms that are heterozygotes are semisterile.

Reciprocal translocations can result from single breaks in nonhomologous chromosomes. These produce cross-shaped figures at synapsis and result in semisterility. The *Bar* eye phenotype of *Drosophila* is an example of a duplication that causes a position effect.

STUDY OBJECTIVE 2: To observe the nature and consequences of variation in chromosome numbers in human and nonhuman organisms 190–199

Changes in chromosome number can involve whole sets (euploidy) or partial sets (aneuploidy) of chromosomes. Aneuploidy usually results from nondisjunction or chromosomal lagging. Several medical syndromes, such as Down, Turner, and Klinefelter syndromes, and the XYY karyotype are caused by aneuploidy.

Polyploidy leads to difficulties in chromosomal sex-determining mechanisms, general chromosomal imbalance, and problems during meiotic segregation. It has been more successful in plants than in animals because plants generally lack chromosomal sex-determining mechanisms. Plants can also avoid meiotic problems by propagating vegetatively. In both animals and plants, even-numbered poly-

ploids do better than odd-numbered polyploids because they have a better chance of producing balanced gametes during meiosis. Somatic doubling provides each chromosome in a hybrid organism with a homologue, and thus makes possible tetrad formation at meiosis. New species have arisen by polyploidy.

SOLVED PROBLEMS

PROBLEM 1: What are the consequences of an inversion?

Answer: In an inversion homozygote, the consequences are change in linkage arrangements, including new orders and map distances, and the possibility of position effects if a locus is placed into or near heterochromatin. In an inversion heterozygote, crossover suppression causes semisterility because zygotes that carry genic imbalances are lost. Inversion heterozygotes can be seen as meiotic loop structures or loops formed in endomitotic chromosomes such as those found in the salivary glands of fruit flies. In an evolutionary sense, inversions result in supergenes, locking together allelic combinations.

PROBLEM 2: What are the consequences of a monosomic chromosome in human beings?

Answer: In human beings, monosomy is rare, meaning that, with few exceptions, it is lethal. In fact, monosomics are also rare in spontaneous abortions, indicating that most monosomic fetuses are lost before the woman is aware of the pregnancy. The only monosomics known to

be viable in human beings are Turner syndrome (45,X) and monosomics of chromosomes 21 and 22, the two smallest autosomal chromosomes.

PROBLEM 3: Ebony body (*e*) in flies is an autosomal recessive trait. A true-breeding ebony female (*ee*) is mated with a true-breeding wild-type male that has been irradiated. Among the wild-type progeny is a single ebony male. Explain this observation.

Answer: The cross is $ee \times e^+e^+$, and all F_1 s should be e^+e (wild-type). The use of irradiation alerts us to the possibility of chromosomal breaks, as well as simple mutations. What type of chromosomal aberration would allow a recessive trait to appear unexpectedly? A deletion, which creates pseudodominance when there is no second allele, is a good possibility. The male in question could have gotten the ebony allele from its mother and no homologous allele from its father. Alternatively, the wild-type allele from the father could have mutated to an ebony allele.

EXERCISES AND PROBLEMS*

VARIATION IN CHROMOSOMAL STRUCTURE

1. What kind of figure is observed in meiosis of a reciprocal translocation homozygote?
2. Can a deletion result in the formation of a variegation position effect? If so, how?
3. Does crossover suppression occur in an inversion homozygote? Explain.
4. Which rearrangements of chromosomal structure cause semisterility?
5. What are the consequences of single crossovers during tetrad formation in a reciprocal translocation heterozygote?
6. Give the gametic complement, in terms of acentrics, dicentrics, duplications, and deficiencies, when a three-strand double crossover occurs within a paracentric inversion loop.
7. In studying a new sample of fruit flies, a geneticist noted phenotypic variegation, semisterility, and the nonlinkage of previously linked genes. What probably caused this, and what cytological evidence would strengthen your hypothesis?
8. In a second sample of flies, the geneticist found a position effect and semisterility. The linkage groups were correct, but the order was changed and crossing over was suppressed. What probably caused this, and what cytological evidence would strengthen your hypothesis?

* Answers to selected exercises and problems are on page A-9.

9. Diagram the results of alternate segregation for a three-strand double crossover between a centromere and the cross center in a reciprocal translocation heterozygote.
10. A heterozygous plant $A B C D E/a b c d e$ is testcrossed with an $a b c d e/a b c d e$ plant. Only the following progeny appear:
- $A B C D E/a b c d e$
 $a b c d e/a b c d e$
 $A b c d e/a b c d e$
 $a B C D E/a b c d e$
 $A B C D e/a b c d e$
 $a b c d E/a b c d e$

What is unusual about the results? How can you explain them?

11. White eye color in *Drosophila* is an X-linked recessive trait. A wild-type male is irradiated and mated with a white-eyed female. Among the progeny is a white-eyed female.
- Why is this result unexpected, and how could you explain it?
 - What type of progeny would you expect if this white-eyed female is crossed with a normal, non-irradiated male?
12. You are trying to locate an enzyme-producing gene in *Drosophila*, which you know is located on the third chromosome. You have five strains with deletions for different regions of the third chromosome (a slash — / — indicates a deleted region):

Normal	0	10	20	30	40	50	60	map units
Strain A	/////	_____						
Strain B	_____	//////////	_____					
Strain C	_____	//////////	_____					
Strain D	_____	//////////	_____					
Strain E	_____	//////////	_____					

You cross each strain with wild-type flies and measure the amount of enzyme in the F_1 progeny. The results appear as follows. In what region is the gene located?

Strain Crossed	Percentage of Wild-Type Enzyme Produced in F_1 Progeny
A	100
B	45
C	54
D	98
E	101

13. Consider the following table, which shows the number of viable progeny produced by a plant under standard conditions. Provide an explanation for the results.

P_1 :	Strain A × Strain A	Strain B × Strain B	Strain A × Strain B
F_1 :	765	750	775
F_2 :	712	783	416

14. The map position for three X-linked recessive genes in *Drosophila* (v , vermilion eyes; m , miniature wings; and s , sable body) is:

v	m	s
33.0	36.1	43.0

A wild-type male is X-rayed and mated to a vermilion, miniature, sable female. Among the progeny is a single vermilion-eyed, long-winged, tan-bodied female. The following shows the progeny when this female is mated with a $v m s$ hemizygous male.

Females	Males
87 vermilion, miniature, sable	89 vermilion, miniature, sable
93 vermilion	1 vermilion

Explain these results by drawing a genetic map.

15. In *Drosophila*, recessive genes clot (ct) and black body (b) are located at 16.5 and 48.5 map units, respectively, on the second chromosome. In one cross, wild-type females that are $ct^+ b^+/ct b$ are mated with $ct b/ct b$ males. They produce these progeny:

wild-type	1,250
clot, black	1,200
black	30
clot	20

What is unusual about the results? How can you explain them?

16. You have four strains of *Drosophila* (1–4) that were isolated from different geographic regions. You compare the banding patterns of the second chromosome and obtain these results (each letter corresponds to a band):

- (1) m n r q p o s t u v
 (2) m n o p q r s t u v
 (3) m n r q t s u p o v
 (4) m n r q t s o p u v

If (3) is presumed to be the ancestral strain, in what order did the other strains arise?

17. In *Drosophila*, the recessive gene for white eyes is located near the tip of the X chromosome. A wild-type male is irradiated and mated with a white-eyed female. Among the progeny is one red-eyed male. How can you explain the red-eyed male, and how could you test your hypothesis?

VARIATION IN CHROMOSOME NUMBER

18. Is a tetraploid more likely to show irregularities in meiosis or mitosis? Explain. What about these processes in a triploid?
19. How many chromosomes would a human tetraploid have? How many chromosomes would a human monosomic have?
20. Do autopolyploids or allopolyploids experience more difficulties during meiosis? Do amphidiploids have more or less trouble than auto- or allopolyploids?
21. If a diploid species of $2n = 16$ hybridizes with one of $2n = 12$, and the resulting hybrid doubles its chromosome number to produce an allotetraploid (amphidiploid), how many chromosomes will it have? How many chromosomes will an allotetraploid have if both parent species had $2n = 20$?
22. If nondisjunction of the sex chromosomes occurs in a female at the second meiotic division, what type of eggs will arise?
23. How might an XO/XY human mosaic arise? An XX/XXY mosaic? How might a trisomy 21 individual arise?
24. Plant species P has $2n = 18$, and species U has $2n = 14$. A fertile hybrid is found. How many chromosomes does it have?
25. A woman with normal vision whose father was color-blind mates with a man with normal vision. They have a color-blind daughter with Turner syndrome. In which parent did nondisjunction occur?
26. A color-blind man mates with a woman with normal vision whose father was color-blind. They have a color-blind son with Klinefelter syndrome. In which parent did nondisjunction occur?
27. Describe a genetic event that can produce an XYY man.
28. Chromosomal analysis of a spontaneously aborted fetus revealed that the fetus was 92,XXYY. Propose an explanation to account for this unusual karyotype.

CRITICAL THINKING QUESTIONS

1. Various species in the grass genus *Bromus* have chromosome numbers of 14, 28, 42, 56, 70, 84, 98, and 112. What can you tell about the genetic relationships among these species and how they might have arisen?
2. There was a humorous television commercial in which someone accidentally discovered the desirability of combining chocolate and peanut butter. Could this combination be achieved by crossing peanut and cocoa plants?

Suggested Readings for chapter 8 are on page B-4.

9

CHEMISTRY OF THE GENE

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3. To investigate the way in which DNA replicates 220

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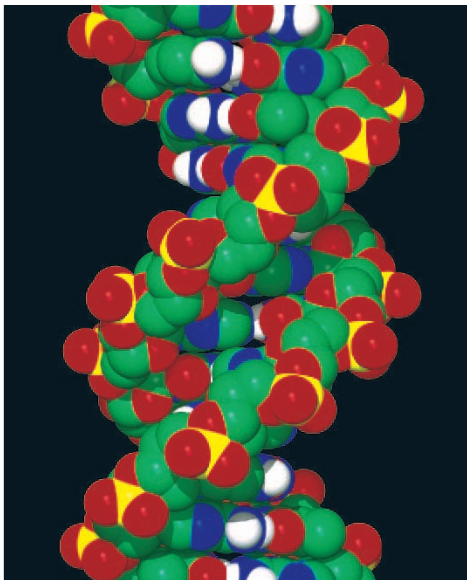
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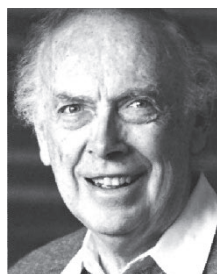
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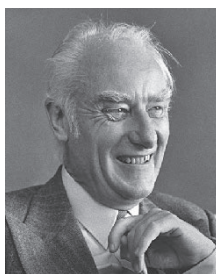


A computer-generated image of deoxyribonucleic acid, DNA. (© Professor K. Seddon & Dr. T. Evans/Queen's University Belfast/SPL/Photo Researchers.)

In 1953, James Watson and Francis Crick published a two-page paper in the journal *Nature* entitled “Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid.” It began as follows: “We wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.” This paper, which first put forth the correct model of DNA structure, is a milestone in the modern era of molecular genetics, compared by some to the work of Mendel and Darwin (box 9.1). (Watson, Crick, and X-ray crystallographer Maurice Wilkins won Nobel Prizes for this work; Rosalind Franklin, also an X-ray crystallographer, was acknowledged, posthumously, to have played a major role in the discovery of the structure of DNA.) Once the structure of the genetic material had been determined, an understanding of its method of replication and its functioning quickly followed.



James D. Watson
(1928–). (Cold Spring
Harbor Laboratory Research
Library Archives. Margot
Bennet, photographer.)



Francis Crick
(1916–). (Reproduced
by permission of Herb
Weitman, Washington
University, St. Louis,
Missouri.)



Maurice H. F. Wilkins
(1916–). (Courtesy of
Dr. Maurice H. F. Wilkins
and Biophysics Department,
King's College, London.)

IN SEARCH OF THE GENETIC MATERIAL

This chapter begins a sequence of nine chapters on the molecular structure of the genetic material, its replication, its expression, and the control of its expression. In this chapter, we look at the evidence that DNA is the genetic material, the chemistry of DNA, and the way in which DNA replicates, including the general enzymatic processes. We look first at prokaryotic, then at eukaryotic, DNA replication. Note that we concentrate on the molecular structure of DNA because, generally, structure reveals function: molecules have shapes that define how they work.

Required Properties of a Genetic Material

We begin with a look at the properties that a genetic material must have and review the evidence that nucleic acids make up the genetic material. To comprise the genes, DNA must carry the information to control the synthesis of the enzymes and proteins within a cell or organism; self-replicate with high fidelity, yet show a low level of mutation; and be located in the chromosomes.

Control of the Proteins

The growth, development, and functioning of a cell are controlled by the proteins within it, primarily its enzymes. Thus, the nature of a cell's phenotype is controlled by the protein synthesis within that cell. The genetic material must therefore determine the need for and effective amounts of the enzymes in a cell. For example, given inorganic salts and glucose, an *E. coli* cell can synthesize, through its enzyme-controlled biochemical pathways, all of the compounds it needs for growth, survival, and reproduction. In contrast, a mammalian red blood cell primarily produces hemoglobin.

At this point we need to review some basic information regarding enzymes. An enzyme is a protein that acts as a catalyst for a specific metabolic process without itself being markedly altered by the reaction. Most reactions that enzymes catalyze could occur anyway, but only under conditions too extreme to take place within living systems. For example, many oxidations occur naturally at high temperatures. Enzymes allow these reactions to occur within the cell by lowering the **free energy of activation** ($\Delta G^\ddagger$) of a particular reaction. In other words, an enzyme allows a reaction to take place without needing the boost in energy that heat usually supplies (fig. 9.1).

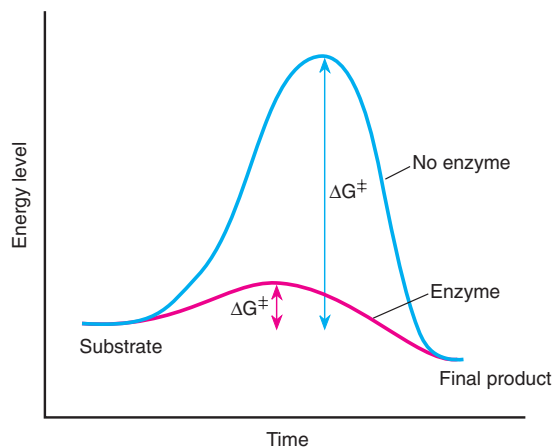


Figure 9.1 An enzyme lowers the free energy of activation ($\Delta G^\ddagger$) for a particular reaction.

BOX 9.1

We wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of considerable biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey.¹ They kindly made their manuscript available to us in advance of publication. Their model consists of three intertwined chains, with the phosphates near the fibre axis, and the bases on the outside. In our opinion, this structure is unsatisfactory for two reasons: (1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid. Without the acidic hydrogen atoms it is not clear what forces would hold the structure together, especially as the negatively charged phosphates near the axis will repel each other. (2) Some of the van der Waals distances appear to be too small.

Another three-chain structure has also been suggested by Fraser (in the press). In his model the phosphates are on the outside and the bases on the inside, linked together by hydrogen bonds. This structure as described is rather ill-defined, and for this reason we shall not comment on it.

We wish to put forward a radically different structure for the salt of deoxyribose nucleic acid. This structure has two helical chains each coiled round the same axis (see diagram

Historical Perspectives

Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid

[fig. 1]). We have made the usual chemical assumptions, namely, that each chain consists of phosphate di-ester groups joining β -D-deoxyribofuranose residues with 3', 5' linkages. The two chains (but not their bases) are related by a dyad perpendicular to the fibre axis. Both chains follow right-handed helices, but owing to the dyad the sequences of the atoms in the two chains run in opposite directions. Each chain loosely resembles Furberg's² model No. 1; that is, the bases are on the inside of the helix and the phosphates on the outside. The configuration of the sugar and the atoms near it is close to Furberg's 'standard configuration,' the sugar being roughly perpendicular to the attached base. There is a residue on each chain every 3.4 Å in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 Å. The distance

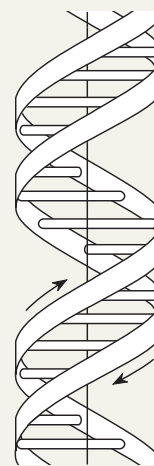


Figure 1 This figure is purely diagrammatic. The two *ribbons* symbolize the two phosphate-sugar chains, and the *horizontal rods* represent the pairs of bases holding the chains together. The *vertical line* marks the fiber axis. (Reprinted with permission from *Nature*, Vol. 171, No. 4356. Watson and Crick, "Molecular Structure of Nucleic Acids," pp. 737-738. Copyright © 1953 Macmillan Magazines Limited.)

of a phosphorus atom from the fibre axis is 10 Å. As the phosphates are on the outside, cations have easy access to them.

Most metabolic processes, such as the biosynthesis or degradation of molecules, occur in pathways, with enzyme facilitating each step in the pathway (see chapter 2). The metabolic pathway for the conversion of threonine into isoleucine (two amino acids) appears in figure 9.2. Each reaction product in the pathway is altered by an enzyme that converts it to the next product. The enzyme threonine dehydratase, for example, converts threonine into α -ketobutyric acid. Enzymes are composed of folded polymers of amino acids. The average protein is three hundred to five hundred amino acids long; only twenty naturally occurring amino acids are used in constructing

these proteins. The sequence of amino acids determines the final structure of an enzyme. (We discuss the structure of proteins in more detail in chapter 11.) The genetic material determines the sequence of the amino acids.

The three-dimensional structure of enzymes permits them to perform their function. An enzyme combines with its substrate or substrates (the molecules it works on) at a part of the enzyme called the **active site** (fig. 9.3). The substrates "fit" into the active site, which has a shape that allows only the specific substrates to enter. This view of the way an enzyme interacts with its substrates is called the *lock-and-key model* of enzyme functioning.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical *z*-coordinates. One of the pair must be a purine and the other a pyrimidine for bonding to occur. The hydrogen bonds are made as follows: purine position 1 to pyrimidine position 1; purine position 6 to pyrimidine position 6.

If it is assumed that the bases only occur in the structure in the most plausible tautomeric forms (that is, with the keto rather than the enol configurations), it is found that only specific pairs of bases can bond together. These pairs are: adenine (purine) with thymine (pyrimidine), and guanine (purine) with cytosine (pyrimidine).

In other words, if an adenine forms one member of a pair, on either chain, then on these assumptions the other member must be thymine; similarly for guanine and cytosine. The sequence of bases on a single chain does not appear to be restricted in

any way. However, if only specific pairs of bases can be formed, it follows that if the sequence of bases on one chain is given, then the sequence on the other chain is automatically determined.

It has been found experimentally^{3,4} that the ratio of the amounts of adenine to thymine, and the ratio of guanine to cytosine, are always very close to unity for deoxyribose nucleic acid. It is probably impossible to build this structure with a ribose sugar in place of the deoxyribose, as the extra oxygen atom would make too close a van der Waals contact.

The previously published X-ray data^{5,6} on deoxyribose nucleic acid are insufficient for a rigorous test of our structure. So far as we can tell, it is roughly compatible with the experimental data, but it must be regarded as unproved until it has been checked against more exact results. Some of these are given in the following communications. We were not aware of the details of the results presented there when we devised our structure, which rests mainly though not entirely on published experimental data and stereo-chemical arguments.

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

Full details of the structure, including the conditions assumed in

building it, together with a set of coordinates for the atoms, will be published elsewhere.

We are much indebted to Dr. Jerry Donohue for constant advice and criticism, especially on inter-atomic distances. We have also been stimulated by a knowledge of the general nature of the unpublished experimental results and ideas of Dr. M. H. F. Wilkins, Dr. R. E. Franklin and their coworkers at King's College, London. One of us (J. D. W.) has been aided by a fellowship from the National Foundation for Infantile Paralysis.

J. D. Watson
F. H. C. Crick

Medical Research Council Unit for the Study of the Molecular Structure of Biological Systems, Cavendish Laboratory, Cambridge. April 2.

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1. Pauling, L., and Corey, R. B., *Nature*, 171, 346 (1953); *Proc. U.S. Nat. Acad. Sci.*, 39, 84 (1953).
2. Furberg, S., *Acta Chem. Scand.*, 6, 634 (1952).
3. Chargaff, E., for references see Zamenhof, S., Brawerman, G., and Chargaff, E., *Biochim. et Biophys. Acta*, 9, 402 (1952).
4. Wyatt, G. R., *J. Gen. Physiol.*, 36, 201 (1952).
5. Astbury, W. T., *Symp. Soc. Exp. Biol. 1, Nucleic Acid* 66 (Camb. Univ. Press, 1947).
6. Wilkins, M. H. F., and Randall, J. T., *Biochim. et Biophys. Acta*, 10, 192 (1953).

When the substrates are in their proper position in the active site of the enzyme, the particular reaction that the enzyme catalyzes takes place. The reaction products then separate from the enzyme and leave it free to repeat the process. Enzymes can work at phenomenal speeds. Some can catalyze as many as a million reactions per minute.

Not all of the cell's proteins function as catalysts. Some are structural proteins, such as keratin, the main component of hair. Other proteins are regulatory—they control the rate at which other enzymes work. Still others are involved in different functions; albumins, for example, help regulate the osmotic pressure of blood.

Replication

The genetic material must be capable of precisely directing its own replication so that every daughter cell receives an exact copy. Some **mutability**, or the ability to change, is also required, because we know that the genetic material has changed, or evolved, over the history of life on earth. In their 1953 paper, Watson and Crick had already worked out the replication process based on the structure of DNA. The fidelity of the replication process is so great that the error rate is only about one in a billion.

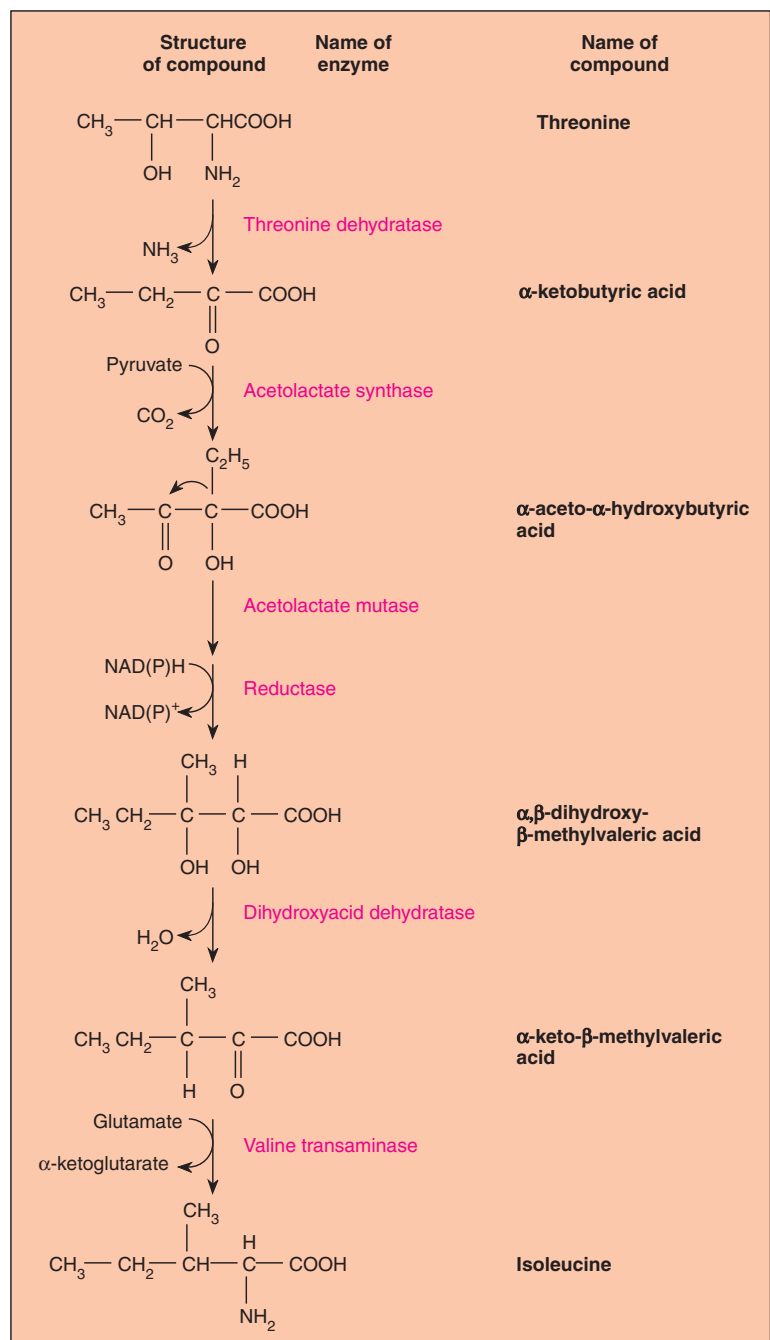


Figure 9.2 Metabolic pathway of conversion of the amino acid threonine into isoleucine.

Location

It has been known since the turn of the century that genes, the discrete functional units of genetic material, are located in chromosomes within the nuclei of eukaryotic cells: the way chromosomes behave during the cel-

lular division stages of mitosis and meiosis mimics the behavior of genes. Thus, the genetic material in eukaryotes must be a part of the chromosomes.

For a long time, proteins were considered the most probable genetic material because they have the neces-

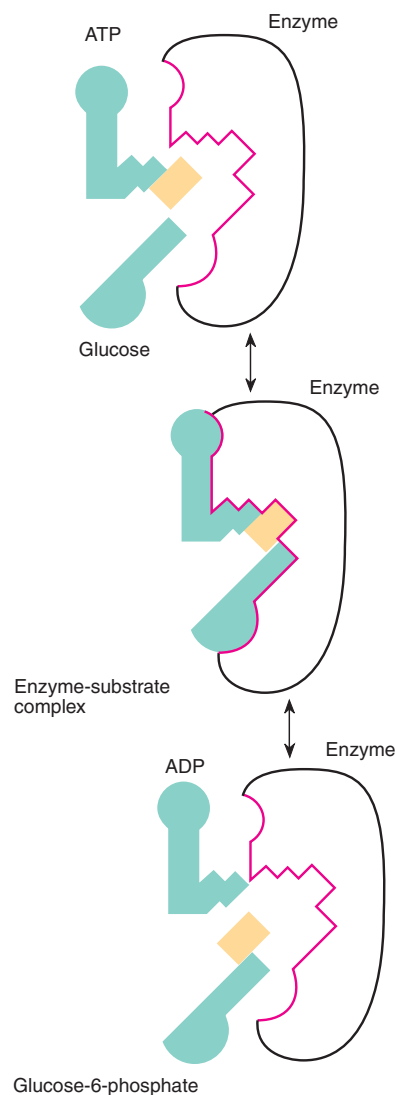


Figure 9.3 The active site of an enzyme recognizes a specific substance. In this case, ATP plus glucose is converted into ADP and glucose-6-phosphate by the enzyme hexokinase. The active site is diagrammed in red. The terminal phosphate group of ATP is tan.

Oswald T. Avery (1877–1955).
(Courtesy of the National Academy of
Sciences.)



sary molecular complexity. The twenty naturally occurring amino acids can be combined in an almost unlimited variety, creating thousands and thousands of different proteins. The first proof that the genetic material is deoxyribonucleic acid (DNA) came in 1944 from Oswald Avery and his colleagues. The Watson and Crick model in 1953 ended a period when many thought DNA was the genetic material, but its structure was unknown.

Evidence for DNA as the Genetic Material

Transformation

In 1928, F. Griffith reported that heat-killed bacteria of one type could “transform” living bacteria of a different type. Griffith demonstrated this transformation using two strains of the bacterium *Streptococcus pneumoniae*. One strain (S) produced smooth colonies on media in a petri plate because the cells had polysaccharide capsules. It caused a fatal bacteremia (bacterial infection) in mice. Another strain (R), which lacked polysaccharide capsules, produced rough colonies on petri plates (fig. 9.4); it did not have a pathological effect on mice. Bacteria of the rough strain are engulfed by the mice’s white blood cells; bacteria of the virulent smooth strain survive because their polysaccharide coating protects them.

Griffith found that neither heat-killed S-type nor live R-type cells, by themselves, caused bacteremia in mice. However, if he injected a mixture of live R-type and heat-killed S-type cells into mice, the mice developed a bacteremia identical to that caused by living S-type cells (fig. 9.5). Thus, something in the heat-killed S cells transformed the R-type bacteria into S-type cells.

In 1944, Oswald Avery and two of his associates, C. MacLeod and M. McCarty, reported the nature of the transforming substance. Avery and his colleagues did their work *in vitro* (literally, in glass), using colony morphology on culture media rather than bacteremia in mice as evidence of transformation. They ruled out proteins, carbohydrates, and lipids by their extraction procedure, by the chemical analysis of the transforming material, and by demonstrating that the only enzymes that destroyed the transforming ability were enzymes that de-

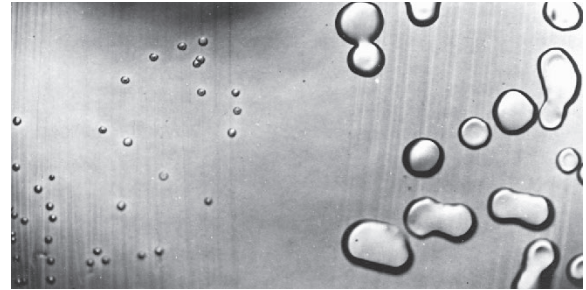


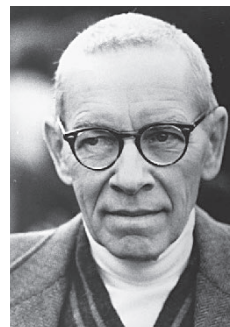
Figure 9.4 Petri plate with smooth and rough colonies of *Streptococcus pneumoniae*. R (rough) strain colonies appear on the left and S (smooth) colonies on the right on the same agar. Magnification 3.5×. (O. T. Avery, C. M. Macleod, and M. McCarty, “Studies on the chemical nature of the substance inducing transformation of pneumococcal types.” Reproduced from the *Journal of Experimental Medicine* 79 (1944):137–58, fig. 1 by copyright permission of the Rockefeller University Press. Reproduced by permission. Photograph made by Mr. Joseph B. Haulenbeek.)

stroyed DNA. This study provided the first experimental evidence that DNA was the genetic material: DNA transformed R-type bacteria into S-type bacteria.

Phage Labeling

Valuable information about the nature of the genetic material has also come from viruses. Of particular value are studies of bacterial viruses—the bacteriophages, or phages. Since phages consist only of nucleic acid surrounded by protein, they lend themselves nicely to the determination of whether the protein or the nucleic acid is the genetic material.

A. D. Hershey and M. Chase published, in 1952, the results of research that supported the notion that DNA is the genetic material and, in the process, helped to



A. D. Hershey
(1908–1997). (Courtesy of
Dr. A. D. Hershey.)



Martha Chase. (Courtesy
of Cold Springs Harbor
Laboratory Archives.)

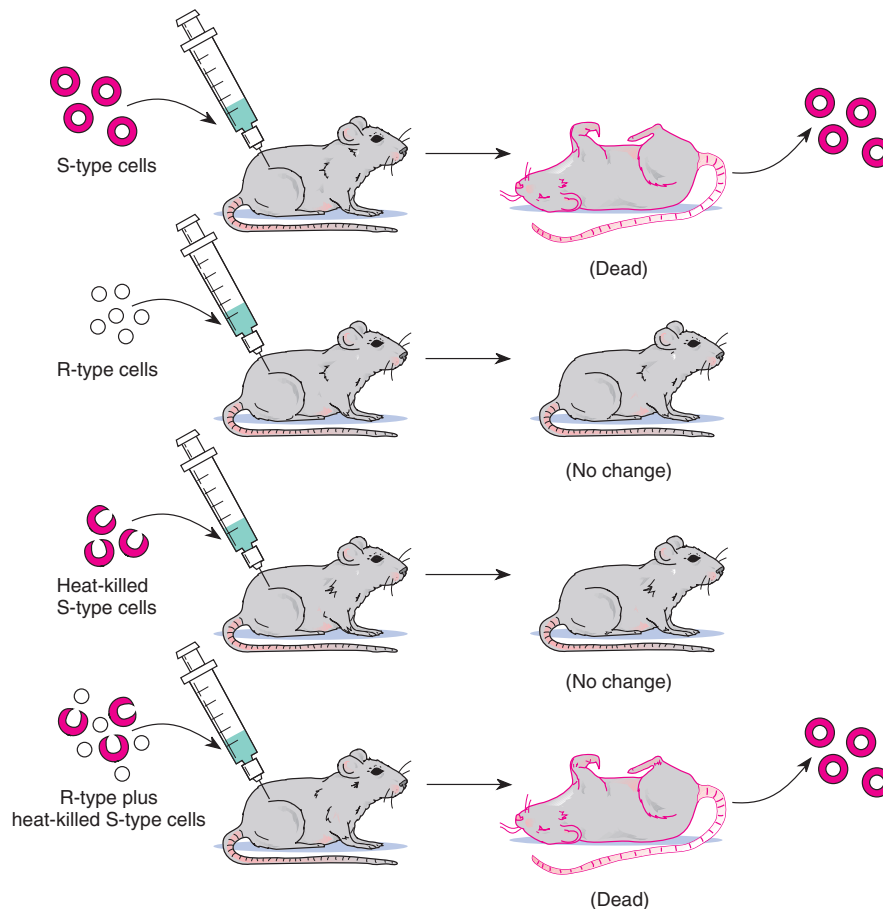


Figure 9.5 Griffith's experiment with *Streptococcus*. S-type cells will kill mice; so will heat-killed S-type cells injected with live R-type cells. S-type cells are recovered from dead mice in both cases.

explain the nature of the viral infection process. Since all nucleic acids contain phosphorus, whereas proteins do not, and since most proteins contain sulfur (in the amino acids cysteine and methionine), whereas nucleic acids do not, Hershey and Chase designed an experiment using radioactive isotopes of sulfur and phosphorus to keep separate track of the viral proteins and nucleic acids during the infection process. They used the T2 bacteriophage and the bacterium *Escherichia coli*. The phages were labeled by having them infect bacteria growing in culture medium containing the radioactive isotopes ^{35}S or ^{32}P . Hershey and Chase then proceeded to identify the material injected into the cell by phages attached to the bacterial wall.

When ^{32}P -labeled phages were mixed with unlabeled *E. coli* cells, Hershey and Chase found that the ^{32}P label entered the bacterial cells and that the next generation of phages that burst from the infected cells carried a significant amount of the ^{32}P label. When ^{35}S -labeled phages

were mixed with unlabeled *E. coli*, the researchers found that the ^{35}S label stayed outside the bacteria for the most part. Hershey and Chase thus demonstrated that the outer protein coat of a phage does not enter the bacterium it infects, whereas the phage's inner material, consisting of DNA, does enter the bacterial cell (fig. 9.6). Since the DNA is responsible for the production of the new phages during the infection process, the DNA, not the protein, must be the genetic material.

RNA as Genetic Material

In some viruses, RNA (ribonucleic acid) is the genetic material. The tobacco mosaic virus that infects tobacco plants consists only of RNA and protein. The single, long RNA molecule is packaged within a rodlike structure formed by over two thousand copies of a single protein. No DNA is present in tobacco mosaic virus particles (fig. 9.7a). In 1955, H. Fraenkel-Conrat and R. Williams

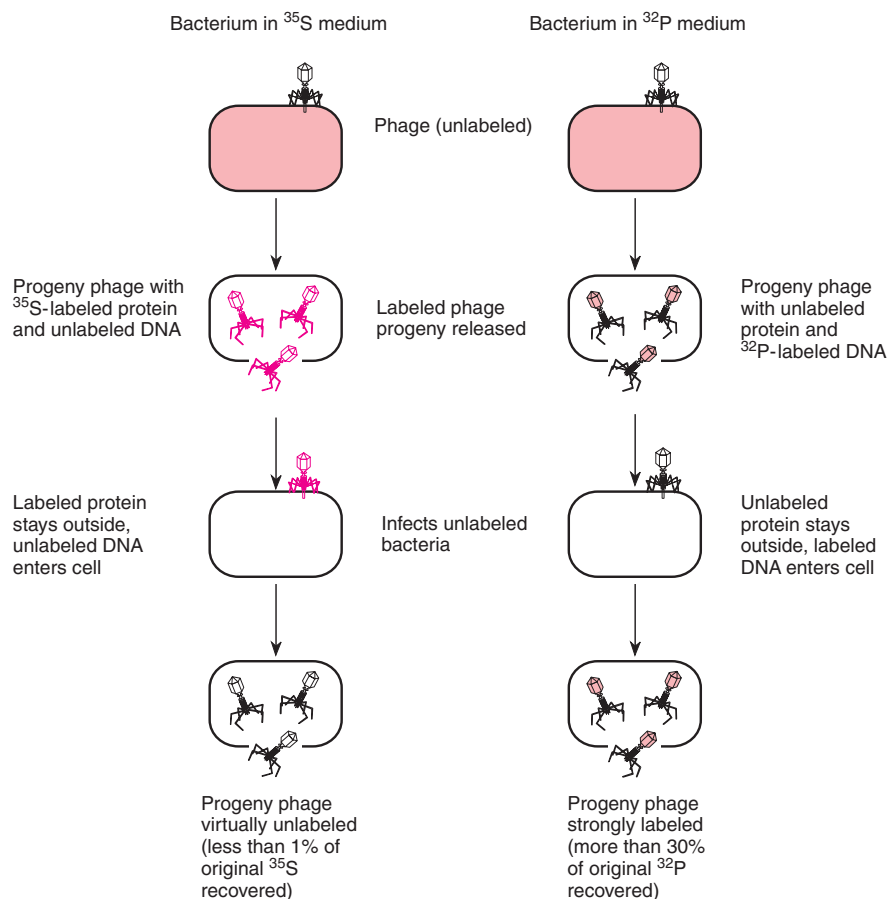


Figure 9.6 The Hershey and Chase experiments using ³⁵S-labeled and ³²P-labeled T2 bacteriophages. The nucleic acid label (³²P) enters the *E. coli* bacteria during infection; the protein label (³⁵S) does not.

showed that a virus can be separated, *in vitro*, into its component parts and reconstituted as a viable virus. This finding led Fraenkel-Conrat and B. Singer to reconstitute tobacco mosaic virus with parts from different strains (fig. 9.7*b*). For example, they combined the RNA from the common tobacco mosaic virus with the protein from the masked (M) strain of tobacco mosaic virus. They then made the reciprocal combination of common-type protein and M-type RNA. In both cases, the tobacco mosaic virus produced during the process of infection was the type associated with the RNA, not with the protein. Thus, it was the nucleic acid (RNA in this case) that was the genetic material. Subsequently, scientists rubbed pure tobacco mosaic virus RNA into plant leaves. Normal infection and a new generation of typical, protein-coated tobacco mosaic virus resulted, confirming RNA as the genetic material for this virus.

We thus conclude that DNA is the genetic material. In the few viruses that do not have DNA, RNA serves as the

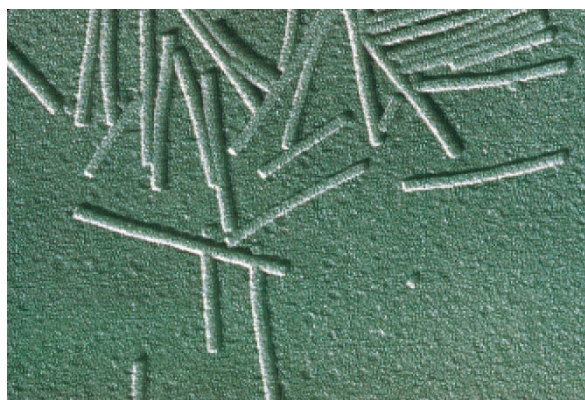
genetic material. The only exception to these statements is one type of disease that is transmitted by a protein without accompanying DNA or RNA (box 9.2).

CHEMISTRY OF NUCLEIC ACIDS

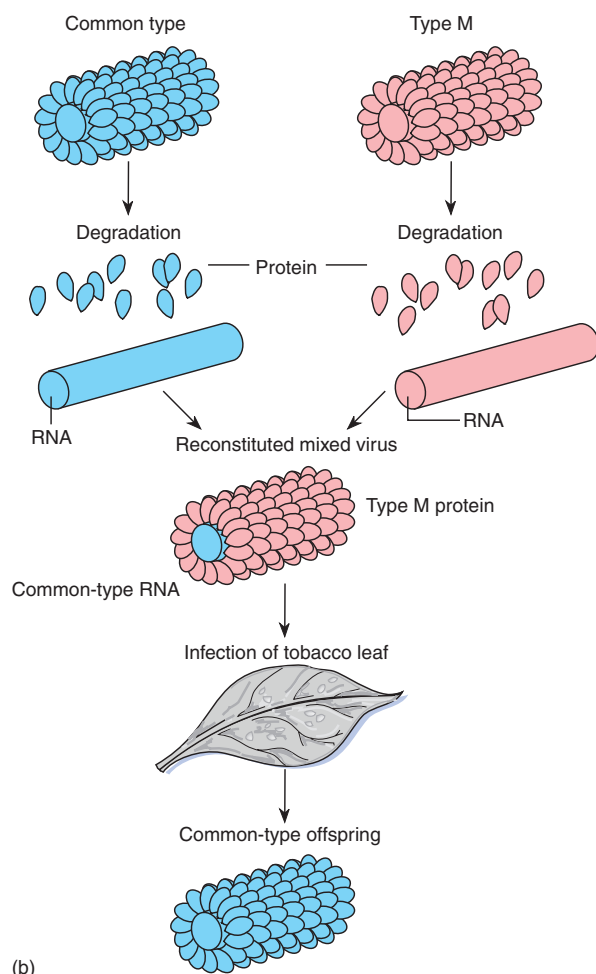


Having identified the genetic material as the nucleic acid DNA (or RNA), we proceed to examine the chemical structure of these molecules. Their structure will tell us a good deal about how they function.

Nucleic acids are made by joining **nucleotides** in a repetitive way into long, chainlike polymers. Nucleotides are made of three components: phosphate, sugar, and a nitrogenous base (table 9.1 and fig. 9.8). When incorporated into a nucleic acid, a nucleotide contains one of each of the three components. But, when free in the cell



(a)



(b)

Figure 9.7 (a) Electron micrograph of tobacco mosaic virus. Magnification 37,428 $\times$. (b) Reconstitution experiment of Fraenkel-Conrat and Singer. The nucleic acid (RNA), not the protein component of the virus, controls inheritance.
([a] © Biology Media/Photo Researchers, Inc.)

Table 9.1 Components of Nucleic Acids

	Phosphate	Sugar	Base	
			Purines	Pyrimidines
DNA	Present	Deoxyribose	Guanine Adenine	Cytosine Thymine
RNA	Present	Ribose	Guanine Adenine	Cytosine Uracil

pool, nucleotides usually occur as triphosphates. The energy held in the extra phosphates is used, among other purposes, to synthesize the polymer. A **nucleoside** is a sugar-base compound. Nucleotides are therefore nucleoside phosphates (fig. 9.9). (Note that ATP, adenosine triphosphate, the energy currency of the cell, is a nucleoside triphosphate.)

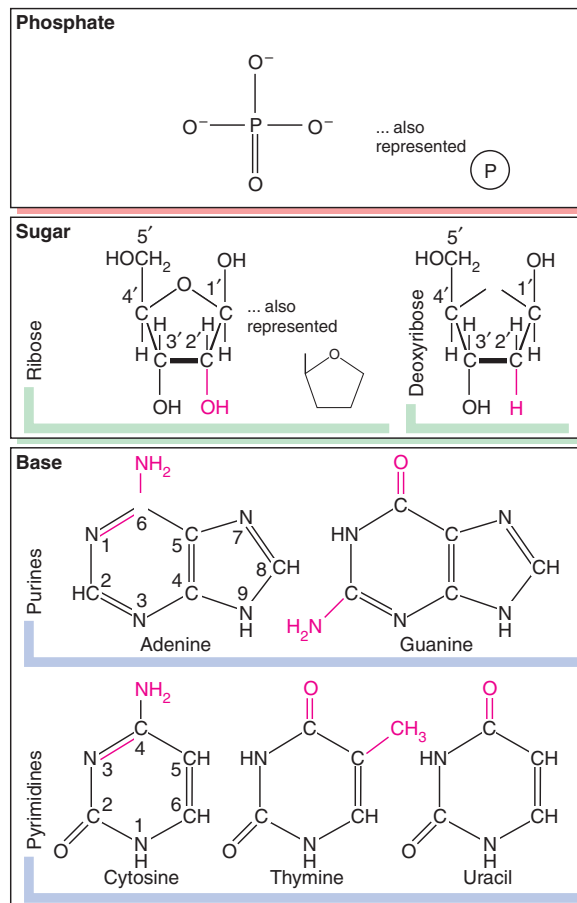


Figure 9.8 Components of nucleic acids: phosphate, sugars, and bases. Primes are used to number the ring positions in the sugars to differentiate them from the ring positions in the bases.

BOX 9.2

Without exception, the genetic material is either DNA or RNA; it is RNA only in a few viruses. Since virtually all transmissible diseases are of bacterial or viral origin, this means that transmissible diseases are also caused by organisms with DNA or RNA as their genetic material. However, in one interesting situation, a transmissible disease appears to be caused by an agent without genetic material. Four human neurological diseases and six similar animal diseases are caused, we believe, by proteins without DNA or RNA. (Two conditions in yeast are probably caused in a similar way.) The human diseases are kuru, Creutzfeldt-Jakob disease, Gerstmann-Sträussler-Scheinker syndrome, and a recently discovered fatal familial insomnia. The animal diseases are scrapie (sheep and goats), four encephalopathies (bovine, feline, ungulate, and mink), and chronic wasting disease (deer and elk). All of these diseases are extremely slow to develop, all are fatal, and all are believed to be caused either by the ingestion of a protein from an infected individual or from a mutation of the normal gene. None of the diseases as yet has a cure, and the mechanism of action is not completely understood.

The diseases appear to be caused by a protein, similar to one normally produced in the brain of healthy individuals. The term **prion** (taken from *proteinaceous infectious particle*) has been given to these agents by Stanley Prusiner at the University of California in San Francisco, a 1997 Nobel laureate. He, along with colleagues,

Biomedical
Applications*Prions: The Biological
Equivalent of Ice-Nine*

isolated the prion protein (PrP) and recently located the gene that codes for the protein on the short arm of chromosome 20. In addition to the infective form, a familial (inherited) form of these diseases can result from a mutation of the gene that codes for the prion protein active in normal individuals (probably at least all mammals). The normal protein is termed PrP^C, and the mutated form is referred to as PrP^{Sc}. Normally, PrP^C is a glycoprotein found on the membrane surface of the cells of the brain and some other tissues.

Although no cures exist for these diseases, kuru, at least, seems to be almost eradicated. It was found only among people in part of New Guinea who practiced cannibalism. Once the people stopped this practice, the spread of the disease also ceased; kuru does not seem to be generated to any major extent by mutation. By controlling feeding practices, it is believed, bovine spongiform encephalopathy will also disappear. In the past, cows were fed protein supplements contaminated by material from infected animals.

In England, a recent epidemic of bovine spongiform encephalopathy

(BSE, or mad cow disease) peaked in 1992-1993, affecting over 160,000 cattle. At least fourteen cases of a variant of Creutzfeldt-Jakob disease in people in England and France were attributed to eating affected beef, creating a panic in England. With a change away from using animal matter in cattle feed and a culling of cattle herds, the epidemic has ended. However, new human cases may show up in the future owing to the long incubation period of this prion disease.

The obvious question is, how does a protein that does not appear to contain genetic material cause a transmissible disease when ingested? Prusiner has suggested several mechanisms that would allow an infective protein to induce copies of the normal protein to become infective. One of these mechanisms involves a cascade in which an infective PrP^{Sc} binds with a normal PrP^C, resulting in two infective PrP^{Sc} proteins. From this, one produces two, two produce four, four produce eight, and so on. As Nancy Touchette, writing in *The Journal of NIH Research*, pointed out, this is the way Kurt Vonnegut described the behavior of the mythical ice-nine in his 1963 book, *Cat's Cradle*. In this fictional account, a single seed caused all of the water on earth, by a chain reaction cascade, to form into a novel type of ice. We have not yet resorted to science fiction to answer the mystery of prion function; however, it seems reasonable to guess that an eventual understanding of the mechanism of prion function will provide us with a biological novelty.

The sugars differ only in the presence (ribose in RNA) or absence (deoxyribose in DNA) of an oxygen in the 2' position. (The carbons of the sugars are numbered 1' to 5'. The primes are used to avoid confusion with the numbering system of the bases; see fig. 9.8.) DNA and RNA both have four bases (two **purines** and two **pyrimidines**) in their nucleotide chains. Both molecules have

the purines **adenine** and **guanine** and the pyrimidine **cytosine**. DNA has the pyrimidine **thymine**; RNA has the pyrimidine **uracil**. Thus, three of the nitrogenous bases are found in both DNA and RNA, whereas thymine is unique to DNA, and uracil is unique to RNA.

A nucleotide is formed in the cell when a base attaches to the 1' carbon of the sugar and a phosphate attaches to

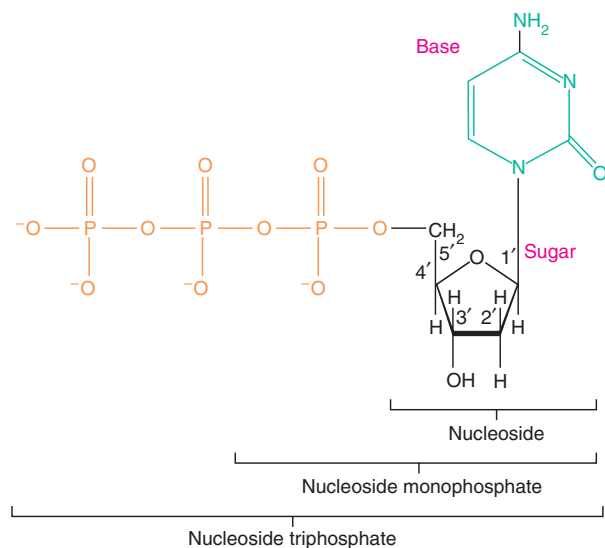


Figure 9.9 The structure of a nucleoside and two nucleotides: a nucleoside monophosphate and a nucleoside triphosphate.

the 5' carbon of the same sugar (fig. 9.10); the nucleotide takes its name from the base (table 9.2). Nucleotides are linked together (**polymerized**) by the formation of a bond between the phosphate at the 5' carbon of one nucleotide and the hydroxyl (OH) group at the 3' carbon of an adjacent molecule. Very long strings of nucleotides can be polymerized by this **phosphodiester bonding** (fig. 9.11).

Biologically Active Structure



Although the identities of the nucleotides that polymerized to form a strand of DNA or RNA were known, the actual structures of these nucleic acids when they function as the genetic material remained unknown until 1953. The general feeling was that the biologically active structure of DNA was more complex than a single string of nucleotides linked together by phosphodiester bonds, and that several interacting strands were involved. In 1953, Linus Pauling, a Nobel laureate who had discovered the

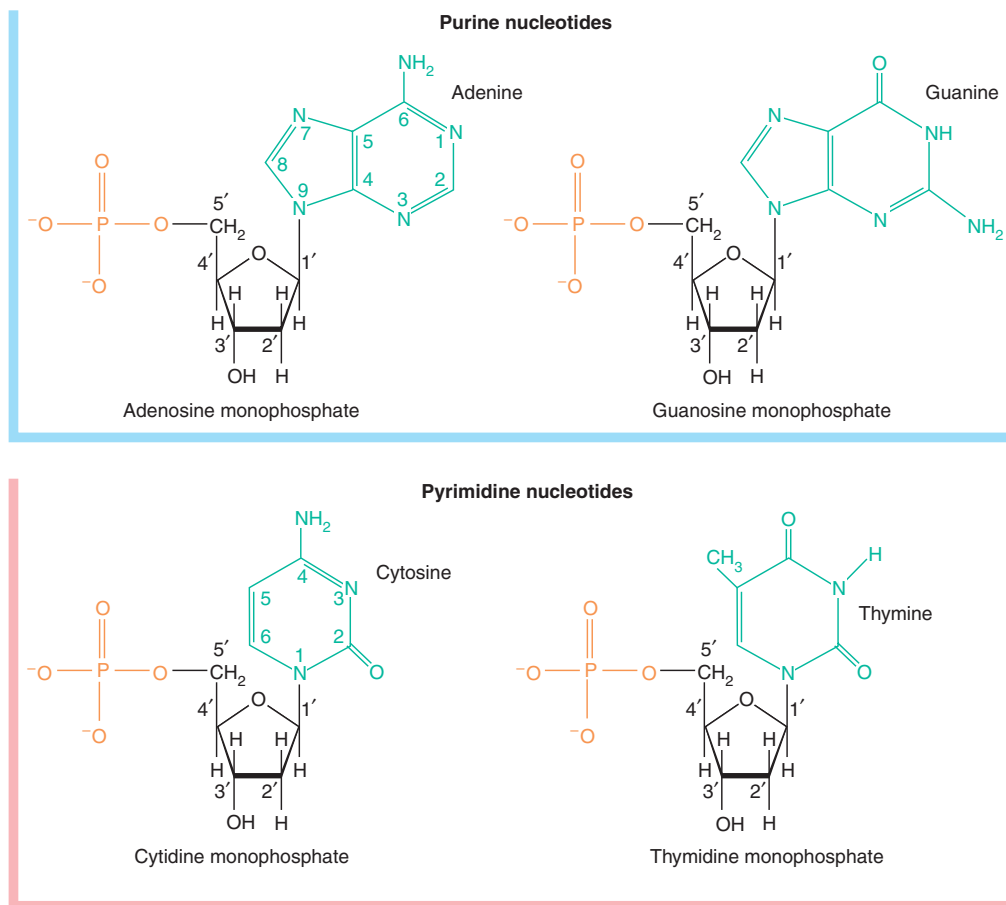
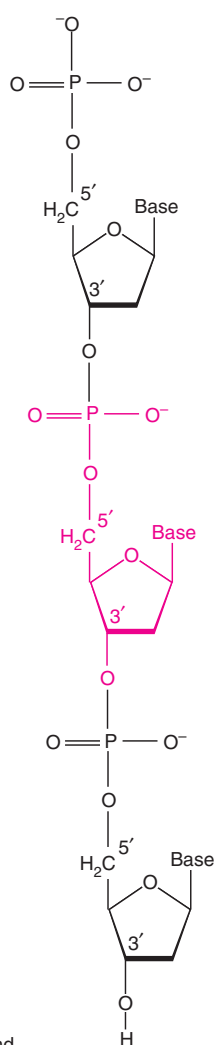


Figure 9.10 Structure of the four deoxyribose nucleotides.

Table 9.2 Nucleotide Nomenclature

Base	Nucleotide (nucleoside monophosphate)	Abbreviation					
		Monophosphate		Diphosphate		Triphosphate	
		Ribose	Deoxyribose	Ribose	Deoxyribose	Ribose	Deoxyribose
Guanine	Guanosine monophosphate	GMP		GDP		GTP	
	Deoxyguanosine monophosphate		dGMP		dGDP		dGTP
Adenine	Adenosine monophosphate	AMP		ADP		ATP	
	Deoxyadenosine monophosphate		dAMP		dADP		dATP
Cytosine	Cytidine monophosphate	CMP		CDP		CTP	
	Deoxycytidine monophosphate		dCMP		dCDP		dCTP
Thymine	Deoxythymidine monophosphate		dTMP		dTDP		dTTP
Uracil	Uridine monophosphate	UMP		UDP		UTP	

5'-PO₄ end**Figure 9.11**

Polymerization of adjacent nucleotides to form a sugar-phosphate strand. There is no limit to the length the strand can be or on the type of base attached to each nucleotide residue.

helical structure of proteins, was investigating a three-stranded structure for the genetic material, whereas Watson and Crick had decided that a two-stranded structure was more consistent with available evidence. Three lines of evidence directed Watson and Crick: the chemical nature of the components of DNA, X-ray crystallography, and Chargaff's ratios.

DNA X-Ray Crystallography

All the time Watson and Crick were studying DNA structure, Maurice Wilkins, Rosalind Franklin, and their colleagues were using **X-ray crystallography** to analyze the structure of DNA. The molecules in a crystal are arranged in an orderly way, so that when a beam of X rays is aimed at the crystal, the beam scatters in an orderly fashion. The scatter pattern can be recorded on photographic film or computer-controlled devices. The nature of this pattern depends on the structure of the crystal. The cross in the center of the photograph in figure 9.12 indicates that the molecule is a helix; the dark areas at the top and bottom come from the bases, stacked perpendicularly to the main axis of



Rosalind E. Franklin
(1920–1958). (Courtesy of Cold
Spring Harbor Laboratory.)

3'-OH end

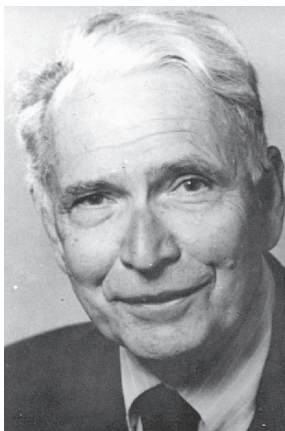
the molecule. This image of the DNA molecule stimulated Watson and Crick's understanding of its structure.

Chargaff's Ratios

Until Erwin Chargaff's work, scientists had labored under the erroneous **tetranucleotide hypothesis**. This hypothesis proposed that DNA was made up of equal quantities of the four bases; therefore, a subunit of this DNA consisted of one copy of each base. Chargaff carefully analyzed the base composition of DNA in various species (table 9.3). He found that although the relative amount of a given nucleotide differs among species, the amount of adenine equaled that of thymine and the amount of guanine equaled that of cytosine. That is, in the DNA of all the organisms studied, a 1:1 correspondence exists between the purine and pyrimidine bases. This is known as **Chargaff's rule**. Chargaff's observations disproved the tetranucleotide hypothesis; the four bases of DNA did not occur in a 1:1:1:1 ratio. His results gave insight to Watson and Crick in the development of their model.

The Watson-Crick Model

With the information available, Watson and Crick began constructing molecular models. They found that a possible structure for DNA was one in which two helices coiled around one another (a **double helix**), with the sugar-



Erwin Chargaff (1905–).
(Courtesy of Dr. Erwin Chargaff.)

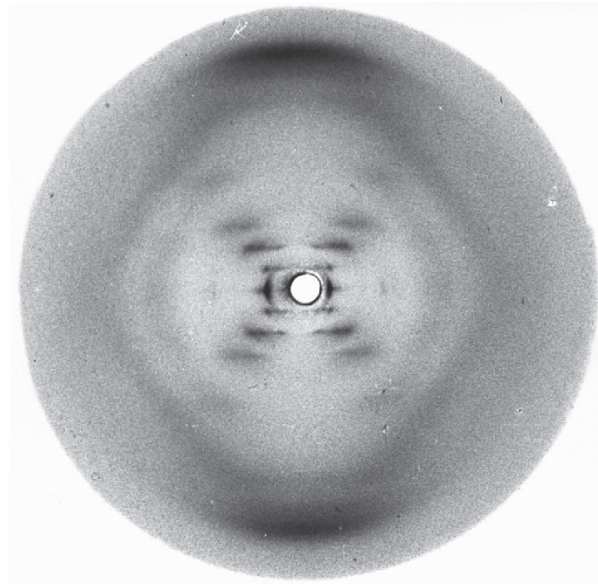


Figure 9.12 Scatter pattern of a beam of X rays passed through crystalline DNA. (Source: Reprinted by permission from R. E. Franklin and R. Gosling, "Molecular configuration in sodium thymonucleate," *Nature* 171:740–41. Copyright 1953 by Macmillan Journals Limited.)

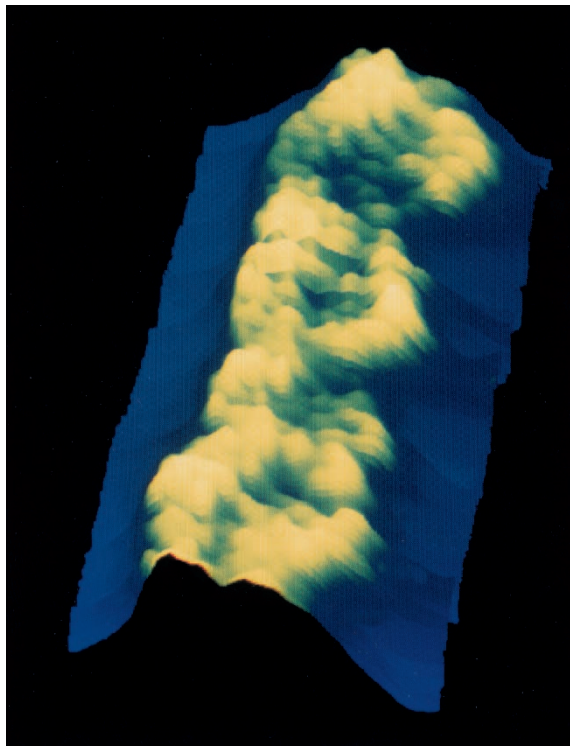
phosphate backbones on the outside and the bases on the inside. This structure would fit the dimensions X-ray crystallography had established for DNA if the bases from the two strands were opposite each other and formed "rungs" in a helical "ladder" (fig. 9.13). The diameter of the helix could only be kept constant at about 20 Å (10 angstrom units = 1 nanometer) if one purine and one pyrimidine base made up each rung. Two purines per rung would be too big, and two pyrimidines would be too small.

After further experimentation with models, Watson and Crick found that the hydrogen bonding necessary to form the rungs of their helical ladder could occur readily between certain base pairs, the pairs that Chargaff found in equal frequencies. (Hydrogen bonds are very weak bonds in which two electronegative atoms, such as O and N, share a hydrogen atom between them. They have 3 to 5% of the strength of a covalent bond.) Thermodynamically stable

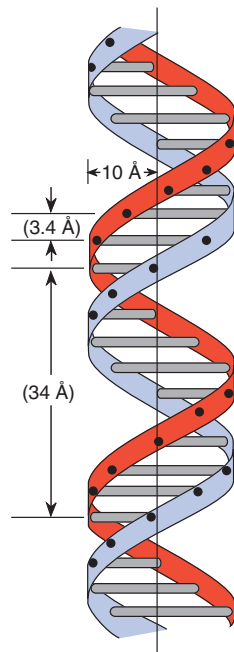
Table 9.3 Percentage Base Composition of Some DNAs

Species	Adenine	Thymine	Guanine	Cytosine
Human being (liver)	30.3	30.3	19.5	19.9
<i>Mycobacterium tuberculosis</i>	15.1	14.6	34.9	35.4
Sea urchin	32.8	32.1	17.7	18.4

Source: From E. Chargaff and J. Davidson, *The Nucleic Acids*, Academic Press, 1955.



(a)



(c)

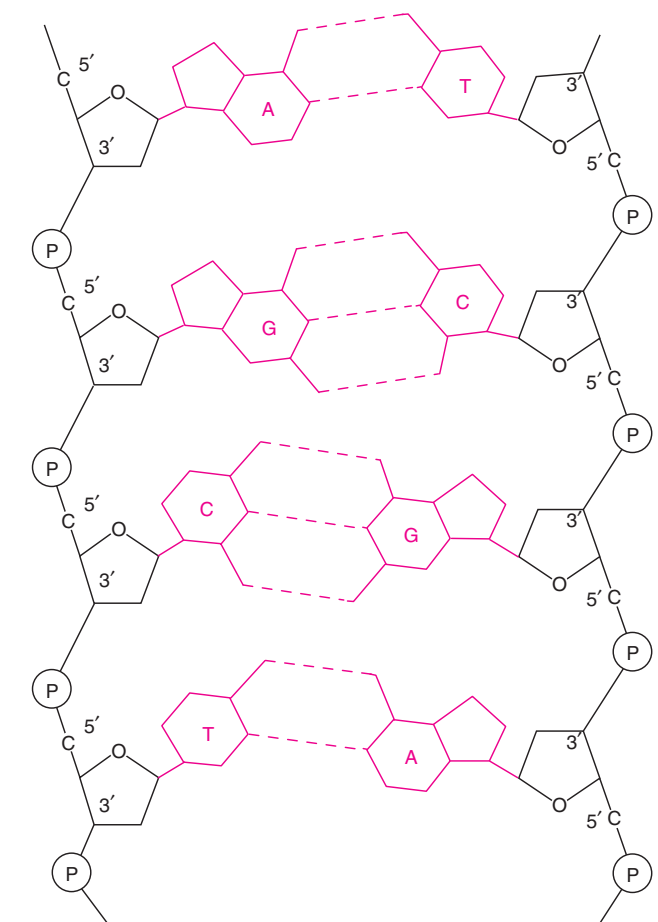
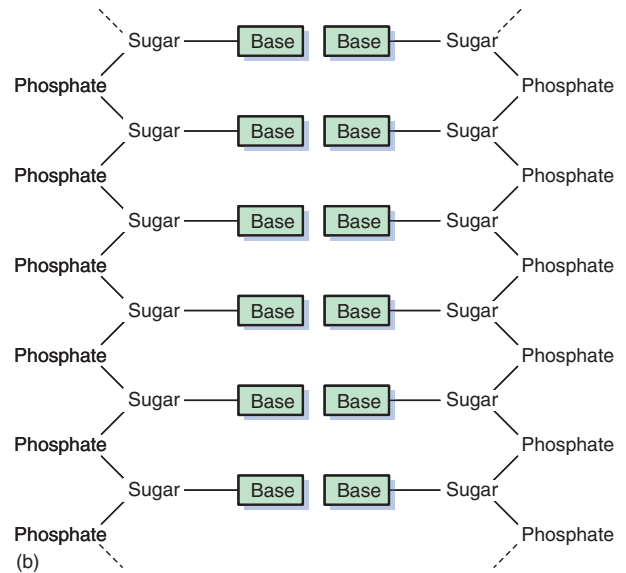


Figure 9.13 Double helical structure of DNA. (a) DNA magnified twenty-five million times by scanning tunneling microscopy. (b) Component parts. (c) Line drawing. ([a] © John D. Baldeschweiler.)

hydrogen bonding occurs between thymine and adenine and between cytosine and guanine (fig. 9.14). The relationship is one of **complementarity**. There are two hydrogen bonds between adenine and thymine and three between cytosine and guanine.

Another point about DNA structure relates to the **polarity** that exists in each strand. That is, one end of a DNA strand has a 5' phosphate and the other end has a 3' hydroxyl group. Watson and Crick found that hydrogen bonding would occur if the polarity of the two strands ran in opposite directions; that is, if the two strands were **antiparallel** (fig. 9.15).

DNA Denaturation

Denaturation studies indicated that the hydrogen bonding in DNA occurs in the way Watson and Crick suggested. Hydrogen bonds, although individually very weak, give structural stability to a molecule in large enough numbers. However, the hydrogen bonds can be broken and the DNA strands separated when the DNA molecule is heated in water. At a certain point, the thermal agitation overcomes the hydrogen bonding, and the molecule becomes **denatured** (or "melts"). It is logical that the more hydrogen bonds DNA contains the higher the temperature needed to denature it. It thus follows that since a G-C (guanine-cytosine)

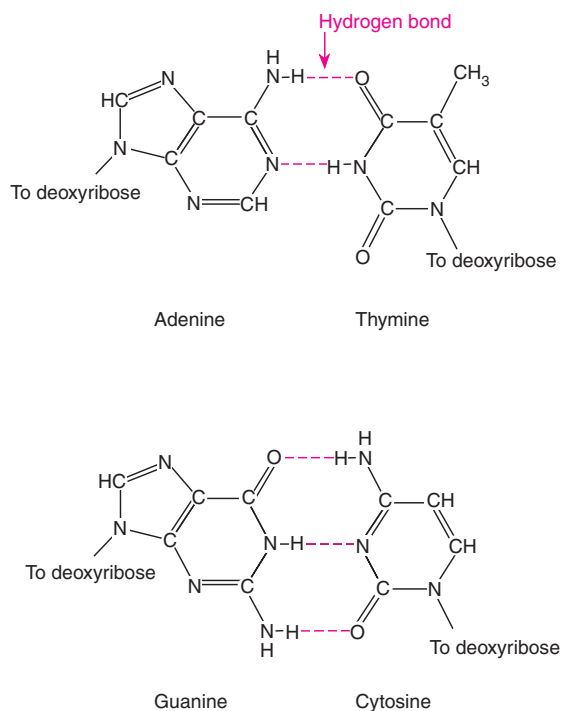


Figure 9.14 Hydrogen bonding between the nitrogenous bases in DNA.

base pair has three hydrogen bonds to every two in an A-T (adenine-thymine) base pair, the higher the G-C content in a given molecule of DNA, the higher the temperature required to denature it. This relationship exists (fig. 9.16).

Requirements of Genetic Material

Let us now return briefly to the requirements we have said a genetic material needs to meet: (1) control of protein synthesis, (2) self-replication, and (3) location on the chromosomes in the nucleus (in organisms with nuclei). Does DNA (or when DNA is absent, RNA) meet these requirements?

Control of Enzymes

In the next several chapters, we examine the details of protein synthesis. We will see that DNA does possess the complexity required to direct protein synthesis. Although complementarity restricts the base opposite a given base in a double helix, there are no restrictions on the sequence of bases on a given strand. Later, we will show that each sequence of three bases in DNA specifies a particular amino acid during protein synthesis. The **ge-**

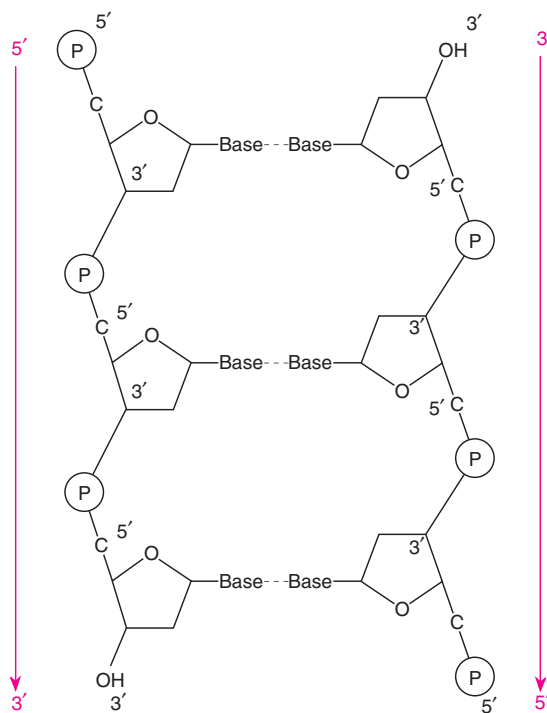


Figure 9.15 Polarity of the DNA strands. Polarity is established by the 3' and 5' carbons of a given sugar. For example, moving down the left strand, the polarity is 5' → 3' (read as *five-prime to three-prime*). Moving down the right strand, the polarity is 3' → 5'.

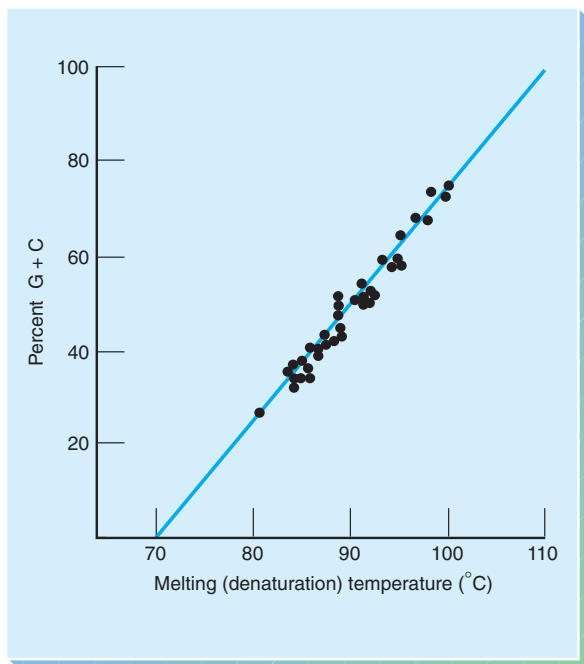


Figure 9.16 Relationship of the number of hydrogen bonds (G-C content) and the thermal stability of DNA from different sources. (From J. Marmur and P. Doty, Jr., "Relationship of the Number of Hydrogen Bonds and the Thermal Stability of DNA from Different Sources," *Journal of Molecular Biology*, 5:109–112. Copyright © 1962 Academic Press LTD.)

netic code gives the relationship of DNA bases to the amino acids in proteins.

Replication

Watson and Crick hinted in their 1953 paper how DNA might replicate. Their observation stemmed from the property of complementarity. Since the base sequence on one strand is complementary to the base sequence on the opposite strand, each strand could act as a template for a new double helix if the molecule simply "unzipped," allowing each strand to specify the sequence of bases on a new strand by complementarity (fig. 9.17). Mutability would occur due to mispairings, other errors in replication, or damage to the DNA.

Location

DNA must reside in the nucleus of eukaryotes, where the genes occur on chromosomes, or in the chromosomes of prokaryotes and viruses. In both prokaryotes and eukaryotes, the majority of the cell's DNA is in the chromosomes. And all viruses contain either DNA or RNA. Thus, DNA fulfills all the requirements of a genetic material. RNA can fulfill the same requirements in RNA viruses and viroids.

Alternative Forms of DNA

The form of DNA we have described so far is called **B DNA**. It is a right-handed helix: it turns in a clockwise manner when viewed down its axis. The bases are stacked almost exactly perpendicular to the main axis, with about ten base pairs per turn (34 Å; see fig. 9.13c). However, DNA can exist in other forms. If the water content increases to about 75%, the A form of DNA (**A DNA**) occurs. In this form, the bases tilt in regard to the axis, and there are more base pairs per turn. However, this and other known forms of DNA are relatively minor variations on the right-handed B form.

In 1979, Alexander Rich and his colleagues at MIT discovered a left-handed helix that they called **Z DNA** because its backbone formed a zigzag structure (fig. 9.18). Z DNA was found by X-ray crystallographic analysis of very small DNA molecules composed of repeating G-C sequences on one strand with the complementary C-G sequences on the other (alternating purines and pyrimidines). Z DNA looks like B DNA with each base rotated 180 degrees, resulting in a zigzag, left-handed structure (fig. 9.19). (The original configuration of the bases is referred to as the *anti* configuration; the rotated configuration is called the *syn* configuration.)

Originally, it was thought that Z DNA would not prove of interest to biologists because it required very high salt concentrations to become stable. However, it was found that Z DNA can be stabilized in physiologically normal conditions if methyl groups are added to the cytosines. Z DNA may be involved in regulating gene expression in eukaryotes. We return to this topic in chapter 16 (box 9.3).

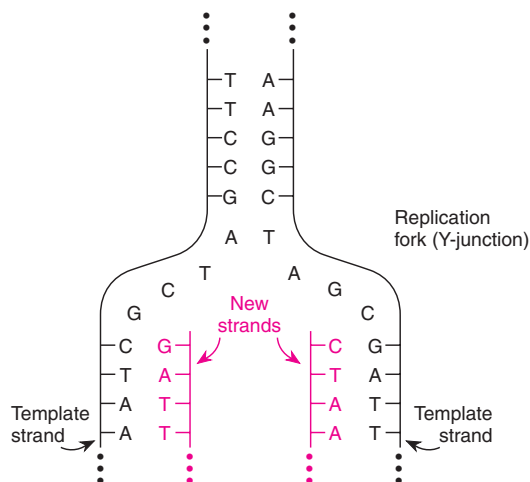


Figure 9.17 Complementarity provides a possible mechanism for accurate DNA replication. The parent duplex opens, and each strand becomes a template for a new duplex.

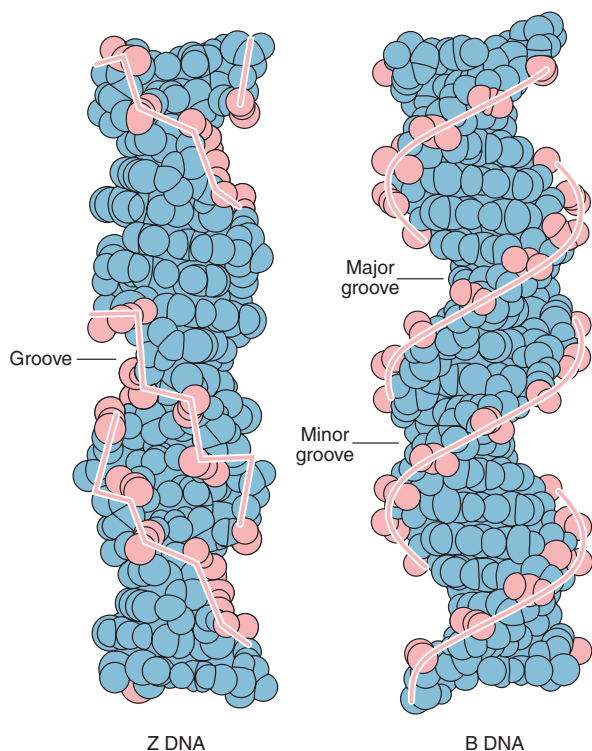


Figure 9.18 Z (left) and B (right) DNA. The lines connect phosphate groups. (Reproduced with permission from the *Annual Review of Biochemistry*, Volume 53, © 1984 by Annual Reviews, Inc.)

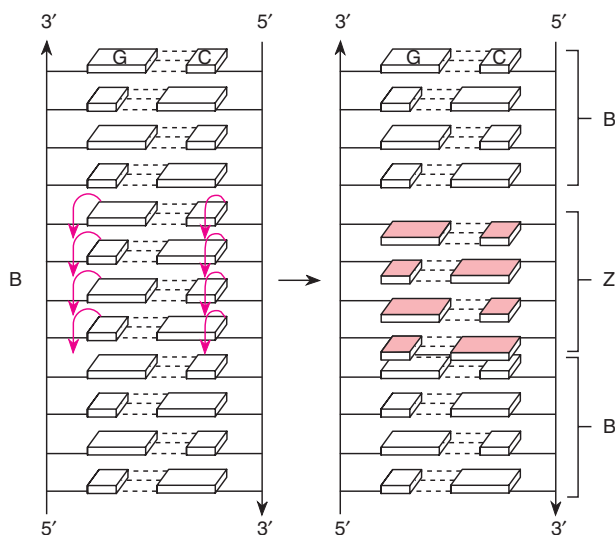


Figure 9.19 B DNA converts to Z DNA by the rotation of bases as indicated by the curved arrows. (Reproduced with permission from *Annual Review of Biochemistry*, Volume 53, © 1984 by Annual Reviews, Inc.)

DNA REPLICATION— THE PROCESS

In their 1953 paper, Watson and Crick hinted that the replication of the double helix could take place as the DNA unwinds, so that each strand would form a new double helix by acting as a **template** for a newly synthesized strand (see fig. 9.17). For example, when a double helix is unwound at an adenine-thymine (A-T) base pair, one unwound strand would carry A and the other would carry T. During replication, the A in the template DNA would pair with a T in a newly replicated DNA strand, giving rise to another A-T base pair. Similarly, the T in the other template strand would pair with an A in the other newly replicated strand, giving rise to another A-T base pair. Thus, one A-T base pair in one double helix would result in two A-T base pairs in two double helices. This process would repeat at every base pair in the double helix of the DNA molecule.

This mechanism is called **semiconservative** replication because, although the entire double helix is not conserved in replication, each strand is. Every daughter DNA molecule has an intact template strand and a newly replicated strand. This is not the only way that replication could occur. The alternative methods are **conservative** and **dispersive**. In conservative replication, in which the whole original double helix acts as a template for a new one, one daughter molecule would consist of the original parental DNA, and the other daughter would be totally new DNA. In dispersive replication, some parts of the original double helix are conserved, and some parts are not. Daughter molecules would consist of part template and part newly synthesized DNA. In reality, the dispersive category is the all-inclusive “other” category, including any possibility other than conservative and semiconservative replication.

The Meselson and Stahl Experiment

In 1958, M. Meselson and F. Stahl reported the results of an experiment designed to determine the mode of DNA



Matthew Meselson
(1930–). (Courtesy of
Dr. Matthew Meselson.
Photograph by Bud Gruce.)



Franklin W. Stahl
(1929–). (Courtesy of
Dr. Franklin W. Stahl.)

BOX 9.3

Biomedical
Applications

Multiple-Stranded DNA

Under natural conditions, single-stranded RNA and double-stranded DNA are the rule. However, under laboratory conditions, it is possible to induce a third strand of DNA to interdigitate itself into the major groove of the double helix of normal DNA in a sequence-specific fashion. That is, the third strand of DNA will not just interdigitate anywhere, but will form a stable triplex at a specific sequence (fig. 1). The rules of binding are a little less precise than normal; not all sequences are recognized, and recognition can depend on surrounding sequences. However, a thymine in the third strand will usually recognize an adenine in an adenine-thymine base pair (T•A-T), and a cytosine in the third strand will recognize a guanine in a guanine-cytosine base pair (C+G-C).

Triple-stranded nucleotide chains were first created in 1957 by three scientists at the National Institutes of Health—Alexander Rich, David Davies, and Gary Felsenfeld—while they were creating artificial nucleic acids. At the time, triple-stranded DNA seemed like a laboratory curiosity. Now it seems of interest because it may have valuable uses both experimentally and clinically. (Rich appar-

ently had the same experience in his codiscovery of Z DNA, which at first seemed like an oddity but now is the focus of some attention—see chapter 16.) Now, researchers are able to form triplexes in naturally occurring DNA. Two applications of this technology are actively being pursued.

Both applications arise because a single strand of DNA is capable of recognizing a relatively long sequence of the double-stranded DNA in a chromosome. Thus, it is possible to selectively locate a particular genic sequence. Once the third strand locates a particular sequence on a chromosome, two things can happen. First, triplex DNA formation can prevent a particular gene from expressing itself. By the same technique, triplex DNA can also be an abortifacient, a safe method for preventing the implantation of a fetus by preventing the expression of genes under the control of the hormone progesterone.

The second use of triplex DNA is to cut DNA at a specific place by adding a cleaving compound to both ends of the third strand of DNA. Once the third strand has interdigitated, it can then break the original double helix. For example, S. Strobel and P. Dervan at the California Institute of Technology have used a chemical complex containing iron attached to both 3' and 5' ends of the third strand of DNA. The addition of a third chemical then initiates the cleavage reaction. The cleavage of the original duplex can be of value in modern recombinant DNA technology (see chapter 13). Whether triplex DNA will ever be of value is not certain at this time. However, it seems to have good potential for therapeutic use and to help in studying and mapping the human genome.

More recently, four-stranded DNA molecules have been found, in which double helices of certain sequences interdigitate to form four-stranded structures. These may be of importance in the formation of crossover sites or in the structures at the ends of eukaryotic chromosomes (see chapter 15).

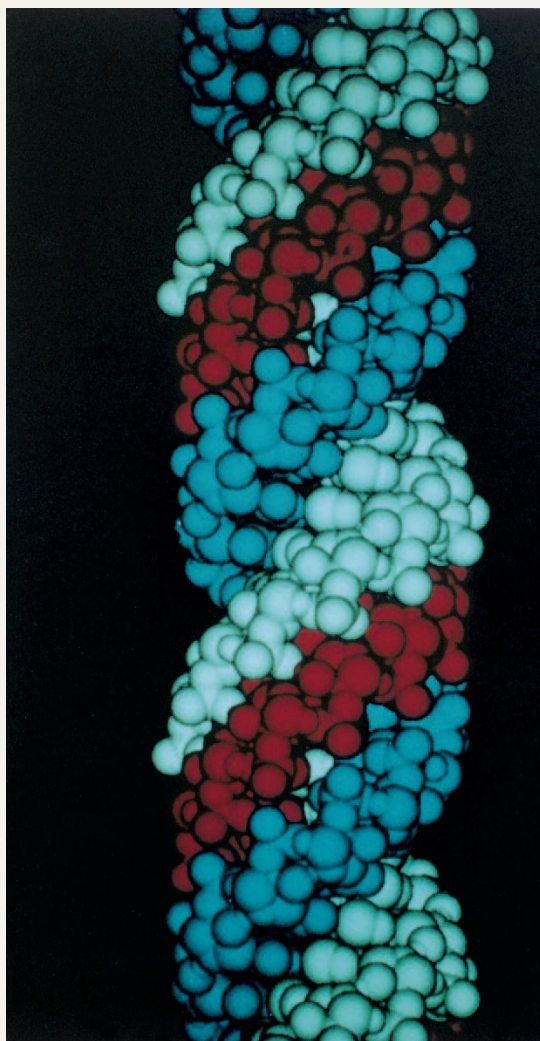
continued

replication. Some historians and philosophers of science consider this the most elegant scientific experiment ever designed. Meselson and Stahl grew *E. coli* in a medium containing a heavy isotope of nitrogen, ^{15}N . (The normal form of nitrogen is ^{14}N .) After growing for several generations on the ^{15}N medium, the DNA of *E. coli* was denser. The researchers determined the density of the strands using a technique known as **density-gradient centrifugation**. In this technique, a cesium chloride (CsCl) solution is spun in an ultracentrifuge at high speed for several hours. Eventually an equilibrium arises between centrifugal force and diffusion, so that a density gradient is established in the tube with an increasing concentration of CsCl from top to bottom. If DNA (or any other substance) is added, it concentrates and forms a band in the tube at the point where its density is the same as that of the CsCl. If several types of DNA with different densities are added, they form several bands. The bands are detectable

under ultraviolet light at a wavelength of 260 nm (nanometers), which nucleic acids absorb strongly.

Meselson and Stahl transferred the bacteria with heavy (^{15}N) DNA to a medium containing only ^{14}N . The new DNA, replicated in the ^{14}N medium, was intermediate in density between light (^{14}N) and heavy (^{15}N) DNA, because the replication was semiconservative (fig. 9.20). If replication had been conservative, two bands would have appeared at the first generation of replication—an original ^{15}N DNA and a new ^{14}N double helix. And, throughout the experiment, if the method of replication had been conservative, the original DNA would have continued to show up as a ^{15}N band. (This, of course, did not happen.) If the method of replication had been dispersive, various multiple-banded patterns would have appeared, depending on the degree of dispersiveness. The results figure 9.20 shows are completely consistent with semiconservative replication and only with semiconservative replication.

BOX 9.3 CONTINUED



(a)



(b)

3'	5'
T	- A
G	- C
C	- G
C	- G
T	- A
A	- T
G	- C
G	- C
* T	• A - T
T	• A - T
T	• A - T
T	• A - T
T	• A - T
C + G	- C
T	• A - T
T	• A - T
T	• A - T
C + G	- C
T	• A - T
T	• A - T
T	• A - T
T	• A - T
C + G	- C
T	• A - T
* T	• A - T
	G - C
	G - C
	C - G
	C - G
	C - G
	A - T
	G - C

(c)

Figure 1 A triple helix of DNA. (a) Space-filling model of (b). (c) Sequence of third strand on *left* and double helix on *right*. The asterisks in (c) and the Fe symbols in (b) refer to the attached iron compound. (© Courtesy Dr. Michael E. Hogan, Triplex Pharmaceutical Corporation.)

Autoradiographic Demonstration of DNA Replication

In 1963, J. Cairns used **autoradiography** to verify the semiconservative method of replication photographically. This technique makes use of the fact that radioactive atoms expose photographic film. The visible silver grains on the film can then be counted to provide an es-

timate of the quantity of radioactive material present. Cairns grew *E. coli* bacteria in a medium containing radioactive thymine, a component of one of the DNA nucleotides. The radioactivity was in tritium (^3H). Cairns then carefully extracted the DNA from the bacteria and placed it on photographic emulsion for a period of time. He developed the emulsion to produce autoradiographs

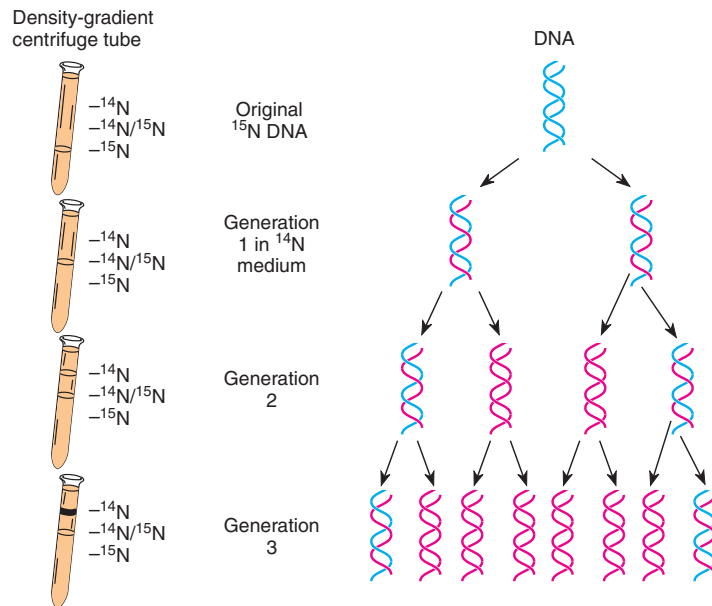
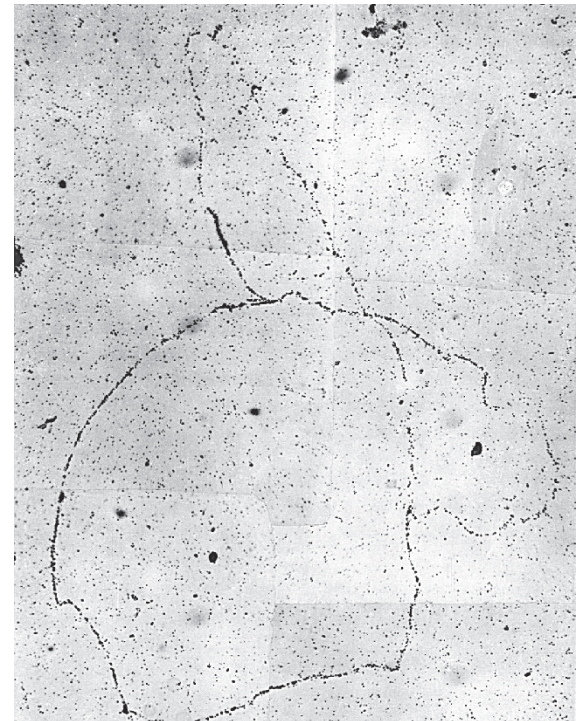


Figure 9.20 The Meselson and Stahl experiment to determine the mode of DNA replication. The bands in the centrifuge tube are visible under ultraviolet light. The pattern of bands (left) comes about from semiconservative DNA replication (right) of ^{15}N DNA (blue) replicating in a ^{14}N medium (red).

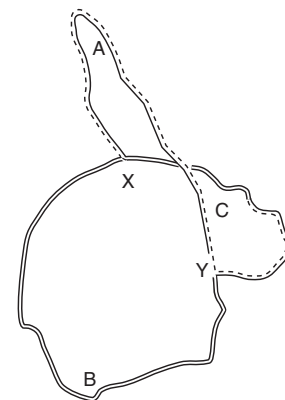


(a)

that he then examined under the electron microscope (fig. 9.21). Each grain of silver represents a radioactive decay. Interpretation of this autoradiograph reveals several points. The first, known at the time, is that the *E. coli* DNA is a circle. The second point is that the DNA is replicated while maintaining the integrity of the circle. That is, the circle does not appear to break during the process of DNA replication; an intermediate **theta structure** forms (topologically similar in shape to the Greek letter theta, θ). Third, replication of the DNA seems to be occurring at one or two moving **Y-junctions** in the circle, which further supports the semiconservative mode of replication. The DNA is unwound at a given point, and replication proceeds at a Y-junction, in a semiconservative manner, in one or both directions (see fig. 9.17).

Figure 9.22 diagrams the way in which the two Y-junctions move along the circle to the final step, forming two new circles. The steps by themselves do not support either a unidirectional or a bidirectional mode of replication. That is, a theta structure will develop if either one or both Y-junctions is active in replication. But with autoradiography, it is possible to determine whether new growth is occurring in only one or in both directions.

In some cases, radioactivity was not applied to the cell until DNA replication had already begun. In these cases, the radioactive label appeared after the theta structure



(b)

Figure 9.21 (a) Autoradiograph of *E. coli* DNA during replication. (b) Diagram has labels on the three segments, A, B, and C, created by the existence of two forks, X and Y, in the DNA. Forks are created when the circle opens for replication. Length of the chromosome is about 1,300 μm . ([a] From J. Cairns, "The chromosome of *E. coli*", *Cold Spring Harbor Symposia on Quantitative Biology*, 28. Copyright © 1963 by Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY. Reprinted by permission.)

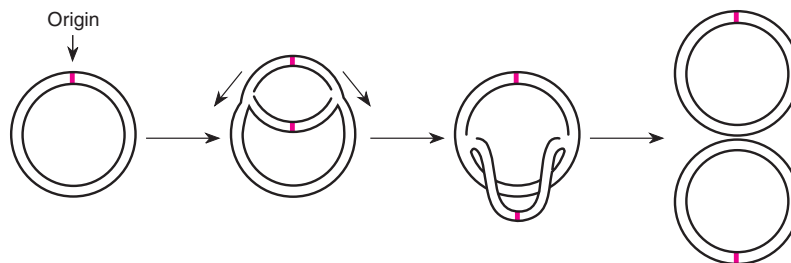


Figure 9.22 Observable stages in the DNA replication of a circular chromosome, assuming bidirectional DNA synthesis. The intermediate figures are called theta structures.

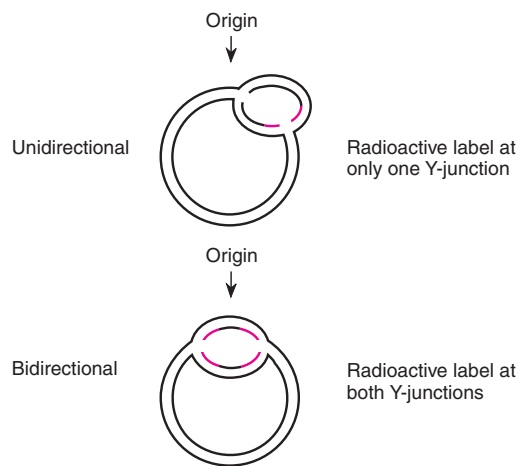
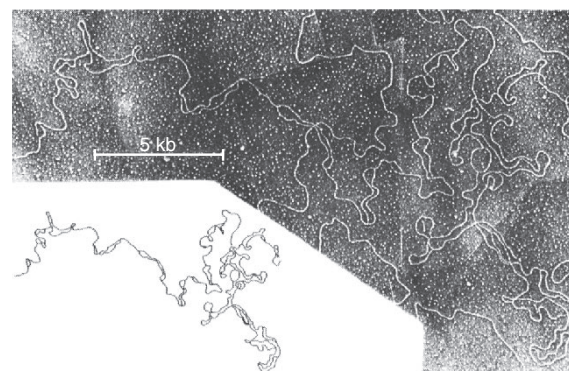
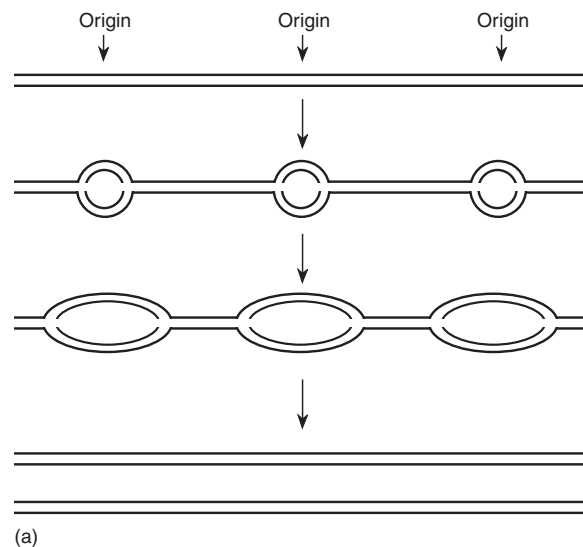


Figure 9.23 Radioactive labels distinguish unidirectional from bidirectional DNA replication. In these hypothetical experiments, DNA replication was allowed to begin, and then a radioactive label was added. After a short period of time, the process was stopped and the autoradiographs prepared. In bidirectional replication (the actual case), the label appears at both Y-junctions.

had already begun forming. Figure 9.23 illustrates hypothetical outcomes for either unidirectional or bidirectional replication. By counting silver grains in autoradiographs, Cairns found growth to be bidirectional. Both autoradiographic and genetic analysis have subsequently verified this finding.

In eukaryotes, the DNA molecules (chromosomes) are larger than in prokaryotes and are not circular; there are also usually multiple sites for the initiation of replication. Thus, each eukaryotic chromosome is composed of many replicating units, or **replicons**—stretches of DNA with a single origin of replication. In comparison, the *E. coli* chromosome is composed of only one replicon. In eukaryotes, these replicating units form “bubbles” (or “eyes”) in the DNA during replication (fig. 9.24).



(b)

Figure 9.24 Replication bubbles. (a) Formation of bubbles (eyes) in eukaryotic DNA because of multiple DNA synthesis sites of origin. (b) Electron micrograph (and explanatory line drawing) of replicating *Drosophila* DNA showing these bubbles. ([b] H. Kreigstein and D. Hogness, “Mechanism of DNA replication in *Drosophila* chromosomes: Structure of replication forks and evidence for bidirectionality,” *Proceeding of the National Academy of Sciences USA*, 71 (1974):135–39. Reproduced by permission.)

DNA REPLICATION— THE ENZYMOLOGY



Let us turn now to the details of the processes that take place during DNA replication. Like virtually all metabolic processes, DNA replication is under the control of enzymes. The evidence for the details we describe comes from physical, chemical, and biochemical studies of enzymes and nucleic acids and from the analysis of mutations that influence the replication processes. More recent techniques of recombinant DNA technology and nucleotide sequencing have allowed us to determine the nucleotide sequences of many of these key regions in DNA and RNA. We will look first at *E. coli*.

There are three major enzymes that will polymerize nucleotides into a growing strand of DNA in *E. coli*. These enzymes are **DNA polymerase I, II, and III**. DNA polymerase I, discovered by Arthur Kornberg, who subsequently won the Nobel Prize for his work, is primarily utilized in filling in small DNA segments during replication and repair processes. DNA polymerase II can serve as an alternative repair polymerase; it can also replicate DNA if the template is damaged. DNA polymerase III is the primary polymerase during normal DNA replication.



Arthur Kornberg (1918–). (Courtesy of Dr. Arthur Kornberg. Photograph by Karsh.)

In the simplest model of DNA replication, new nucleotides would be simultaneously added, according to the rules of complementarity, on both strands of newly synthesized DNA at the replication fork as the DNA opens up. But a problem exists, created by DNA's antiparallel nature; the two strands of a DNA double helix run in opposite directions. Going in one direction on the duplex, for example, one strand is a $5' \rightarrow 3'$ strand, whereas the other is a $3' \rightarrow 5'$ strand. These directions

refer to the numbering of carbon atoms across the sugar. In figure 9.25, going from the bottom of the figure to the top, the left-hand strand is a $3' \rightarrow 5'$ strand, and the right-hand strand is a $5' \rightarrow 3'$ strand. Since DNA replication involves the formation of two new antiparallel strands with the old single strands as templates, one new strand would have to be replicated in the $5' \rightarrow 3'$ direction and the other in the $3' \rightarrow 5'$ direction.

However, all the known polymerase enzymes add nucleotides in only the $5' \rightarrow 3'$ direction. That is, the polymerase catalyzes a bond between the first $5'$ - PO_4 group of a new nucleotide and the $3'$ -OH carbon of the last nucleotide in the newly synthesized strand (fig. 9.25). The polymerases cannot create the same bond with the $5'$ phosphate of a nucleotide already in the DNA and the $3'$ end of a new nucleotide. Thus, the simple model needs some revision.

Continuous and Discontinuous DNA Replication



Autoradiographic evidence leads us to believe that replication occurs simultaneously on both strands. **Continuous replication** is, of course, possible on the $3' \rightarrow 5'$ template strand, which begins with the necessary $3'$ -OH **primer**. (Primer is double-stranded DNA—or, as we shall see, a DNA-RNA hybrid—continuing as single-stranded DNA template. The strand being synthesized has a $3'$ -OH available; fig. 9.26.) A **discontinuous** form of replication takes place on the complementary strand, where it occurs in short segments, moving backward, away from the Yjunction (fig. 9.27). These short segments, called **Okazaki fragments** after R. Okazaki, who first saw them, average about 1,500 nucleotides in prokaryotes and 150 in eukaryotes. The strand synthesized continuously is referred to as the **leading strand**, and the strand synthesized discontinuously is referred to as the **lagging strand**.

Once initiated, continuous DNA replication can proceed indefinitely. DNA polymerase III on the leading-strand template has what is called high **processivity**: once it attaches, it doesn't release until the entire strand is replicated. Discontinuous replication, however, requires the repetition of four steps: primer synthesis, elongation, primer removal with gap filling, and ligation.

Primer Synthesis and Elongation

To synthesize Okazaki fragments, a primer must be created *de novo* (Latin: from the beginning). None of the DNA polymerases can create that primer. Instead, **primase**, an RNA polymerase coded for by the *dnaG* gene, creates the primer, ten to twelve nucleotides, at the site

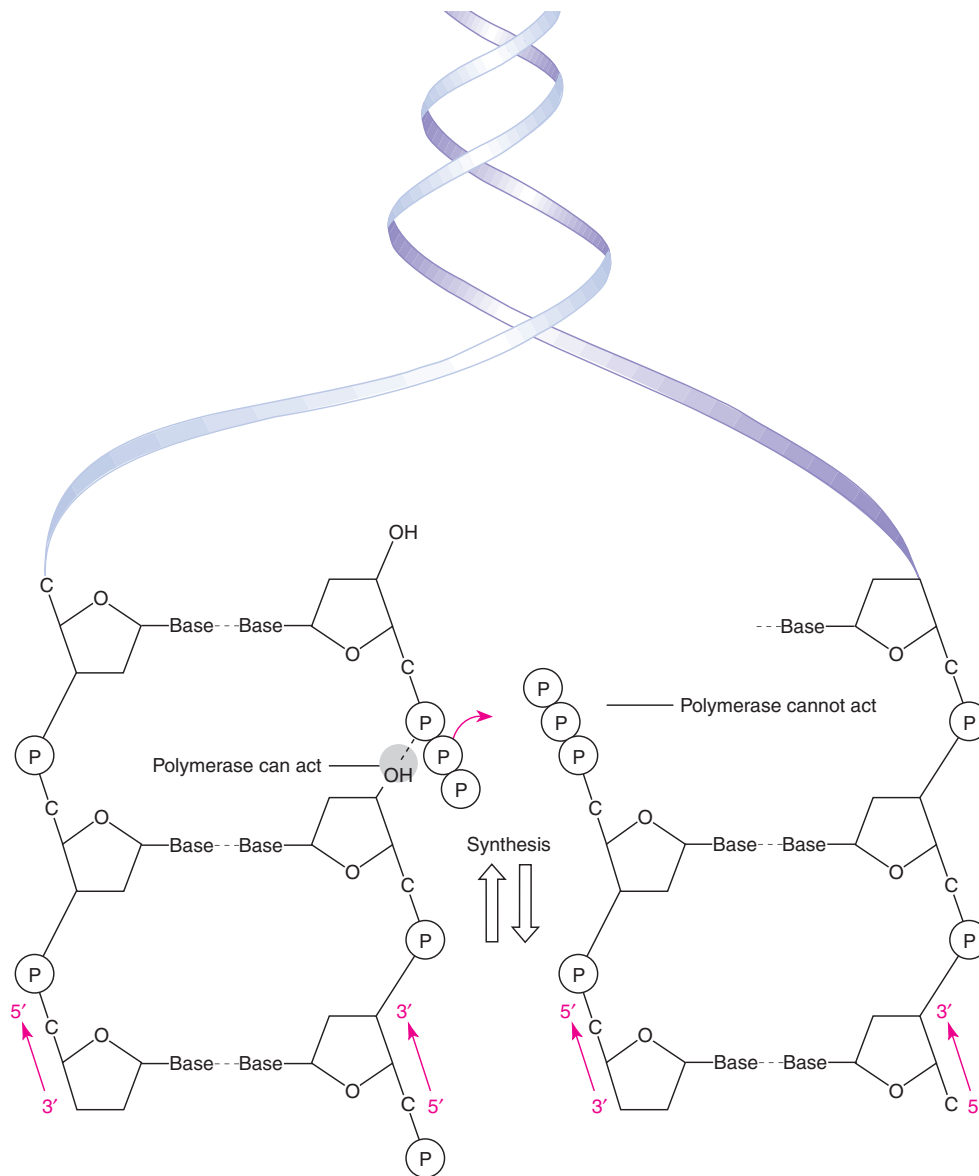


Figure 9.25 New nucleotides can be added to DNA only during replication in the 5' → 3' direction.

of Okazaki fragment initiation (fig. 9.28). The result is a short RNA primer that provides the free 3'-OH group that DNA polymerase III needs in order to synthesize the Okazaki fragment. DNA polymerase III continues until it reaches the primer RNA of the previously synthesized Okazaki fragment. At that point, it stops and releases from the DNA.

All three prokaryotic polymerases not only can add new nucleotides to a growing strand in the 5' → 3' di-

rection, but also can remove nucleotides in the opposite 3' → 5' direction. This property is referred to as 3' → 5' *exonuclease activity*. Enzymes that degrade nucleic acids are nucleases. They are classified as **exonucleases** if they remove nucleotides from the end of a nucleotide strand or as **endonucleases** if they can break the sugar-phosphate backbone in the middle of a nucleotide strand. At first glance, exonuclease activity seems like an extremely curious property for a polymerase to have—